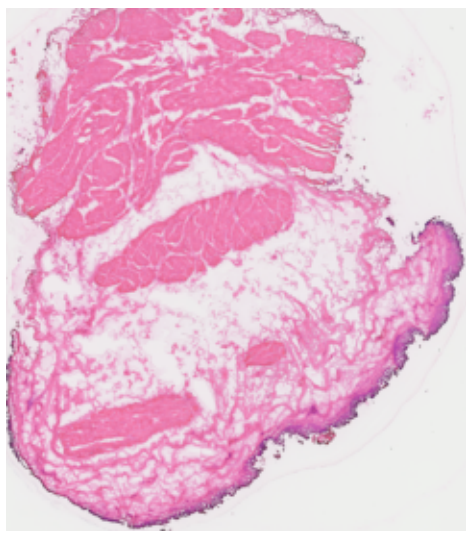
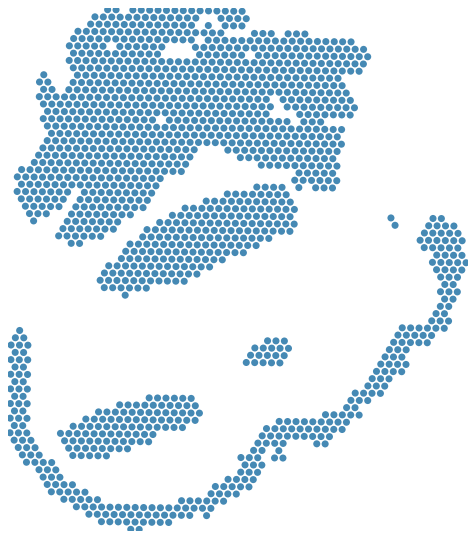
GSE285715_N1 | Tissue context

Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

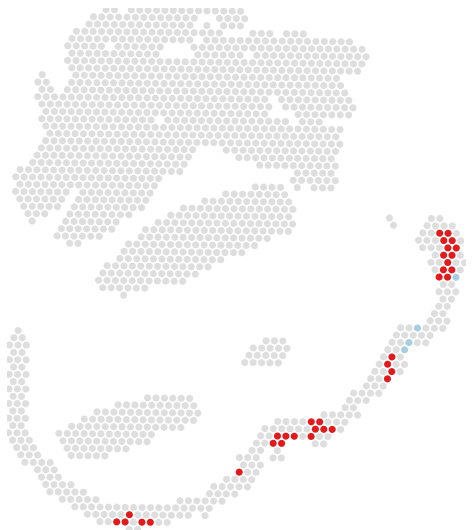
H&E and observed lesion edge



Final candidate location

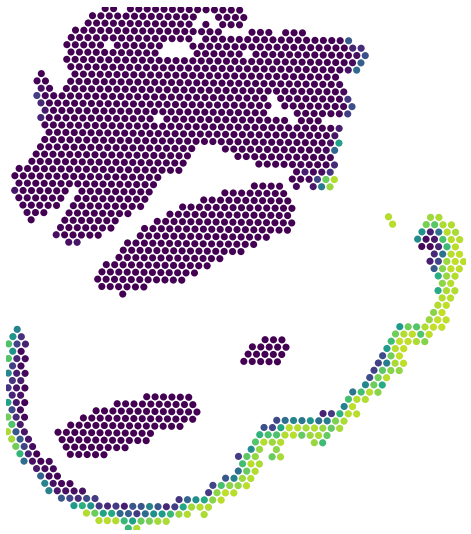


Accepted R12 (original 0–8)

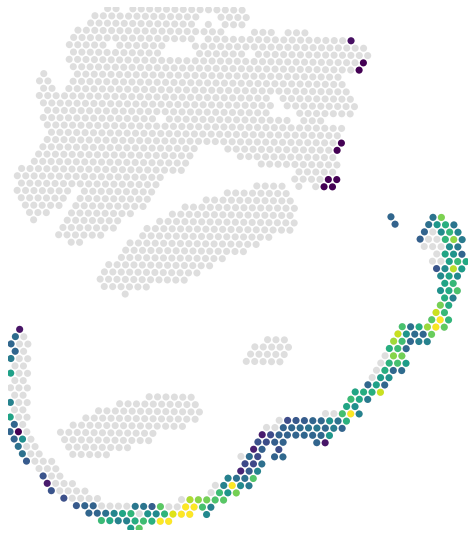


No observed candidate outer edge

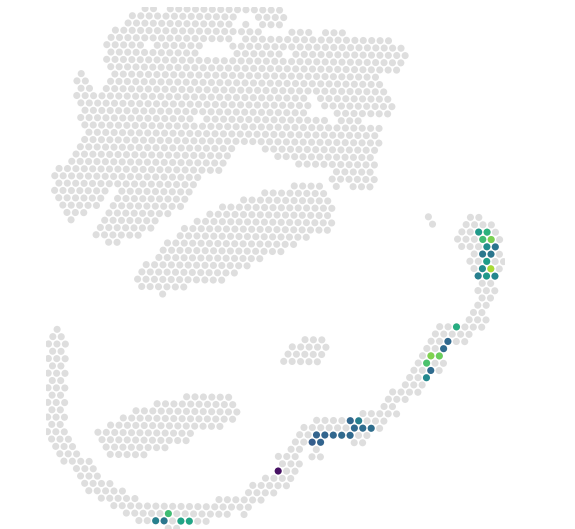
Epithelial support



MP10 in Epi queries

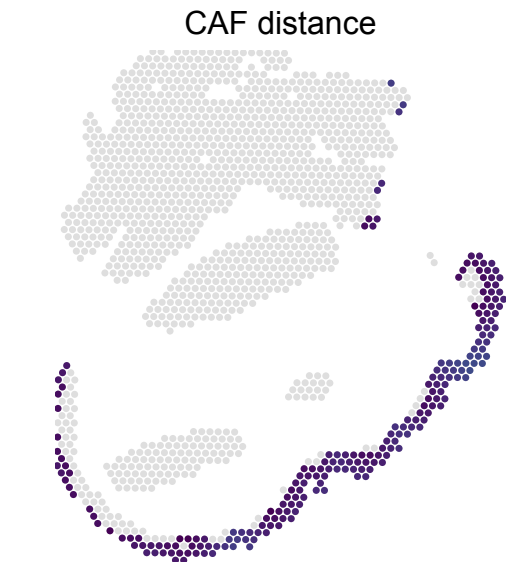


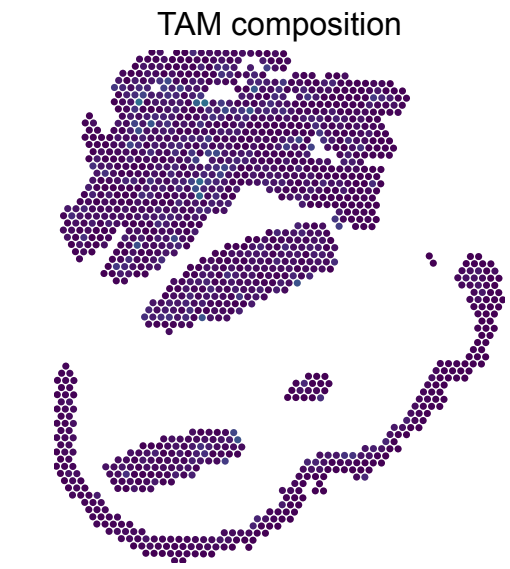
Epi-component MP10 (valid only)

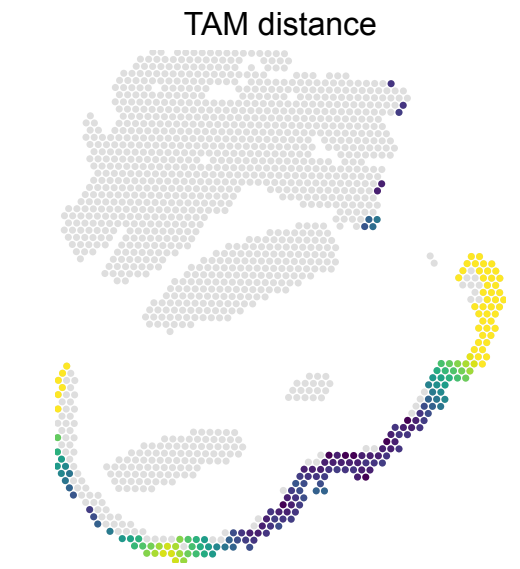


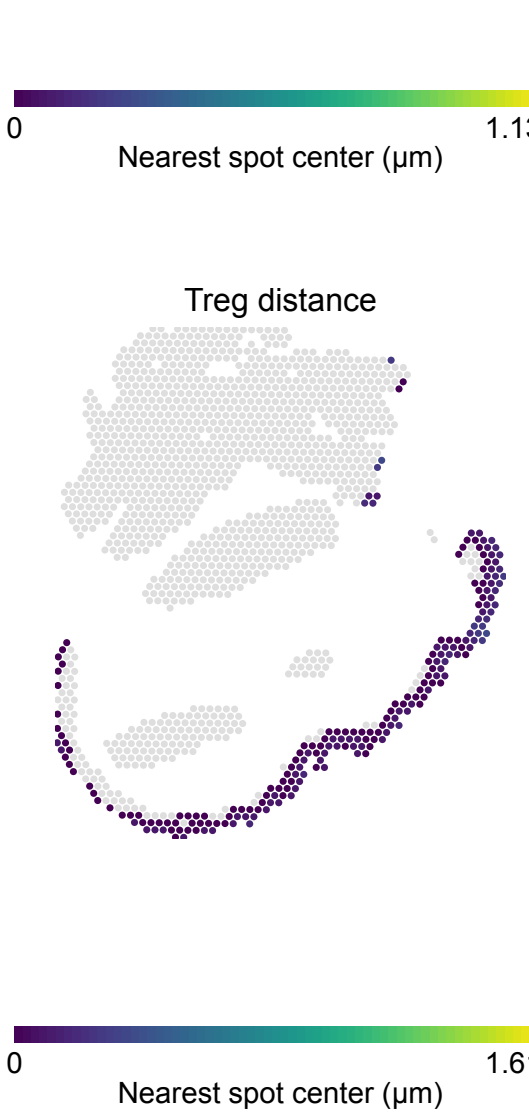
GSE285715_N1 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries



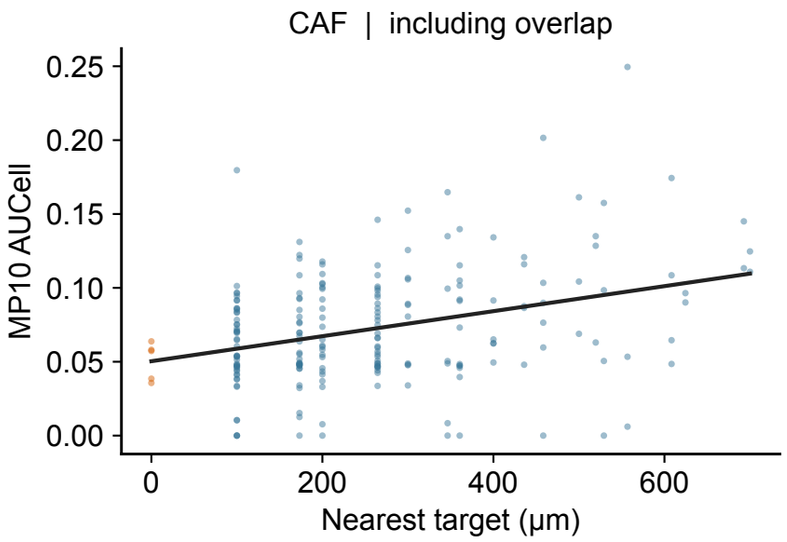




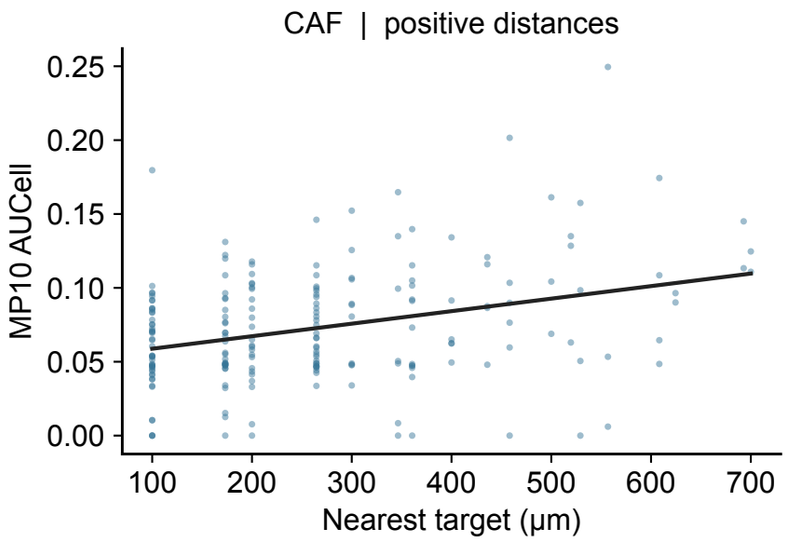


GSE285715_N1 | MP10 versus nearest target

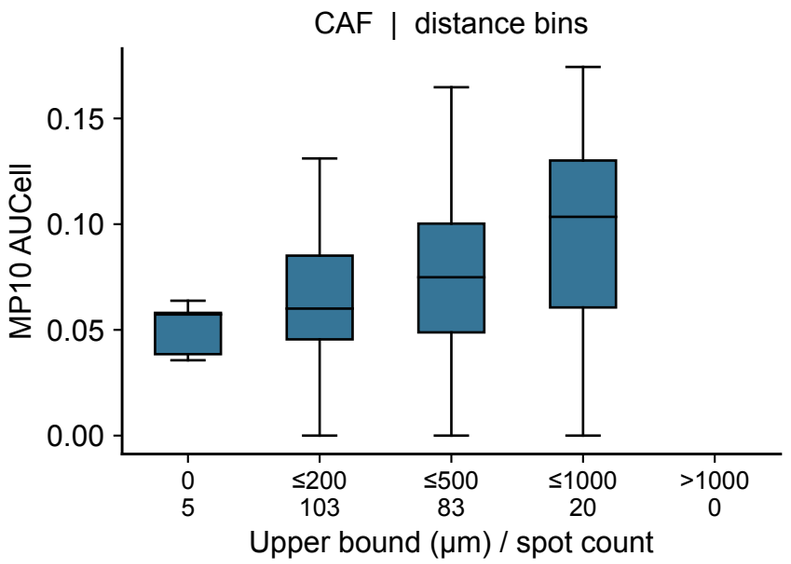
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



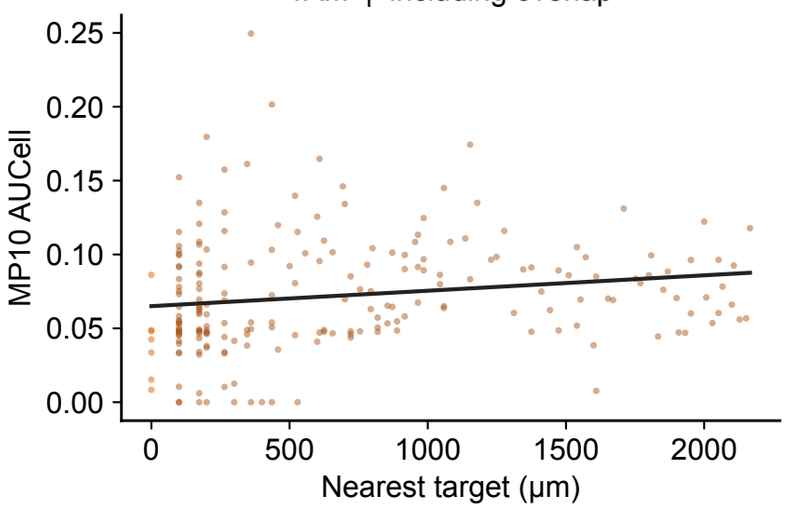
r +0.34 ρ +0.30
n 211/213 queries • 284 targets



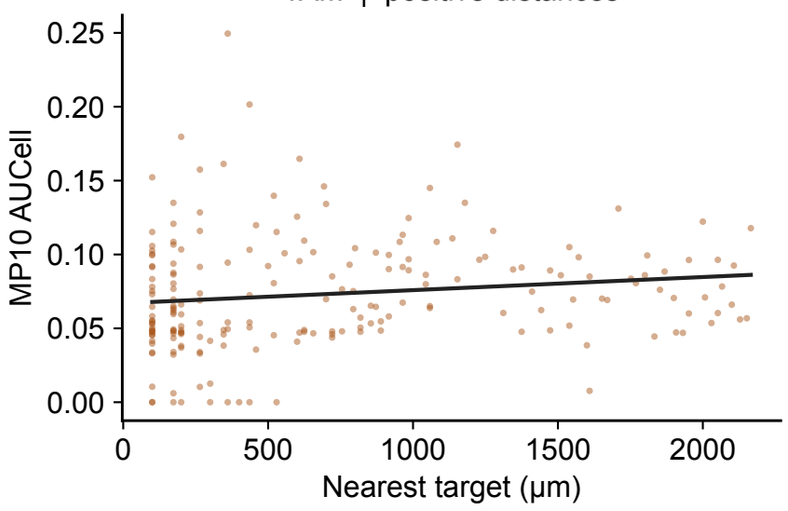
r +0.33 ρ +0.29
n 206/213 queries • 284 targets



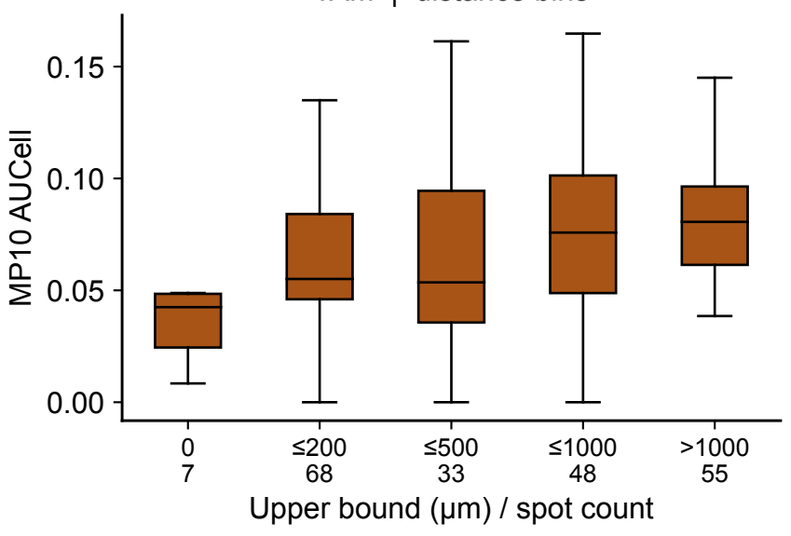
Orange = zero distance
Line = within-section linear fit



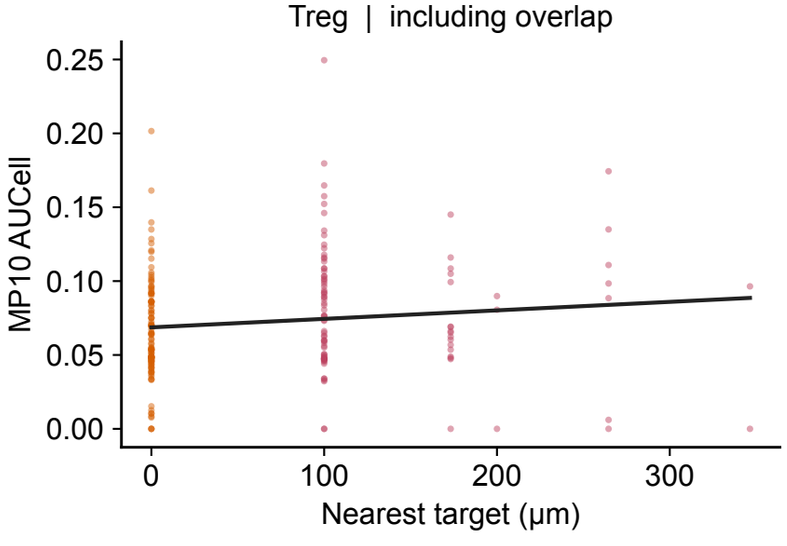
r +0.17 ρ +0.27
n 211/213 queries • 284 targets



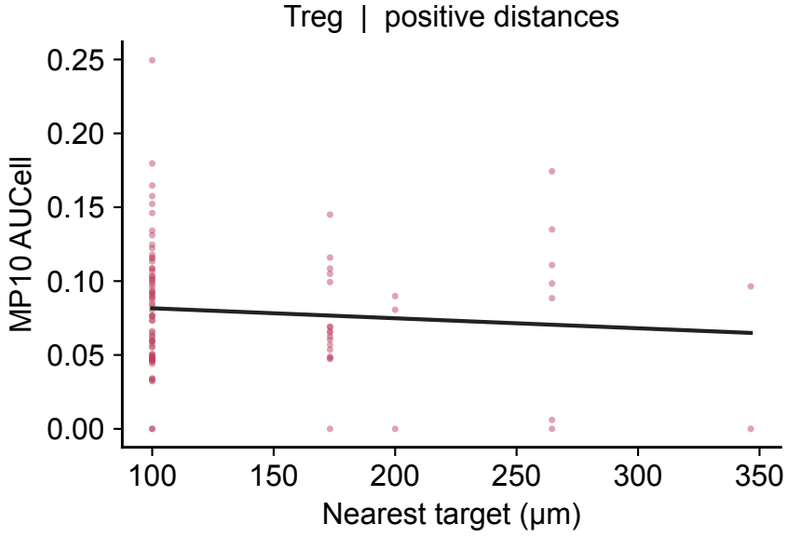
r +0.14 ρ +0.24
n 204/213 queries • 284 targets



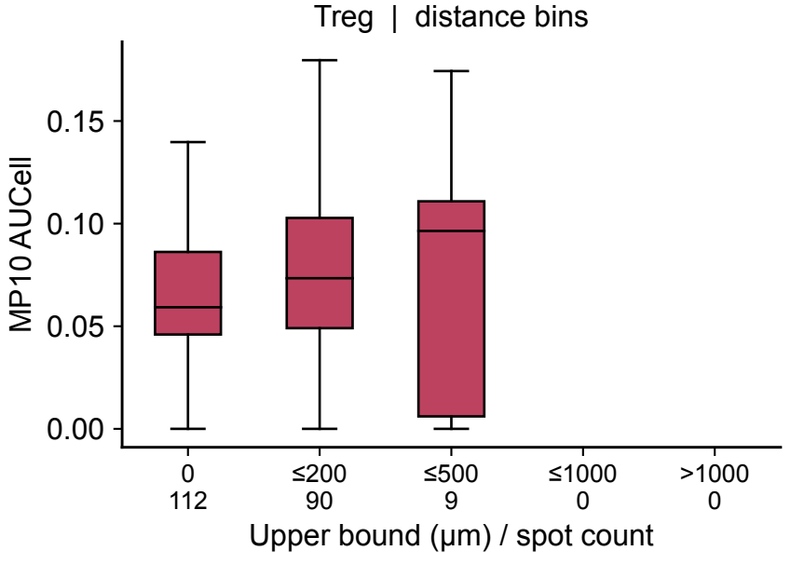
Orange = zero distance
Line = within-section linear fit



r +0.11 ρ +0.16
n 211/213 queries • 284 targets



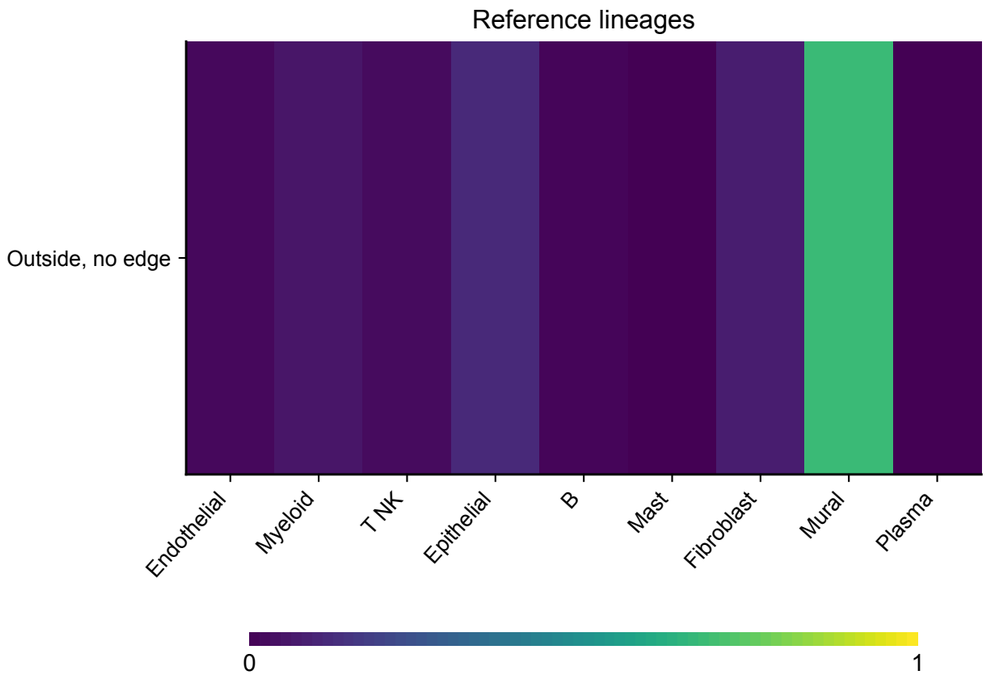
r -0.09 ρ -0.02
n 99/213 queries • 284 targets

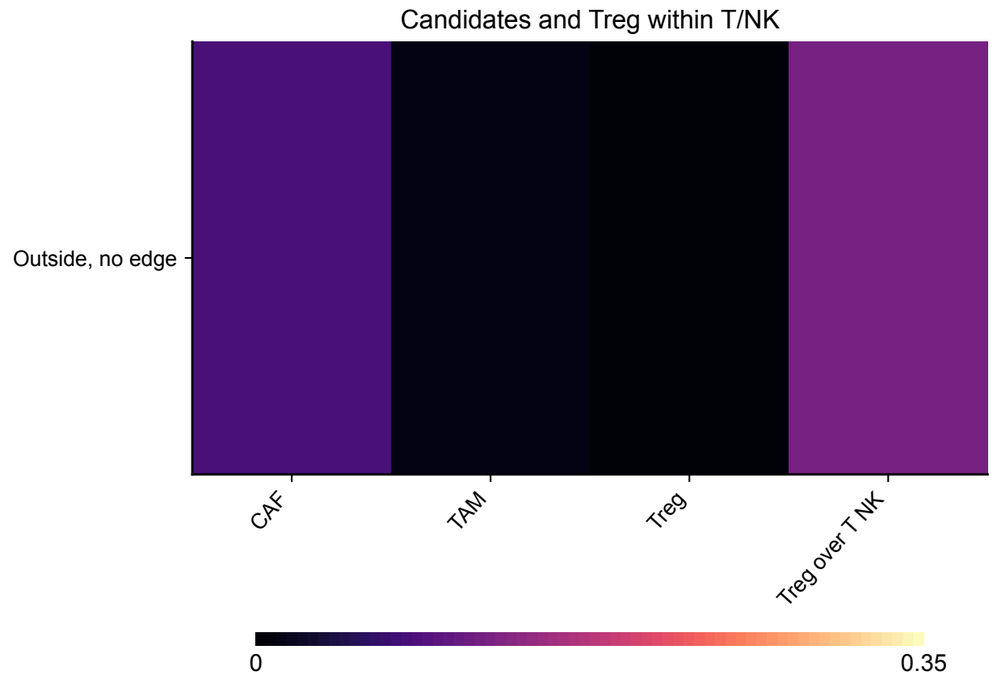


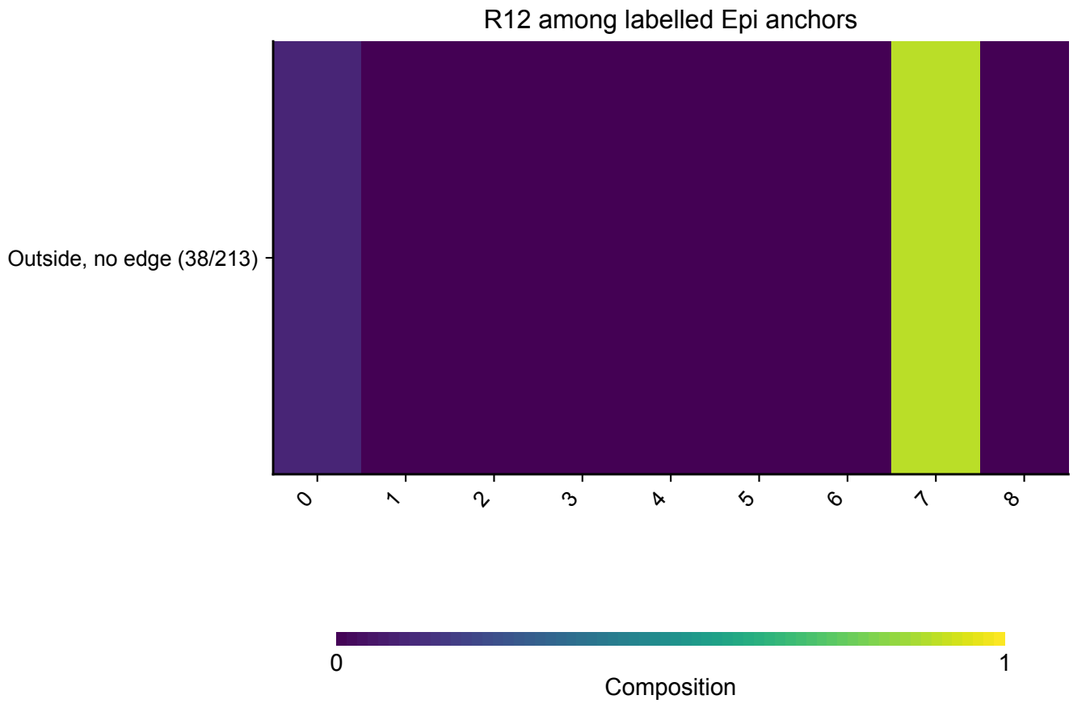
Orange = zero distance
Line = within-section linear fit

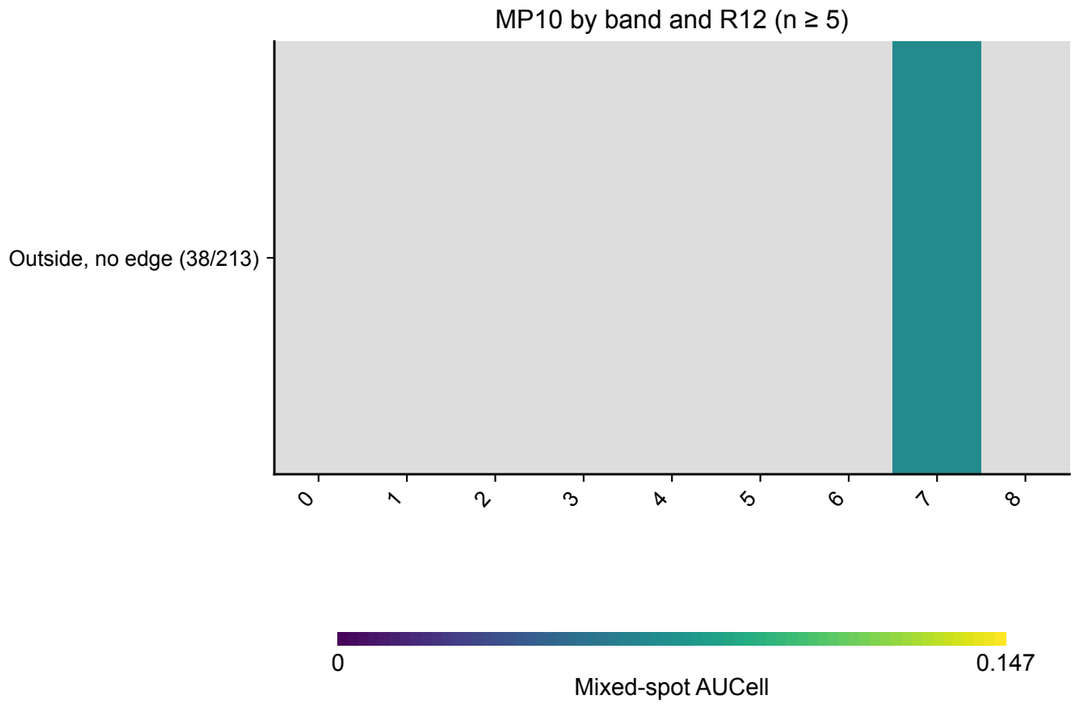
GSE285715_N1 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





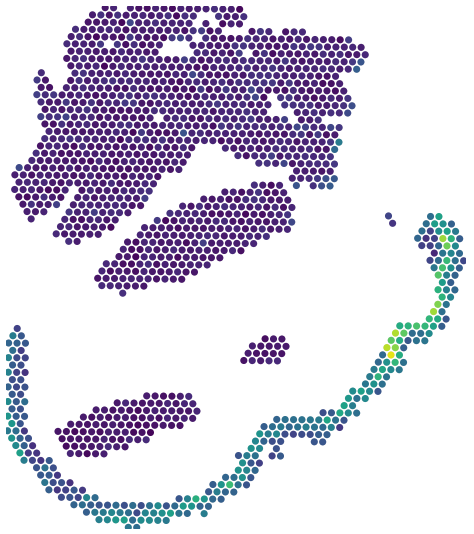




GSE285715_N1 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

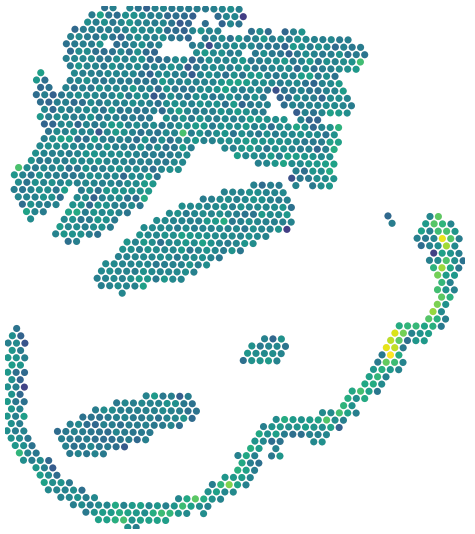
E2F Targets



0 0.152

UCell score

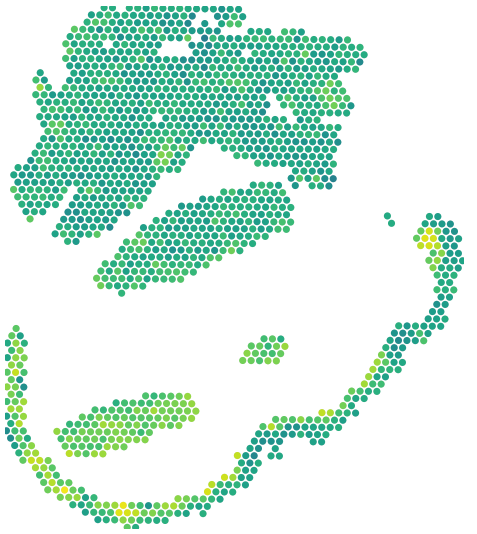
G2M Checkpoint



0 0.134

UCell score

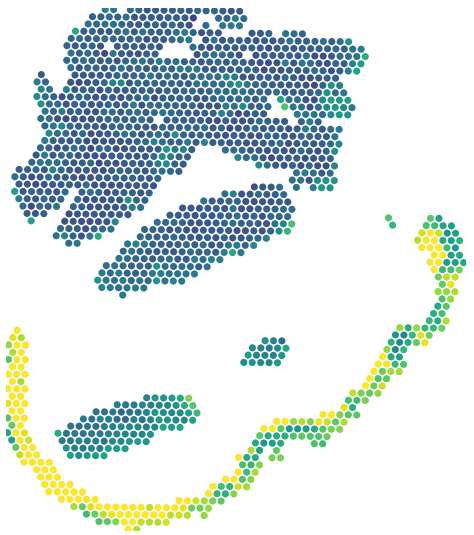
Hypoxia



0 0.211

UCell score

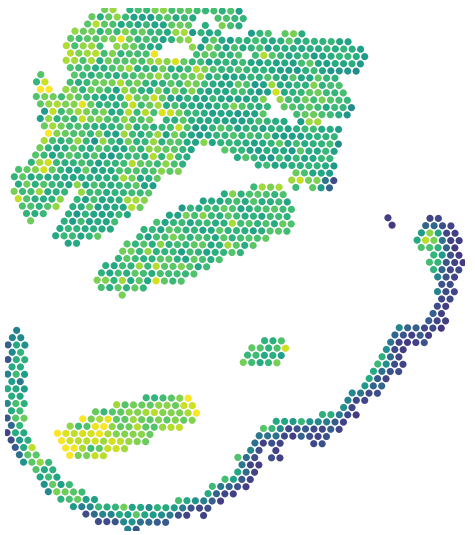
Tnfa Signaling Via Nfkb



0 0.275

UCell score

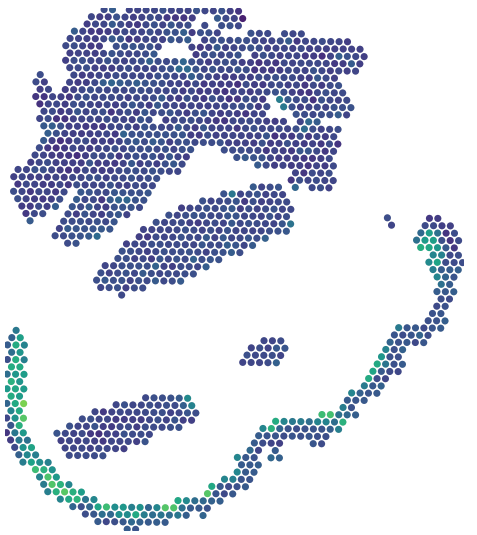
Emt



0 0.386

UCell score

Interferon Gamma Response



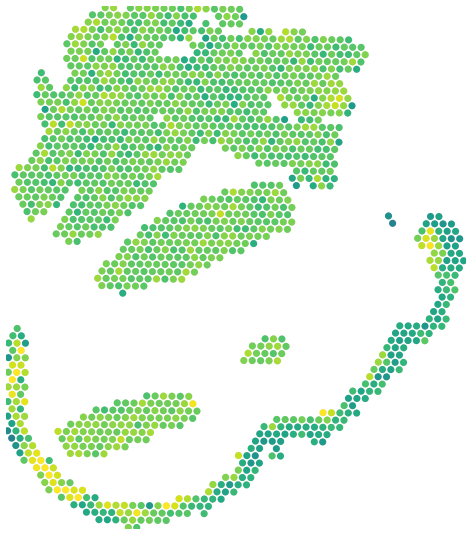
0 0.275

UCell score

GSE285715_N1 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

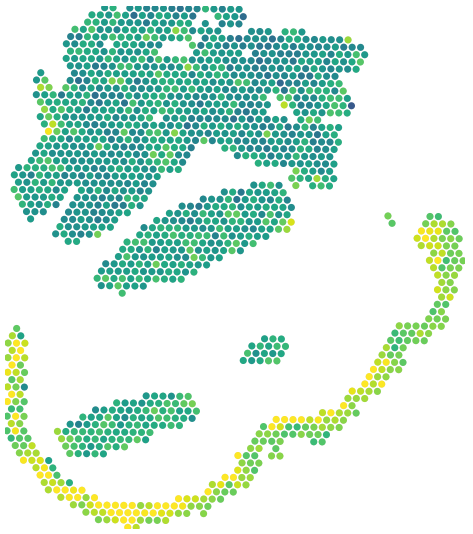
Il2 Stat5 Signaling



0 0.144

UCell score

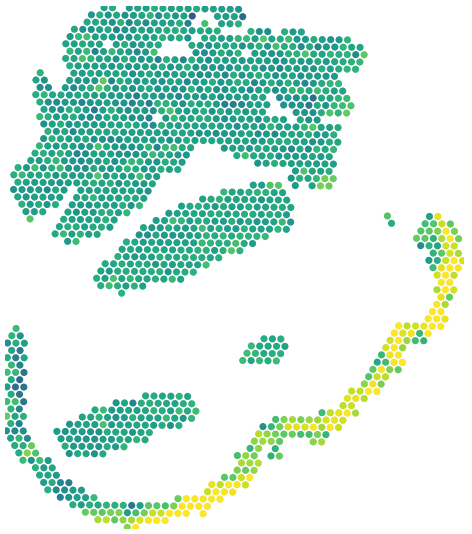
Tgf Beta Signaling



0 0.232

UCell score

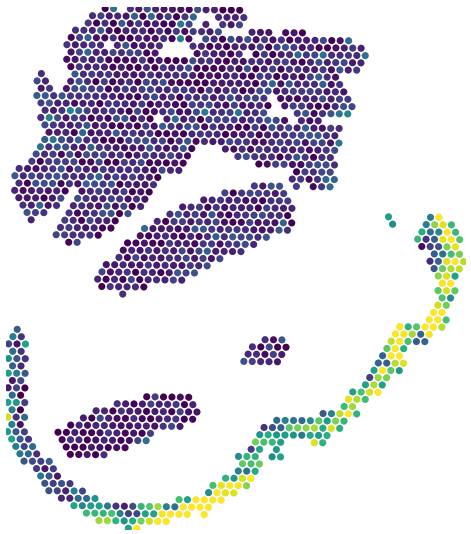
Cholesterol Homeostasis



0 0.21

UCell score

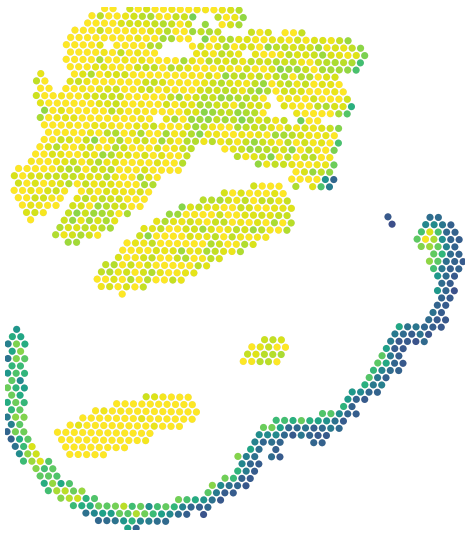
Cholesterol Biosynthesis



0 0.166

UCell score

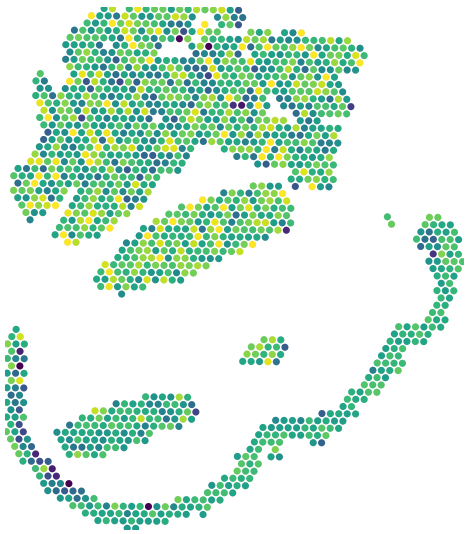
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



0 0.151

UCell score

GSE285715_N1 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Treg composition

0 0.102

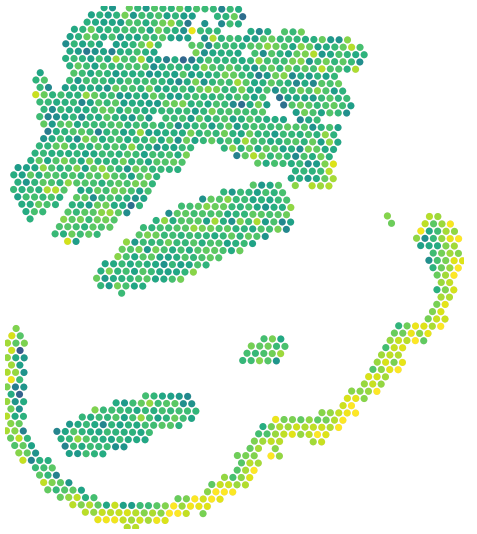
Fraction of all 30

Treg estimated abundance

0 2.85

Posterior mean estimate

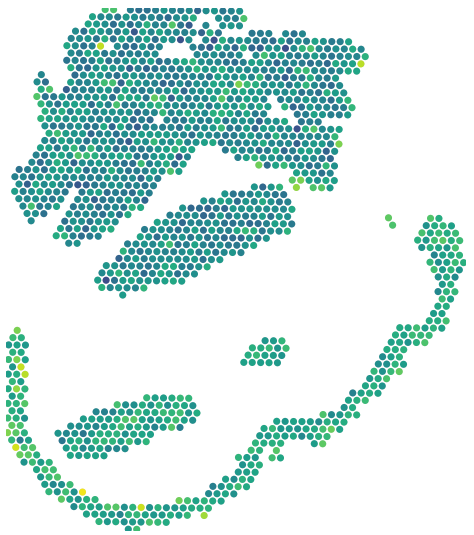
Treg Resting Vs Activated Dn



0 0.0778

UCell score

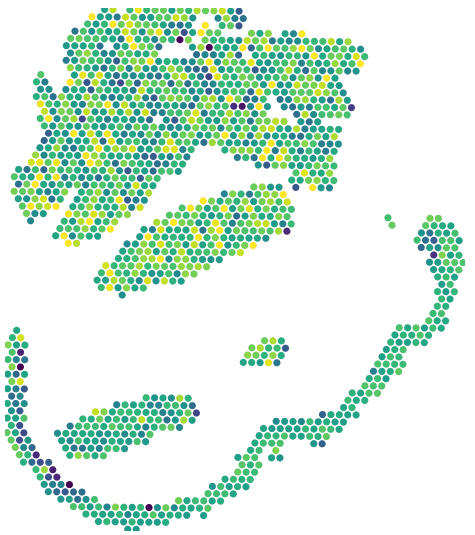
Treg Resting Vs Activated Up



0 0.0609

UCell score

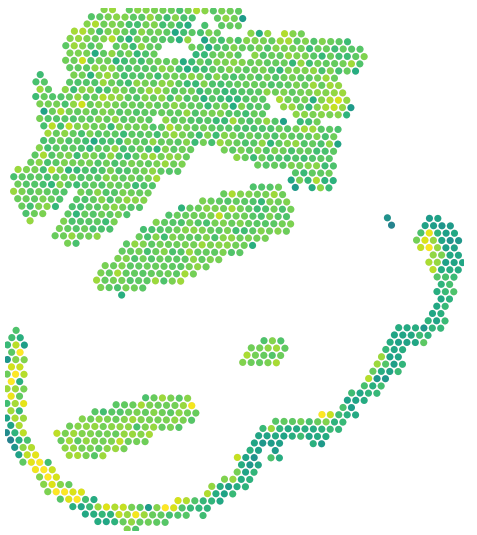
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

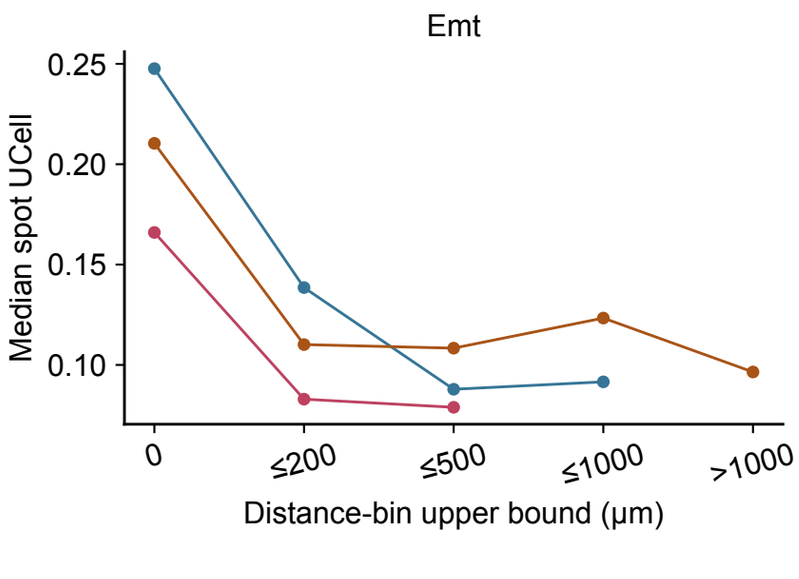


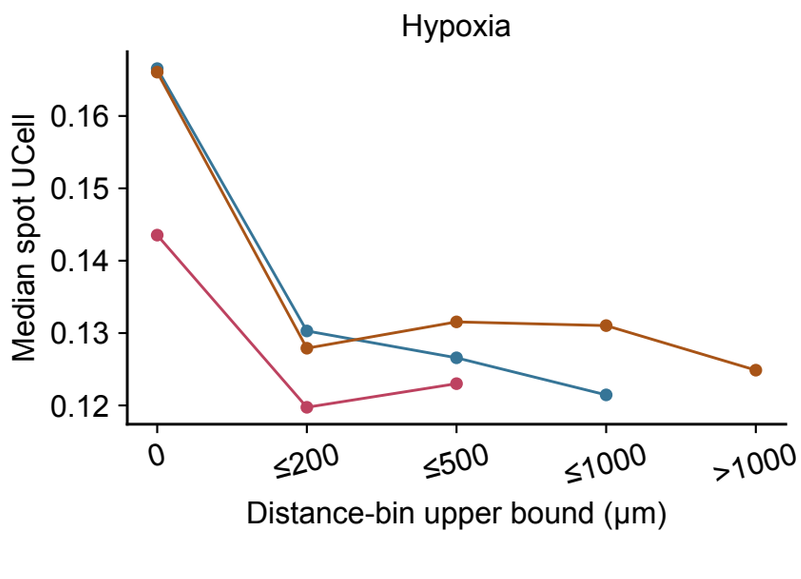
0 0.144

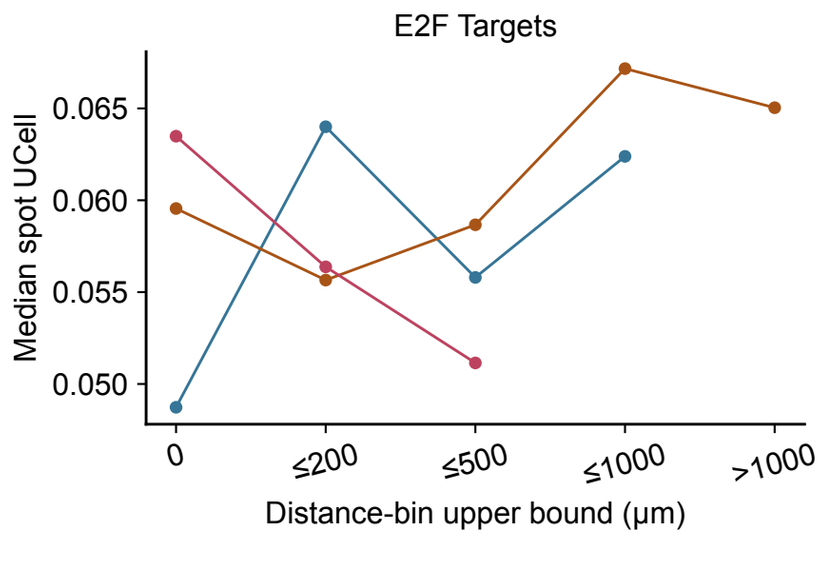
UCell score

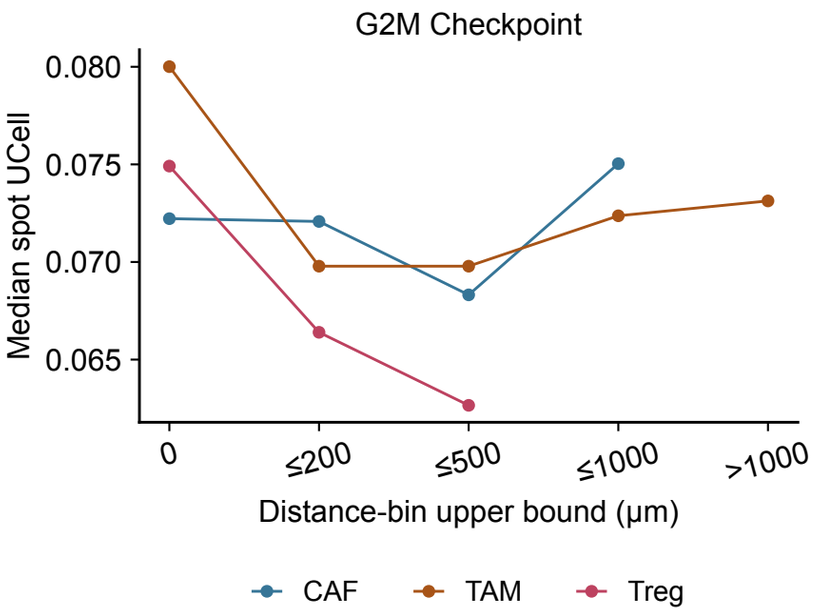
GSE285715_N1 | Programs by target distance

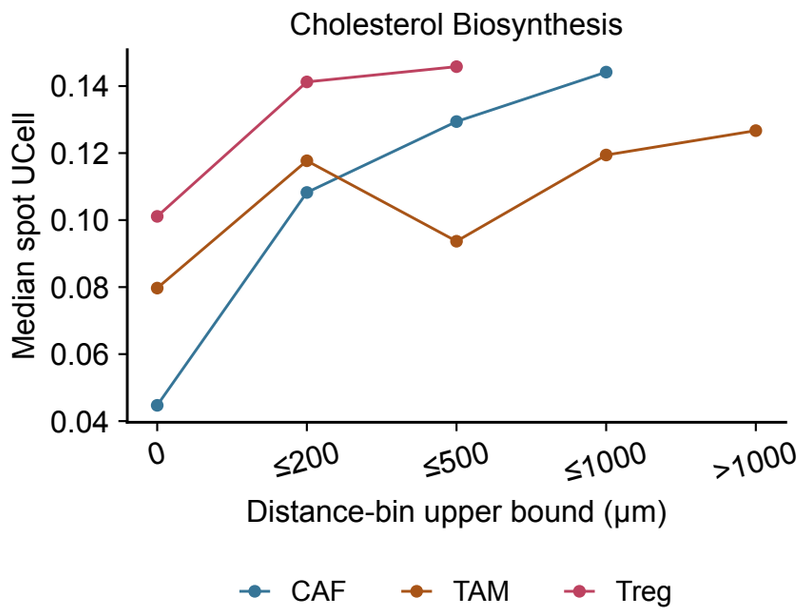
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

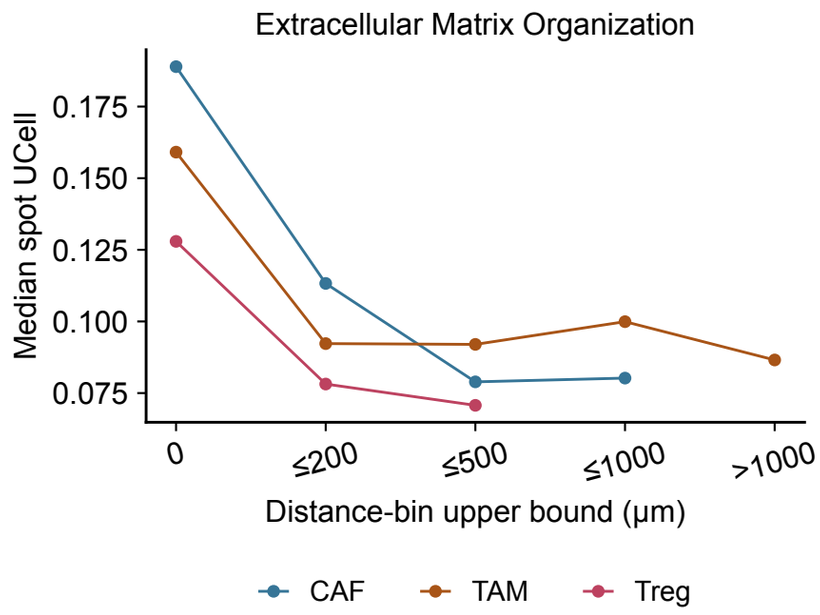






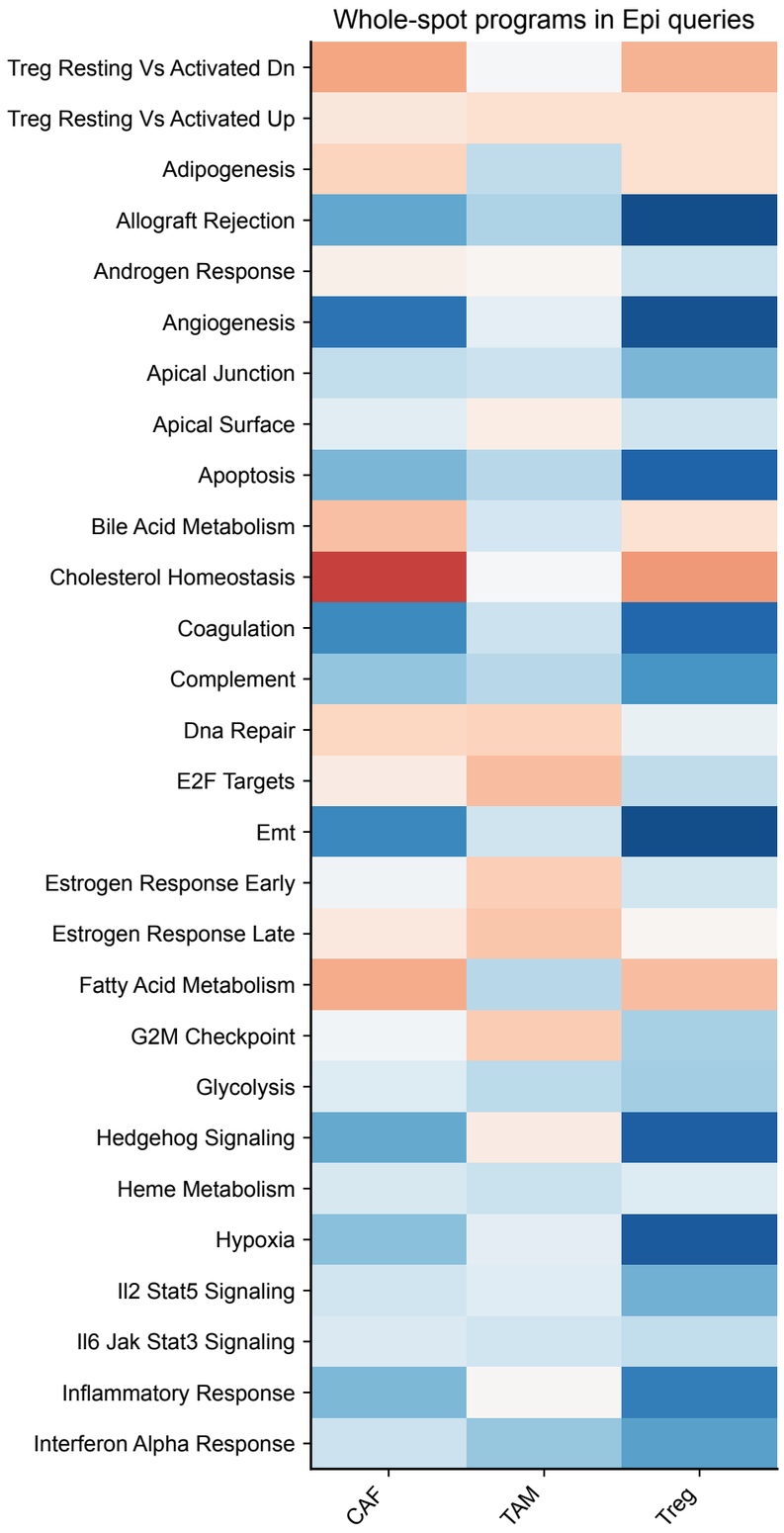


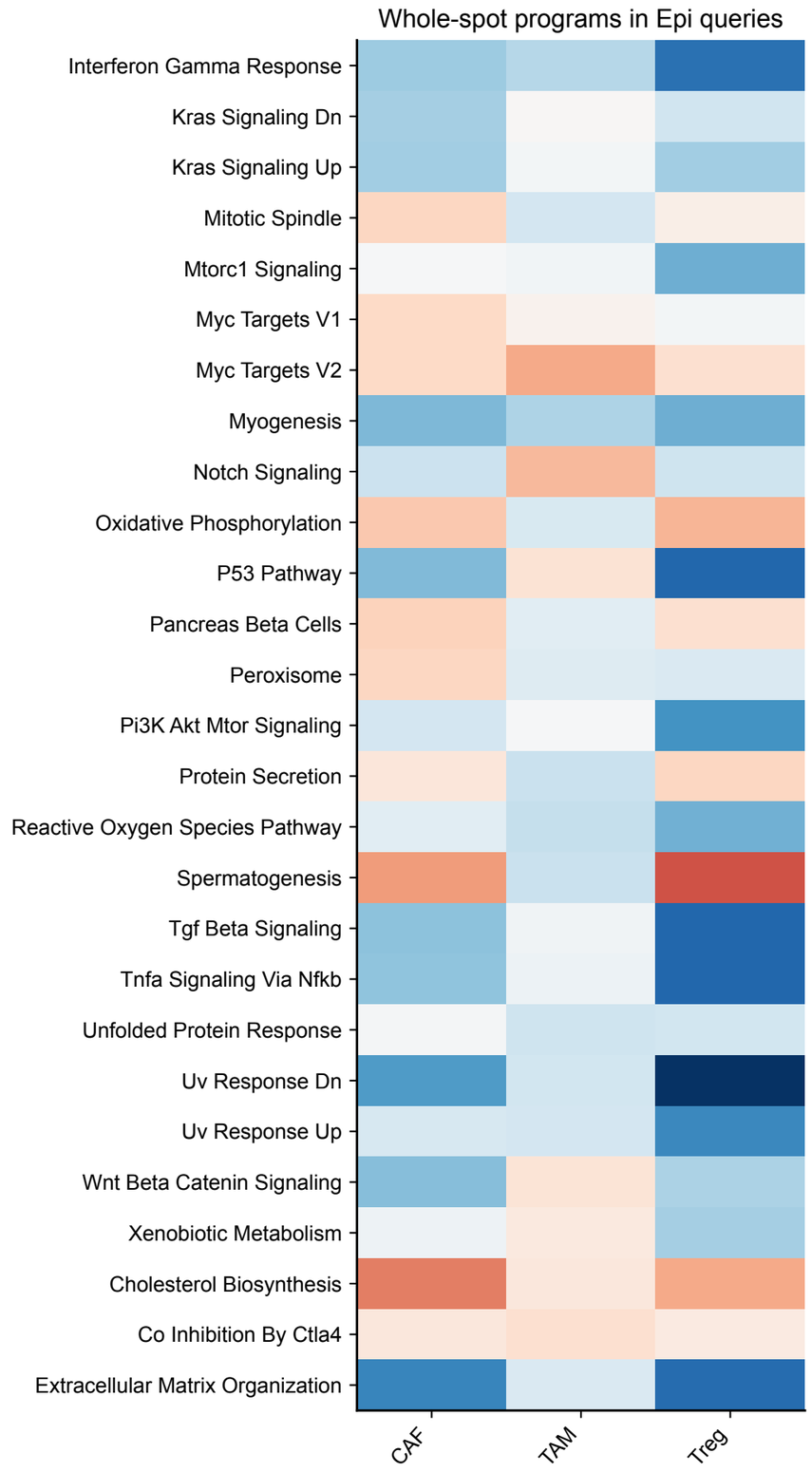




GSE285715_N1 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

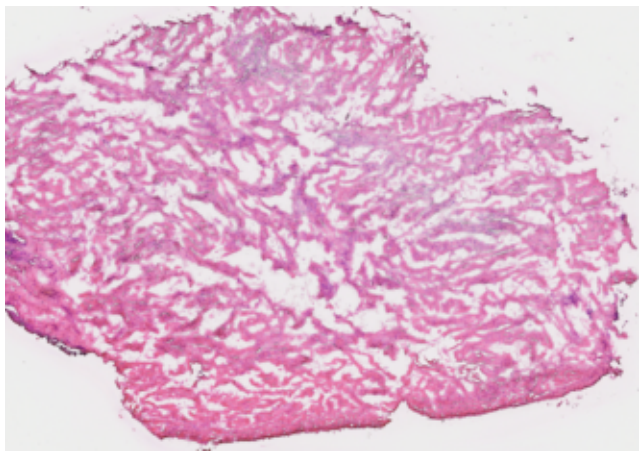




GSE285715_N2 | Tissue context

Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



Final candidate location

Accepted R12 (original 0–8)

No observed candidate outer edge

outside unlocalized

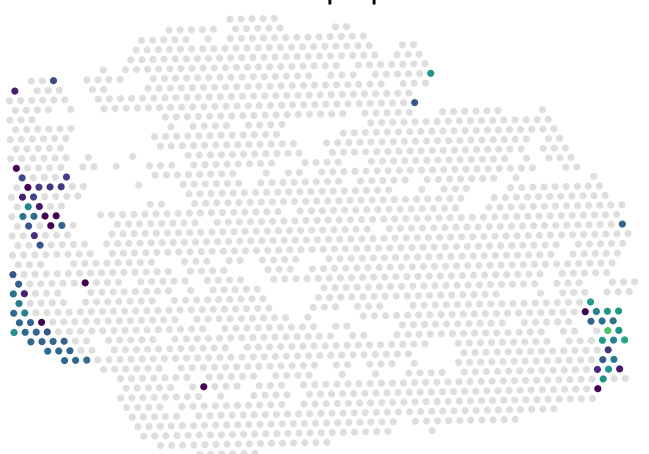
0 5

Epithelial support

0 1

Fraction of all 30

MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

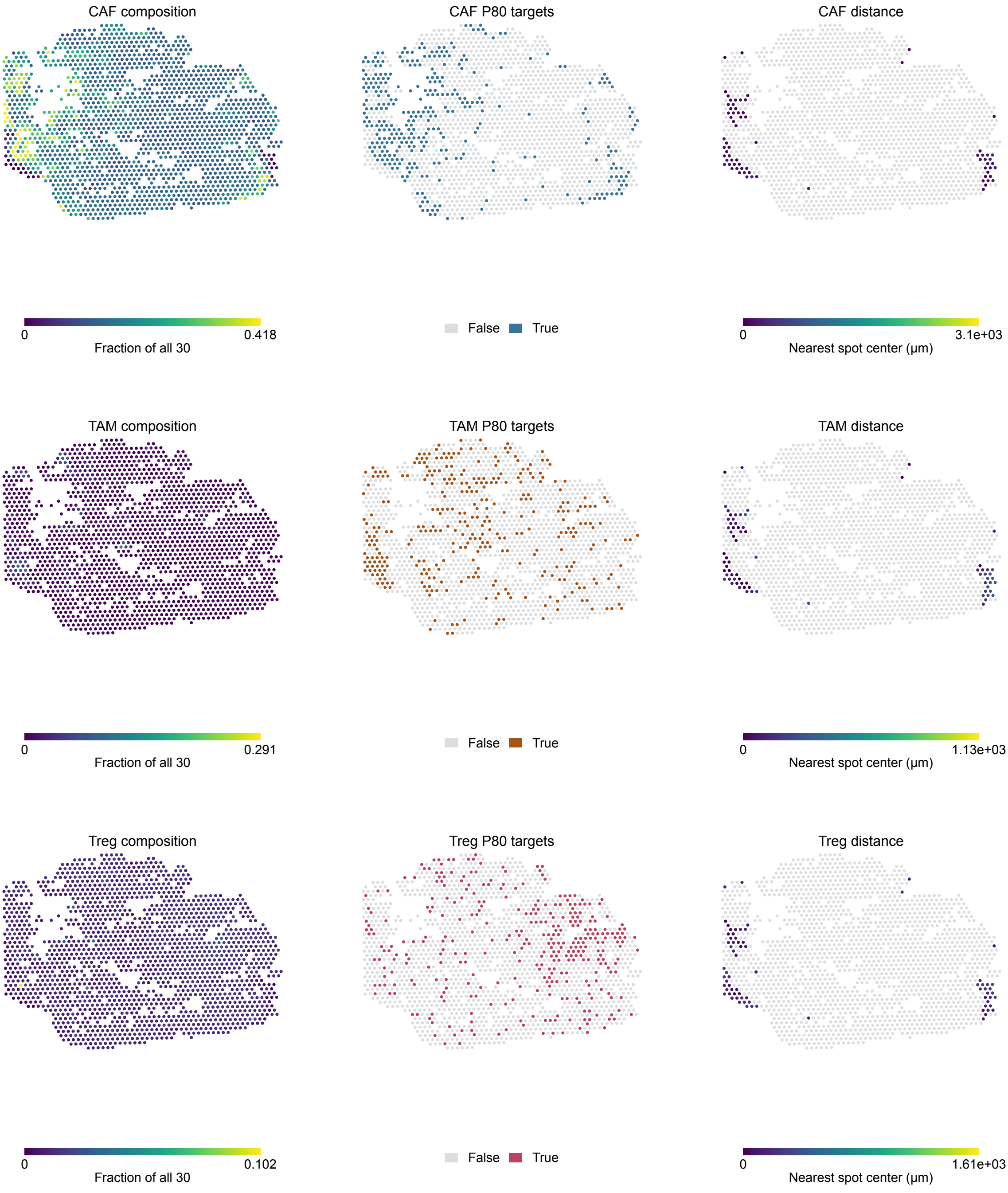
Epi-component MP10 (valid only)

0 0.14

Matched-model AUCell

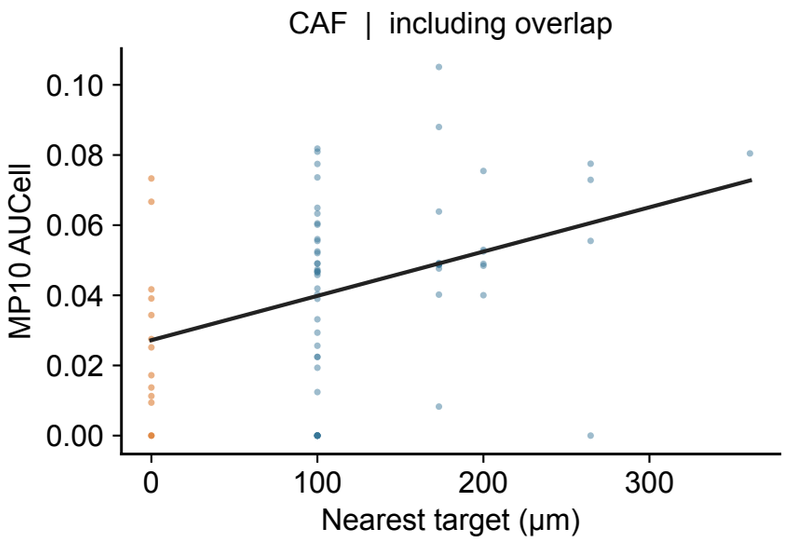
GSE285715_N2 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

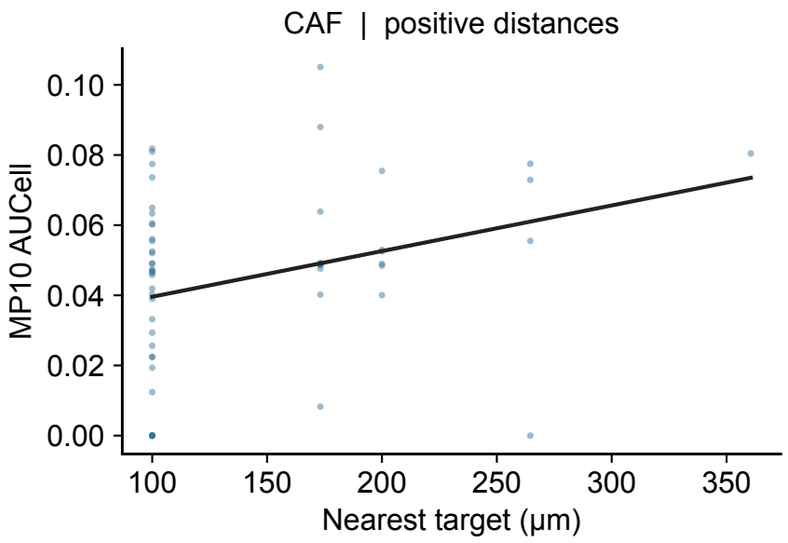


GSE285715_N2 | MP10 versus nearest target

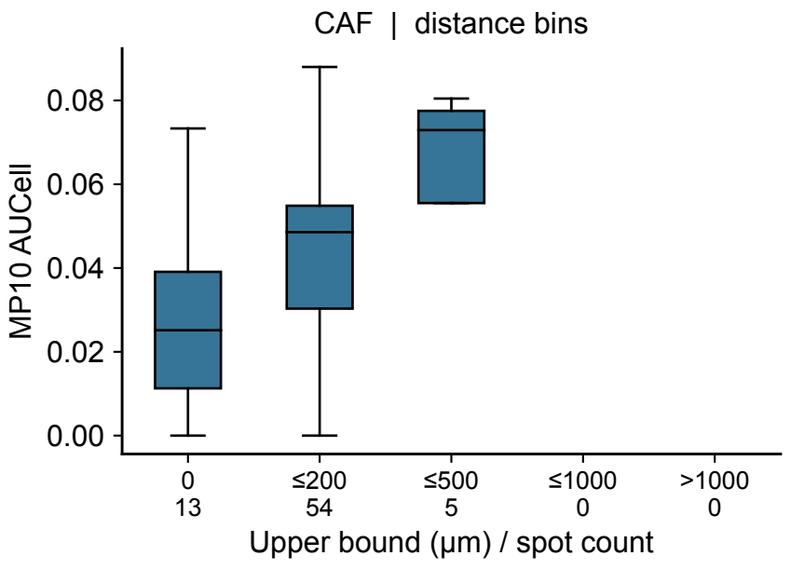
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



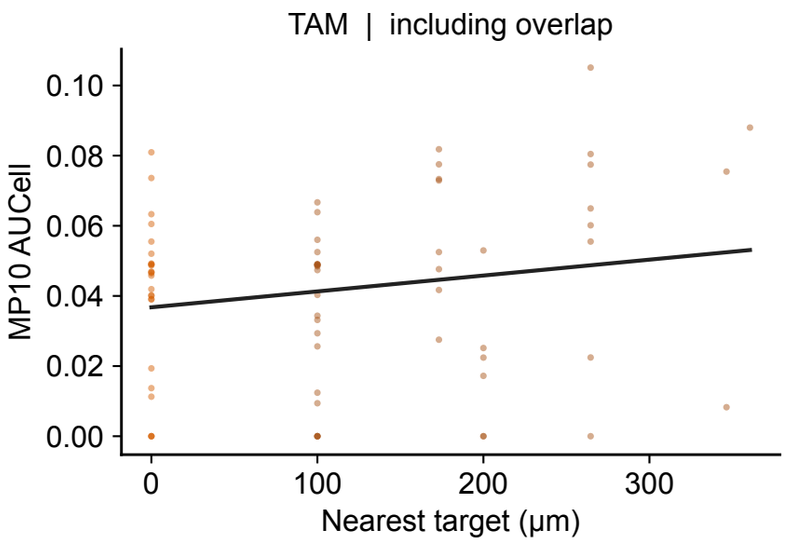
r +0.37 ρ +0.30
n 72/72 queries • 345 targets



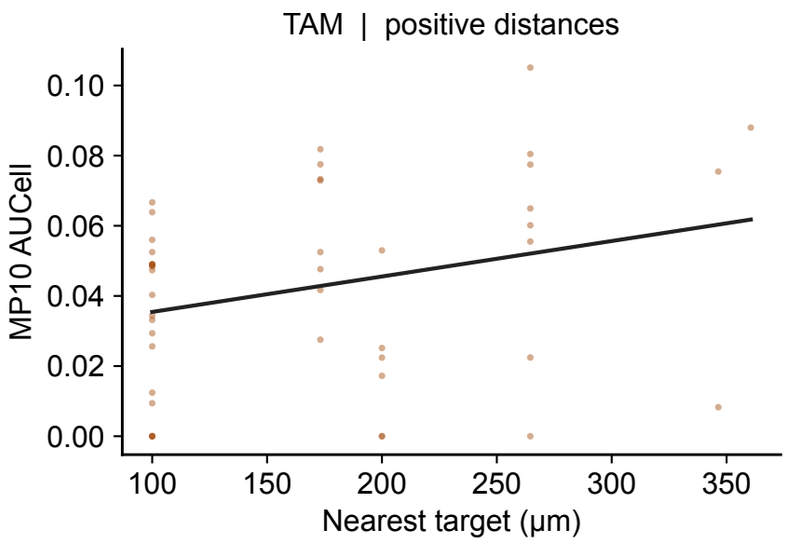
r +0.30 ρ +0.16
n 59/72 queries • 345 targets



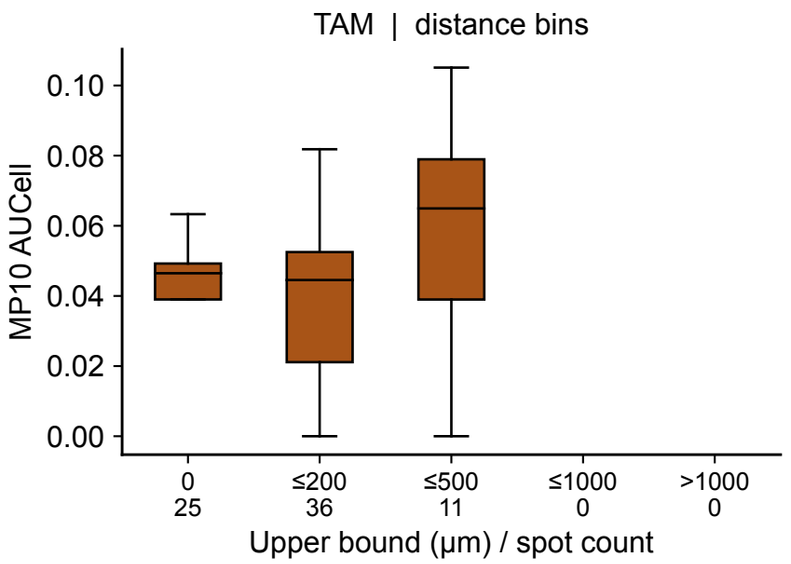
Orange = zero distance
Line = within-section linear fit



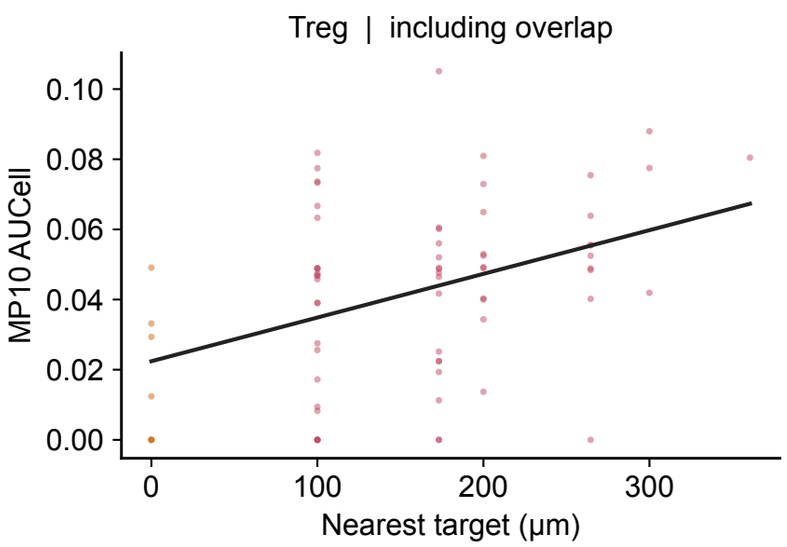
r +0.18 ρ +0.16
n 72/72 queries • 345 targets



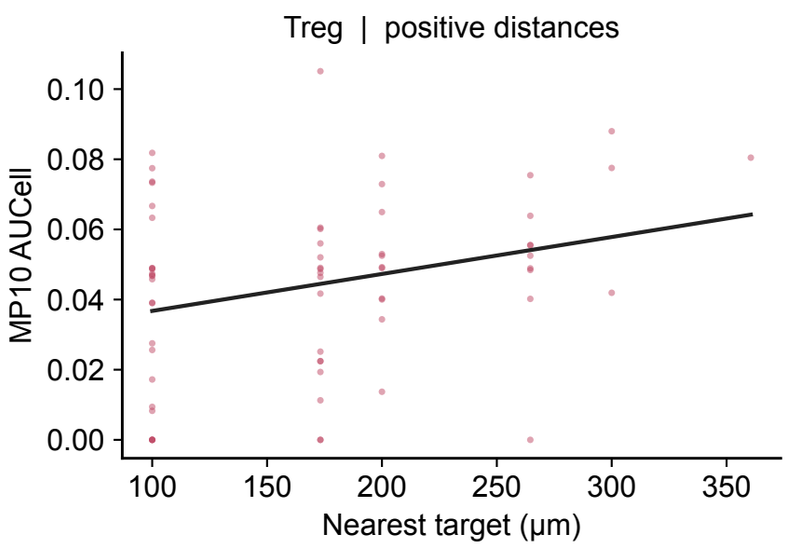
r +0.29 ρ +0.25
n 47/72 queries • 345 targets



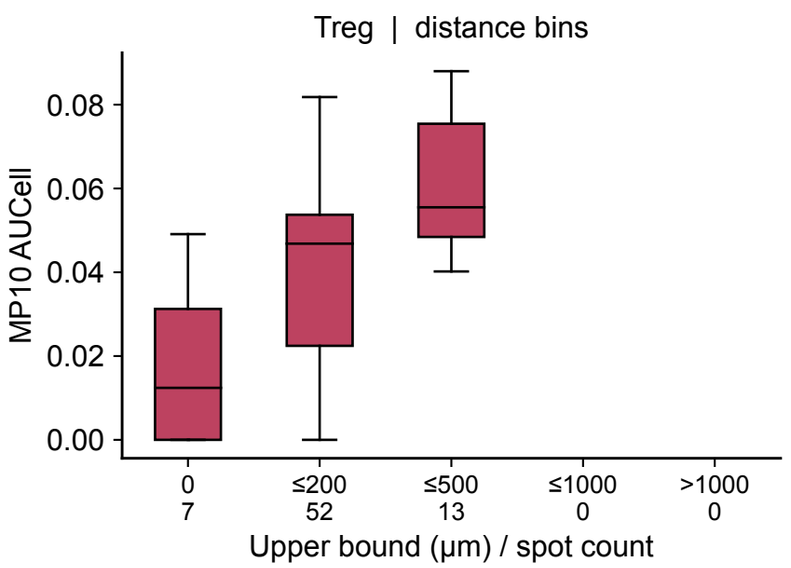
Orange = zero distance
Line = within-section linear fit



r +0.40 ρ +0.38
n 72/72 queries • 345 targets



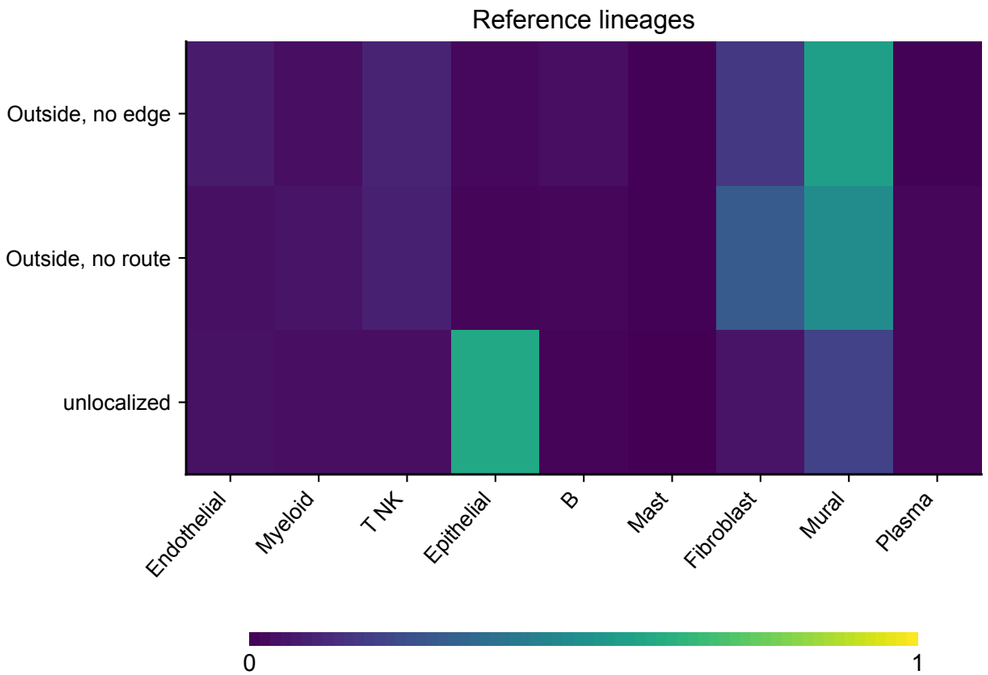
r +0.29 ρ +0.28
n 65/72 queries • 345 targets

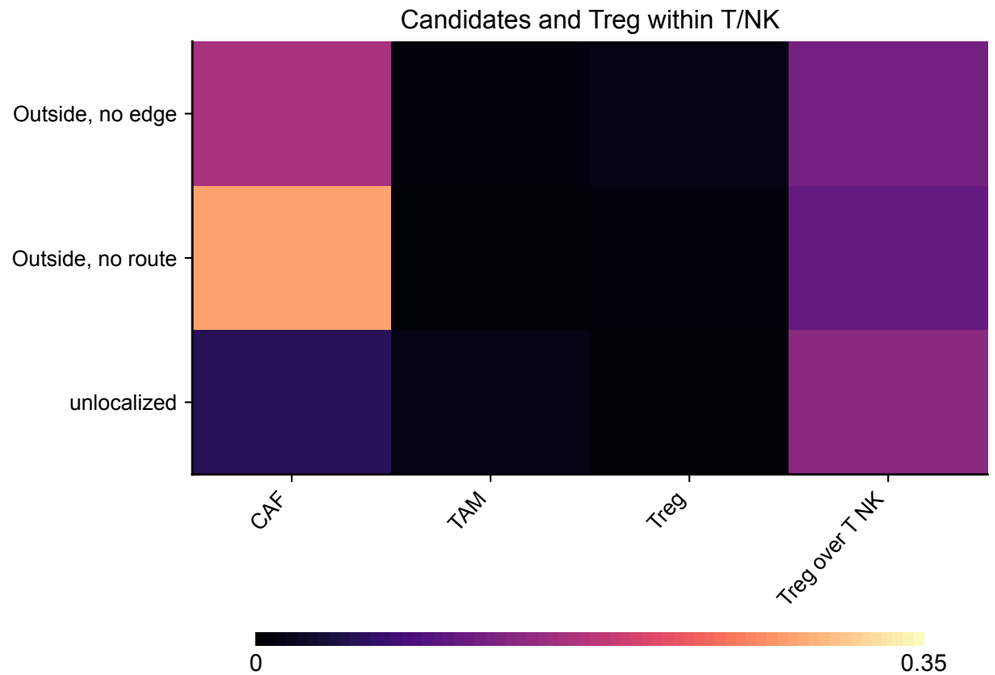


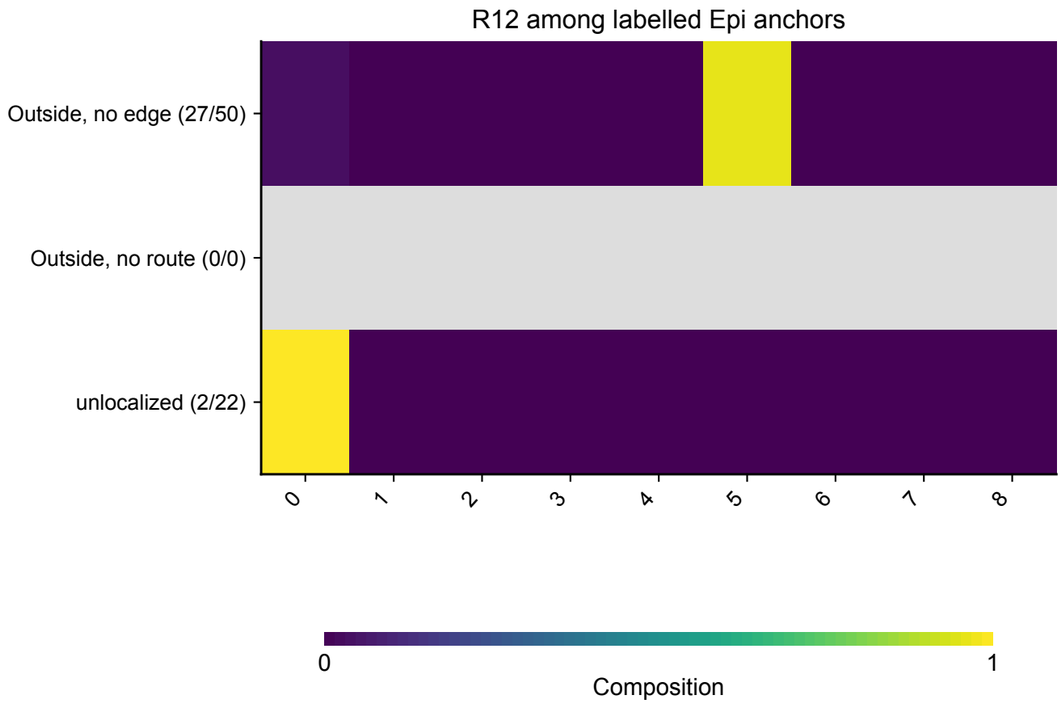
Orange = zero distance
Line = within-section linear fit

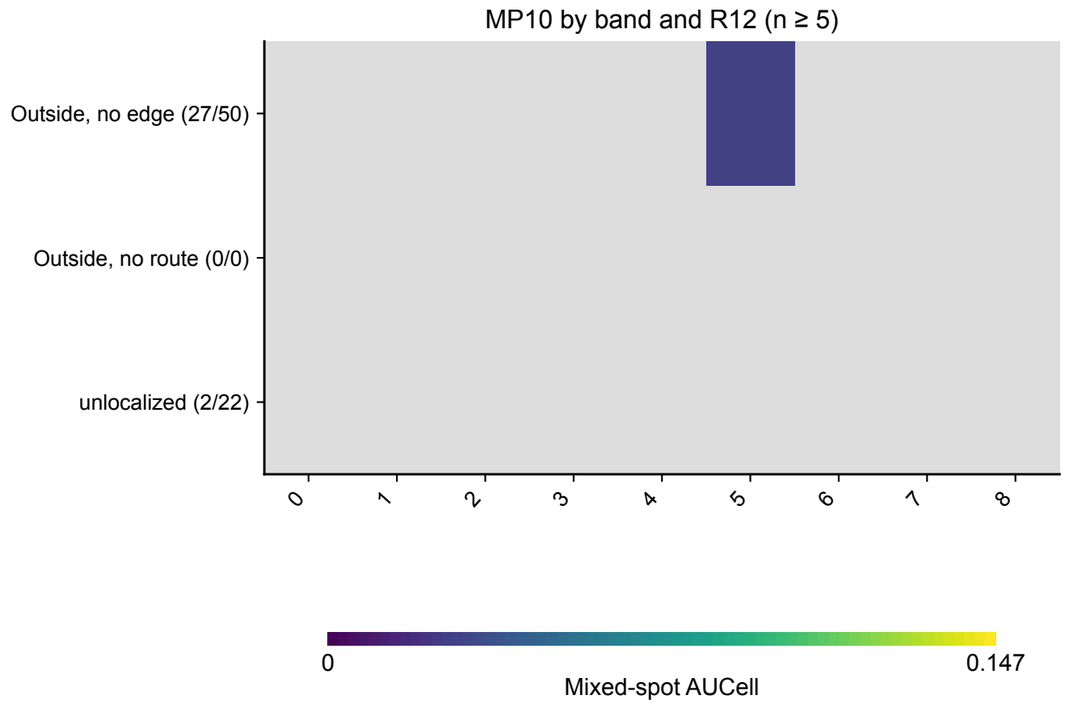
GSE285715_N2 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





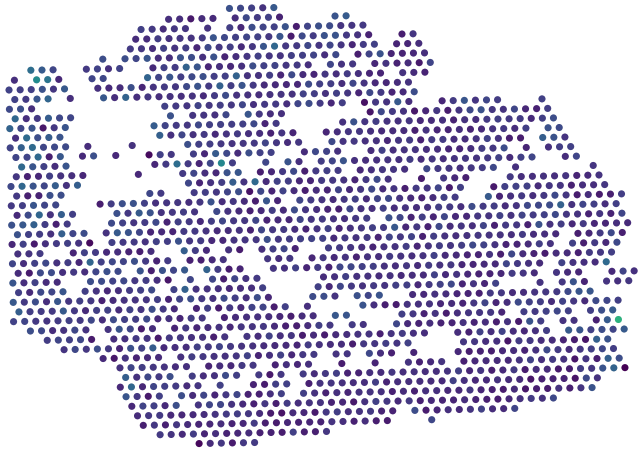




GSE285715_N2 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

E2F Targets



0 0.152

UCell score

G2M Checkpoint

0 0.134

UCell score

Hypoxia

0 0.211

UCell score

Tnfa Signaling Via Nfkb

0 0.275

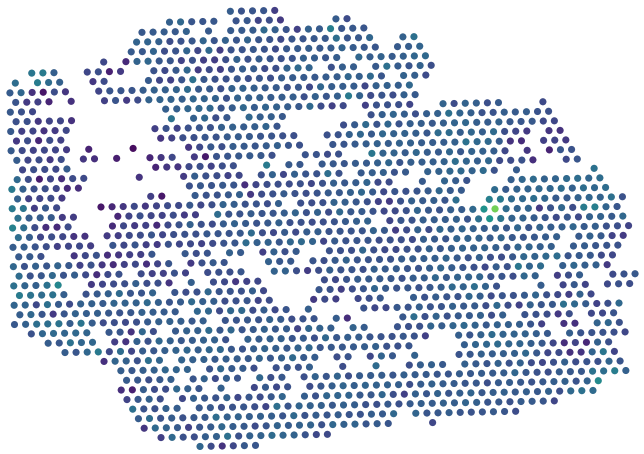
UCell score

Emt

0 0.386

UCell score

Interferon Gamma Response



0 0.275

UCell score

GSE285715_N2 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Il2 Stat5 Signaling

0 0.144

UCell score

Tgf Beta Signaling

0 0.232

UCell score

Cholesterol Homeostasis

0 0.21

UCell score

Cholesterol Biosynthesis

0 0.166

UCell score

Extracellular Matrix Organization

0 0.247

UCell score

Co Inhibition By Ctla4

0 0.151

UCell score

GSE285715_N2 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Treg composition

0 0.102

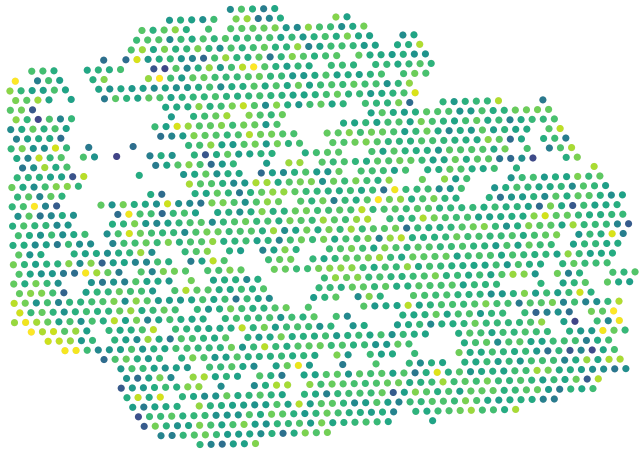
Fraction of all 30

Treg estimated abundance

0 2.85

Posterior mean estimate

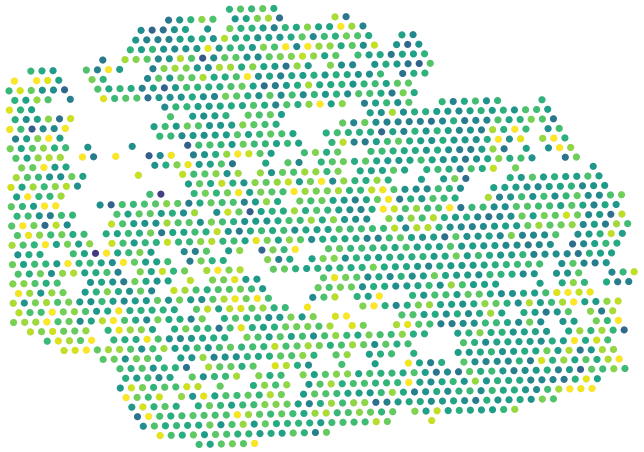
Treg Resting Vs Activated Dn



0 0.0778

UCell score

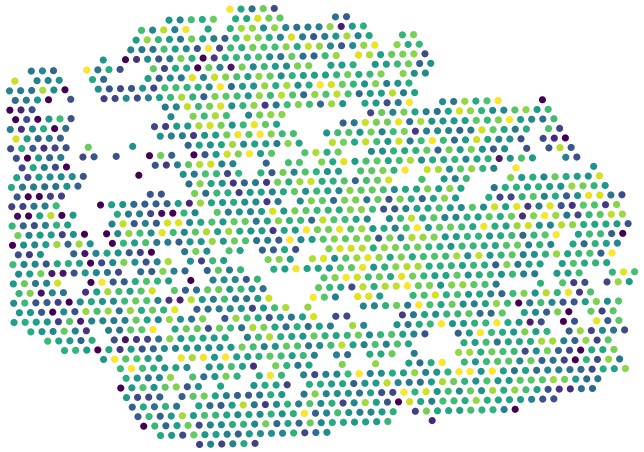
Treg Resting Vs Activated Up



0 0.0609

UCell score

Co Inhibition By Ctla4



0 0.151

UCell score

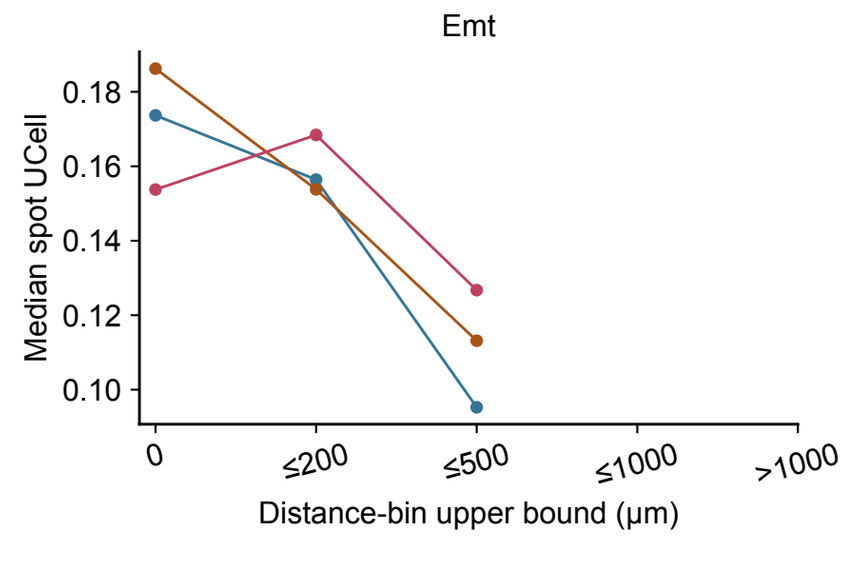
Il2 Stat5 Signaling

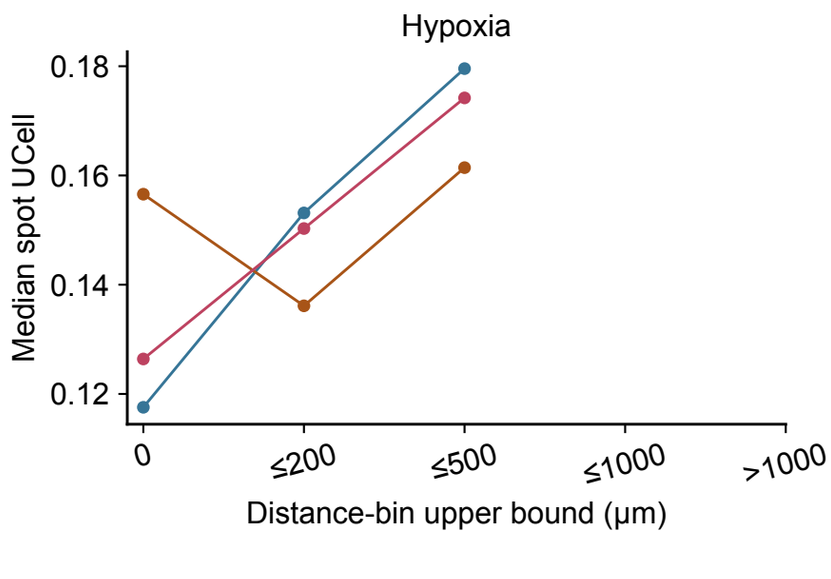
0 0.144

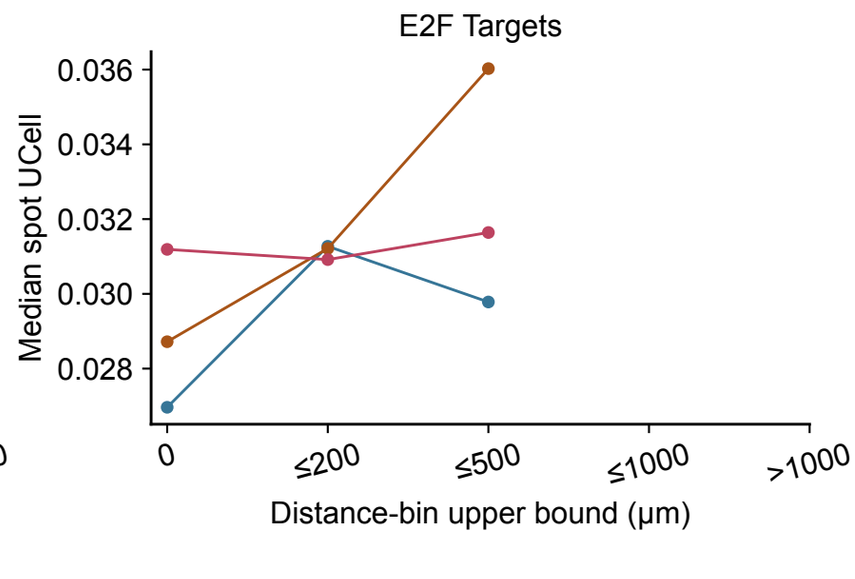
UCell score

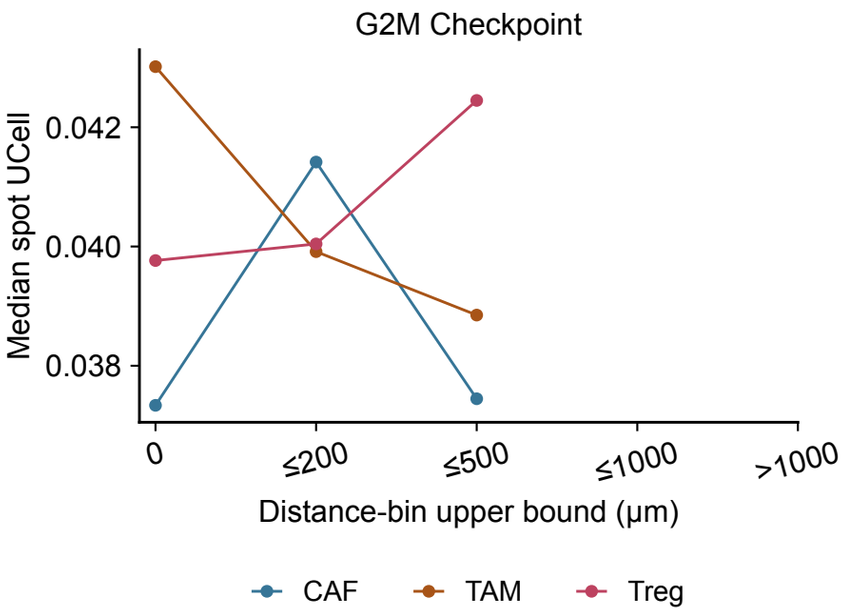
GSE285715_N2 | Programs by target distance

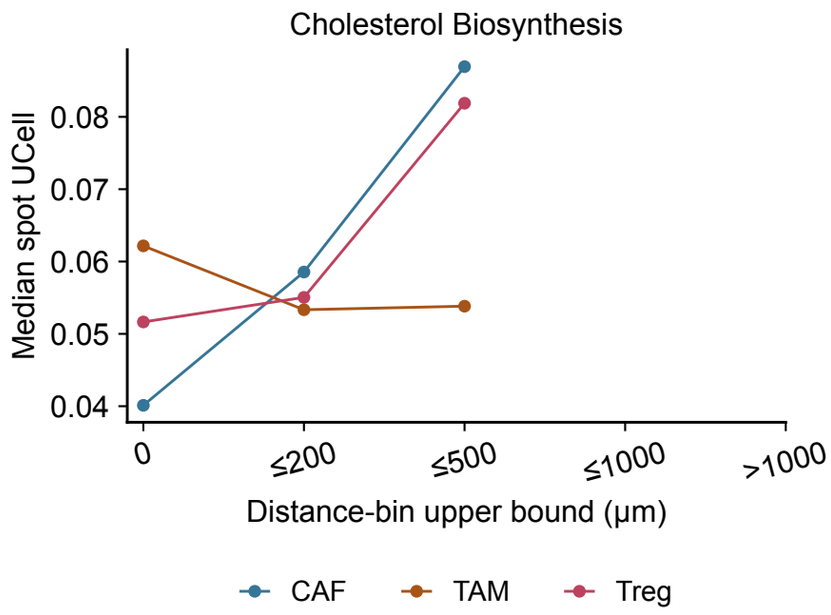
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

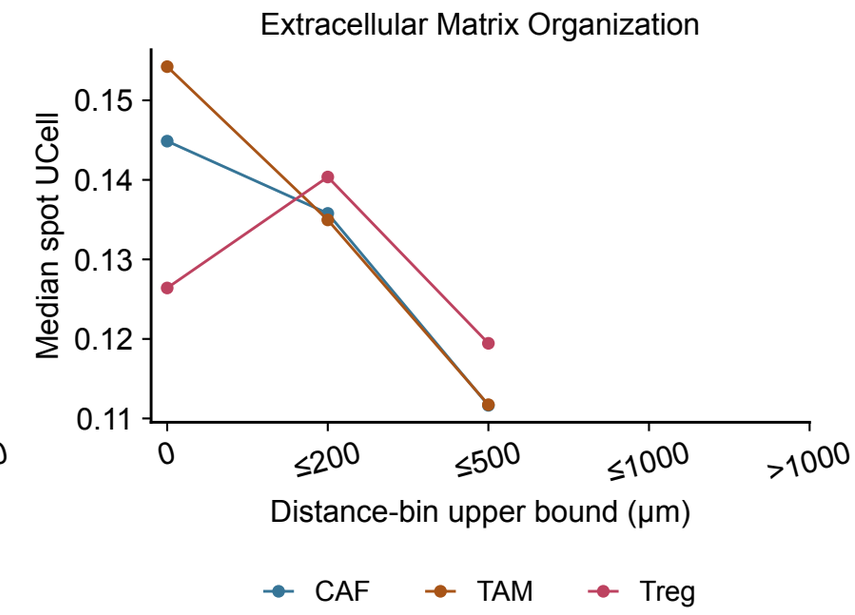






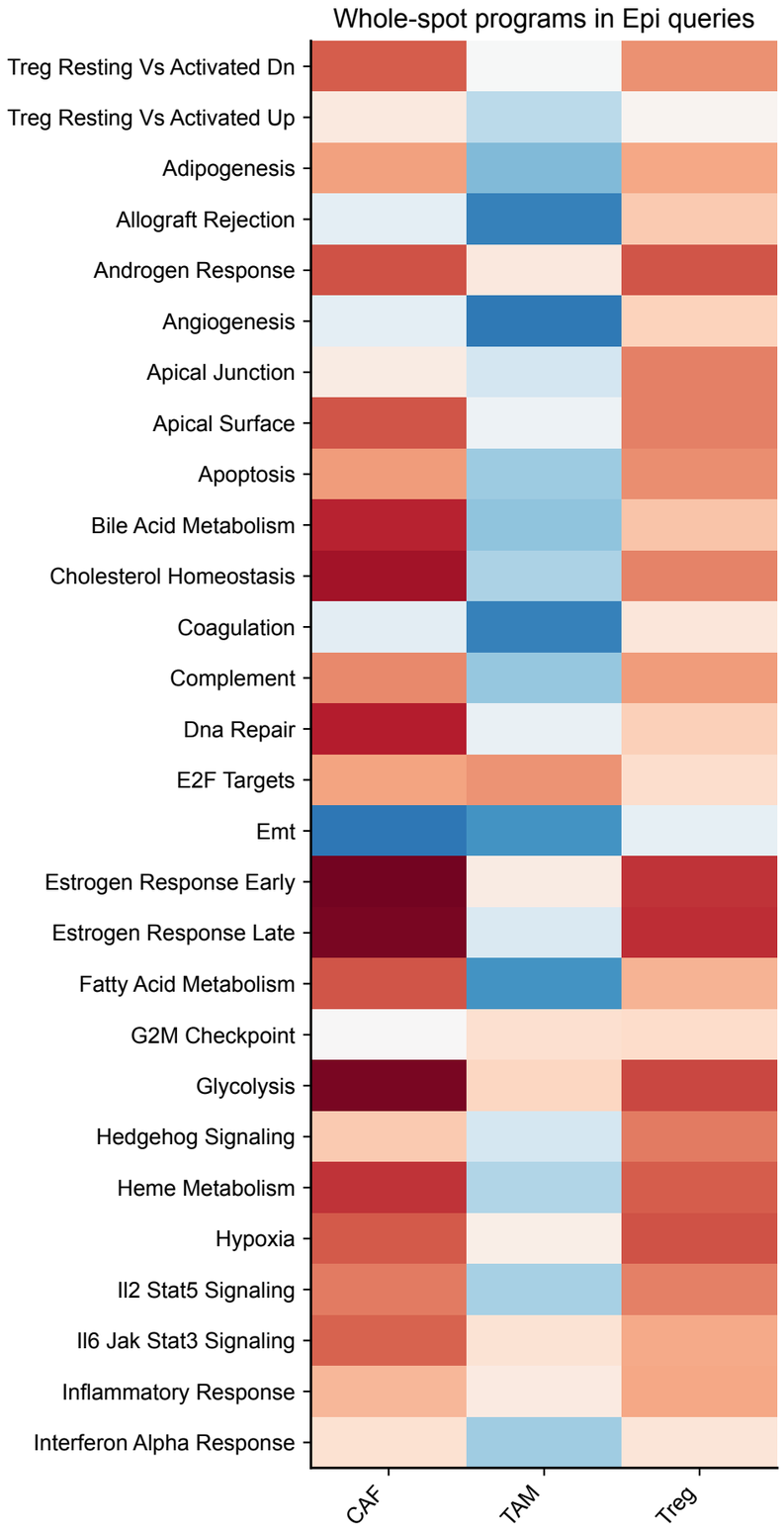


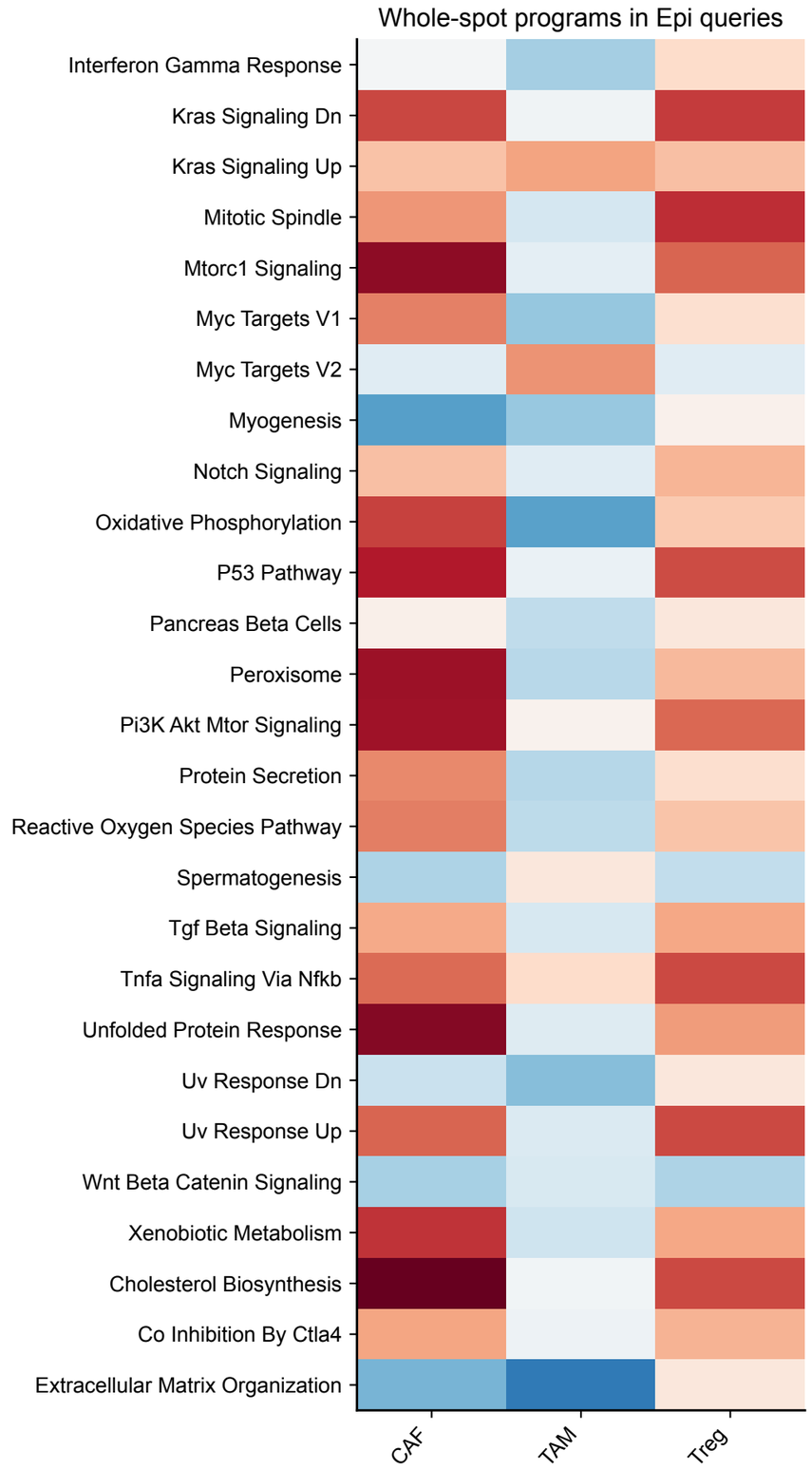




GSE285715_N2 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

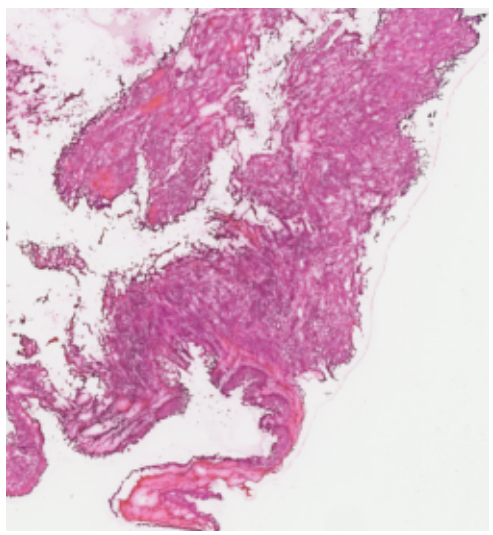




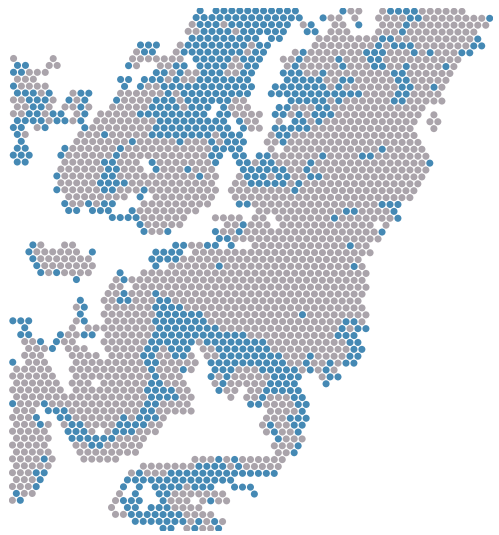
GSE285715_N3 | Tissue context

Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

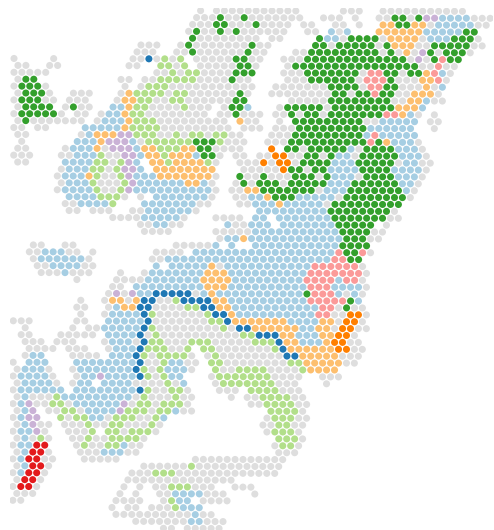
H&E and observed lesion edge



Final candidate location



Accepted R12 (original 0–8)

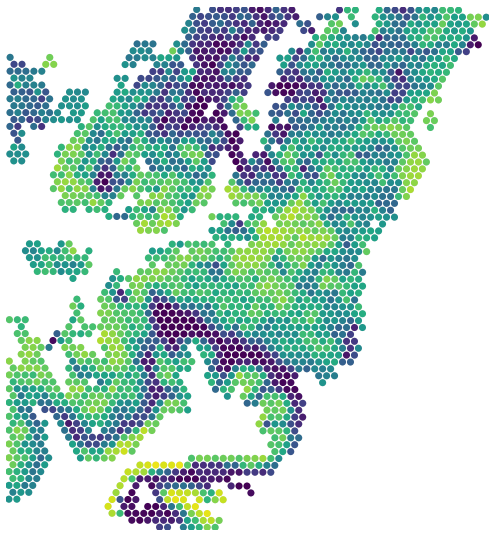


No observed candidate outer edge

outside unlocalized

0 2 4 6 8
1 3 5 7

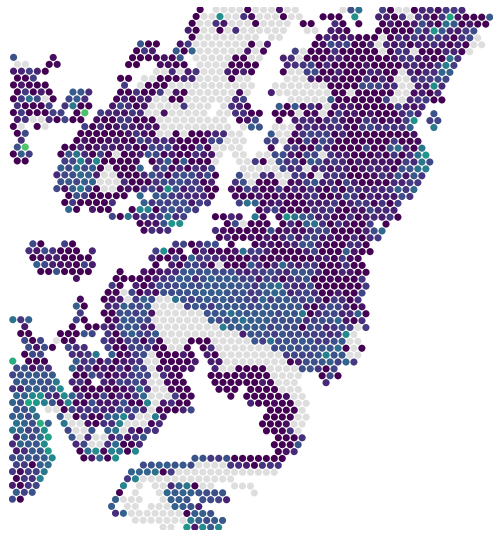
Epithelial support



0 1

Fraction of all 30

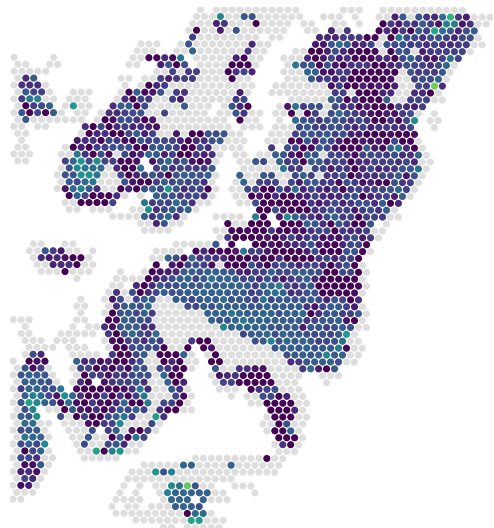
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



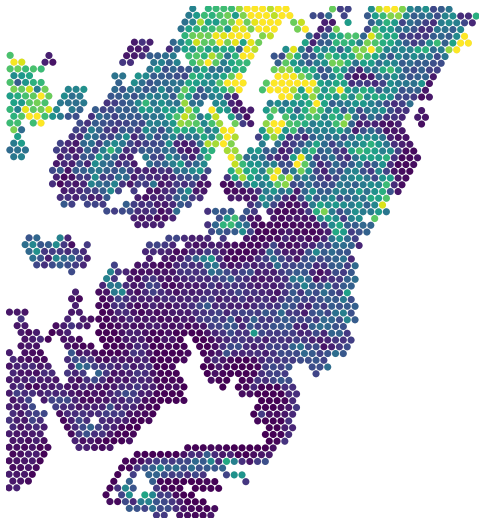
0 0.14

Matched-model AUCell

GSE285715_N3 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

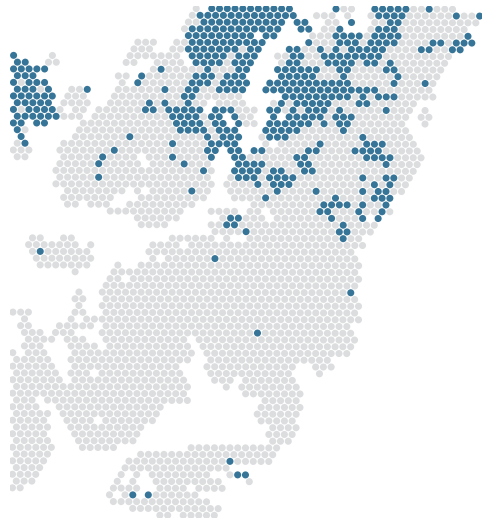
CAF composition



0 0.418

Fraction of all 30

CAF P80 targets



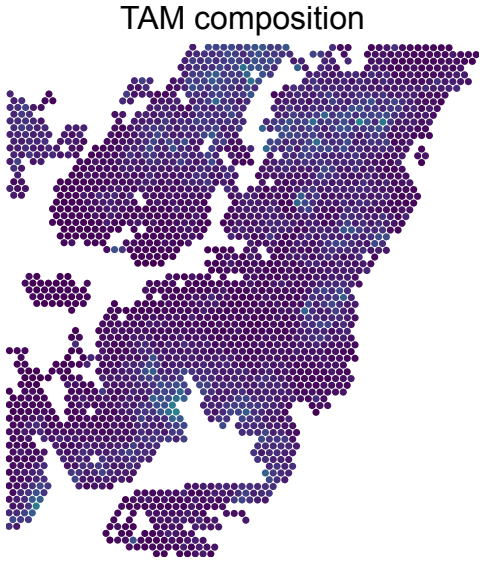
False True

CAF distance

0 3.1e+03

Nearest spot center (µm)

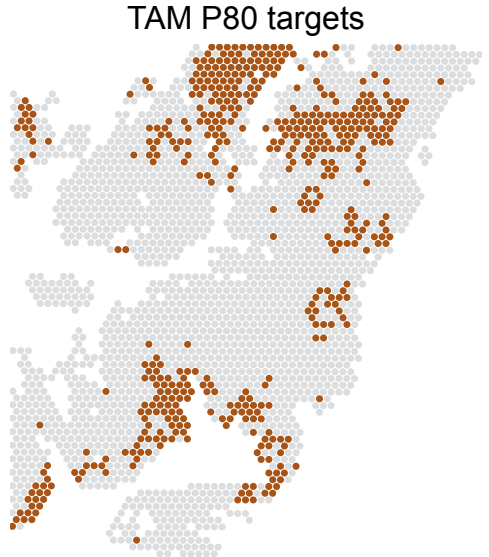
TAM composition



0 0.291

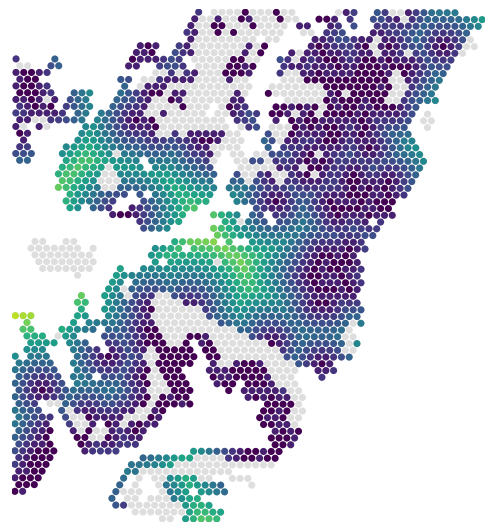
Fraction of all 30

TAM P80 targets



False True

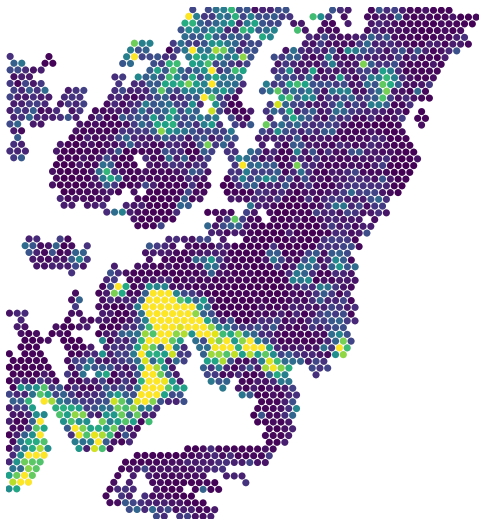
TAM distance



0 1.13e+03

Nearest spot center (µm)

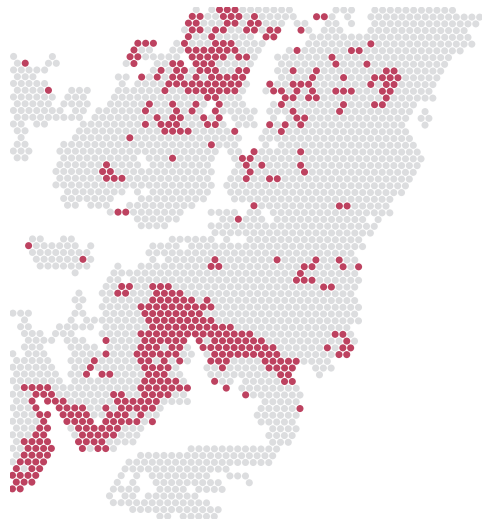
Treg composition



0 0.102

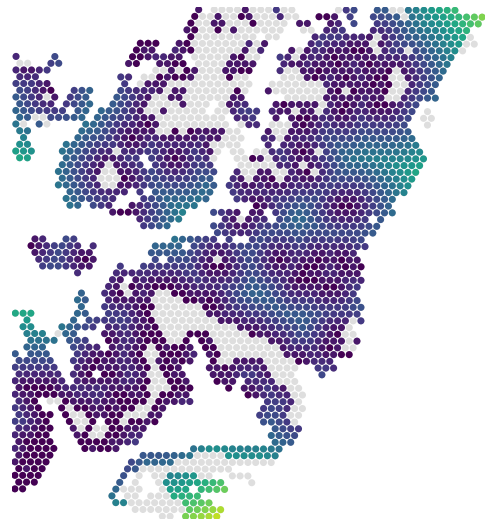
Fraction of all 30

Treg P80 targets



False True

Treg distance

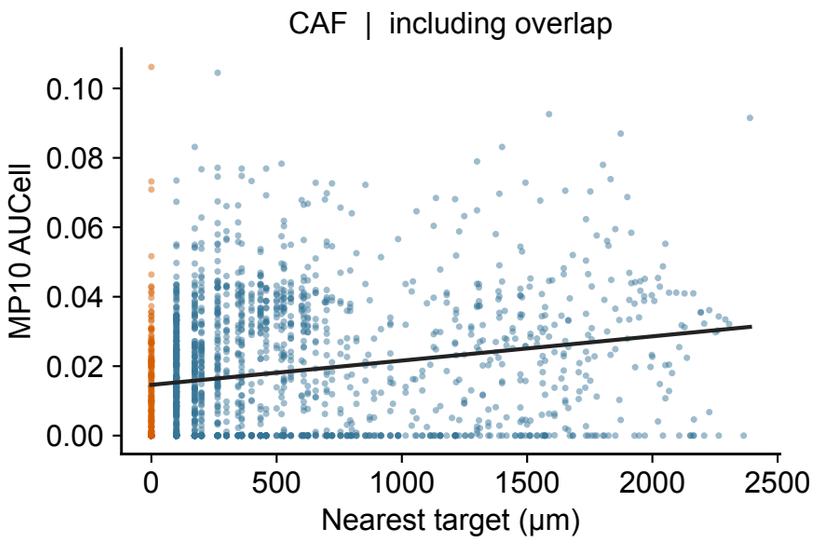


0 1.61e+03

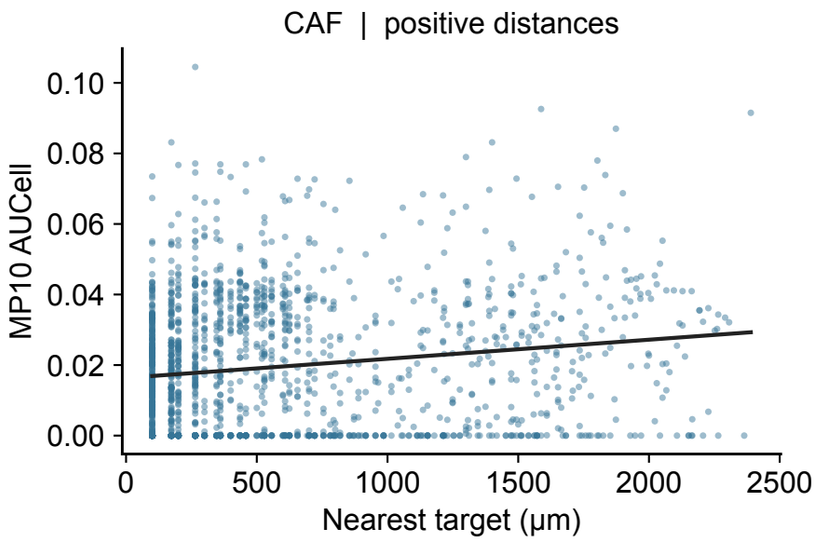
Nearest spot center (µm)

GSE285715_N3 | MP10 versus nearest target

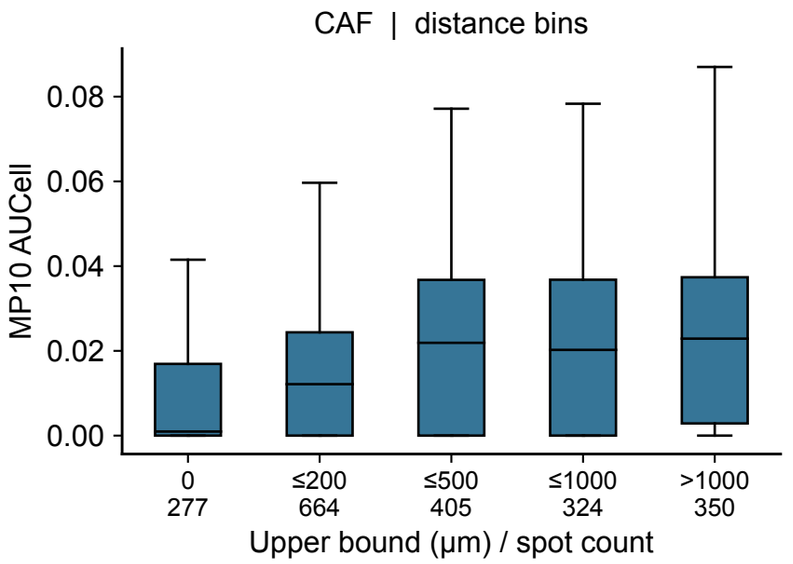
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



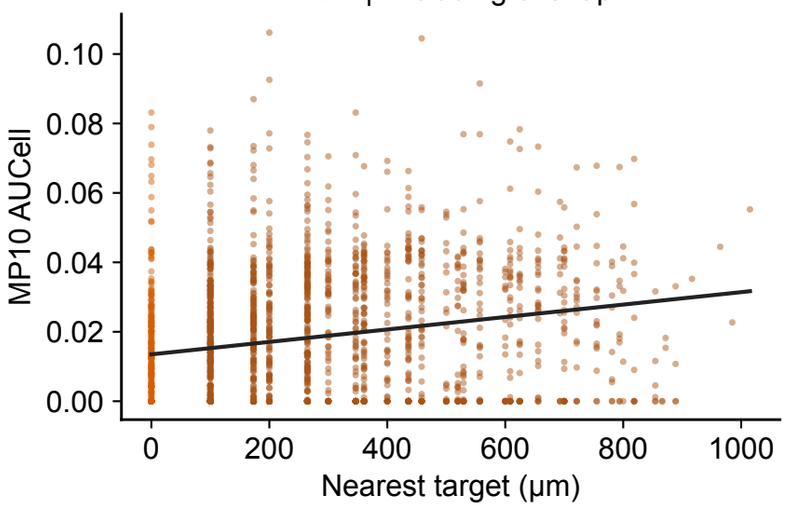
r +0.21 ρ +0.25
n 2020/2024 queries • 498 targets



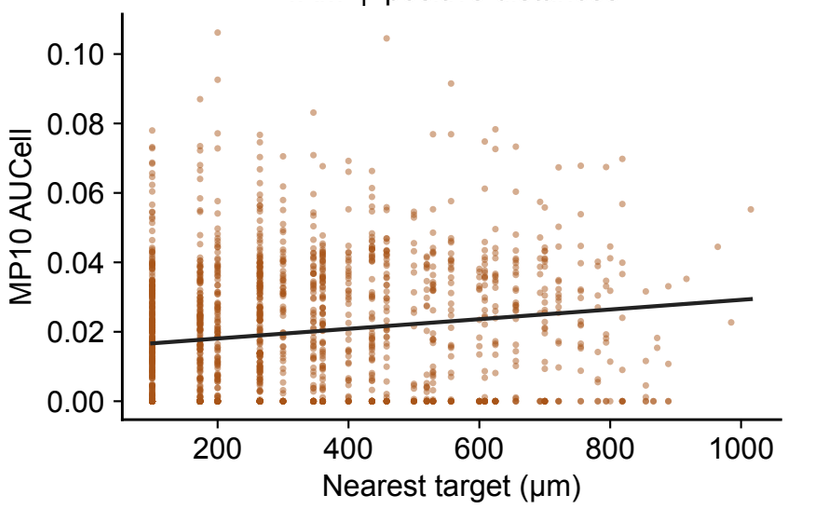
r +0.16 ρ +0.18
n 1743/2024 queries • 498 targets



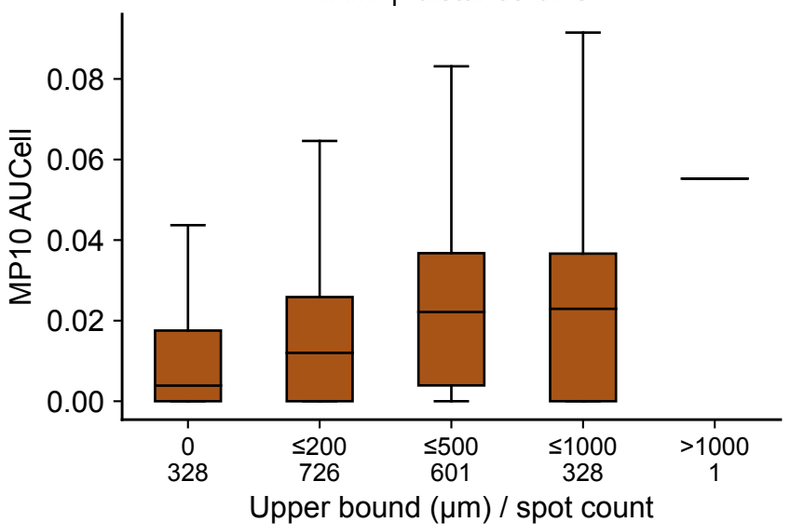
Orange = zero distance
Line = within-section linear fit



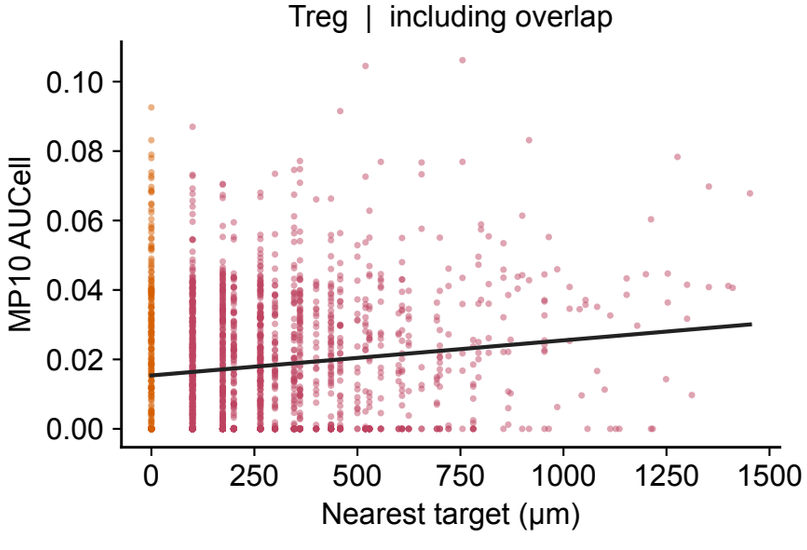
r +0.21 ρ +0.25
n 1984/2024 queries • 498 targets



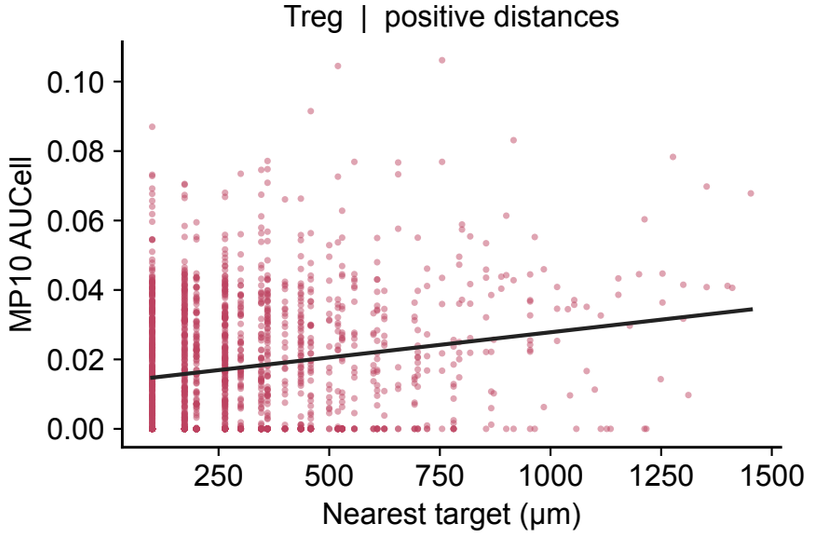
r +0.15 ρ +0.19
n 1656/2024 queries • 498 targets



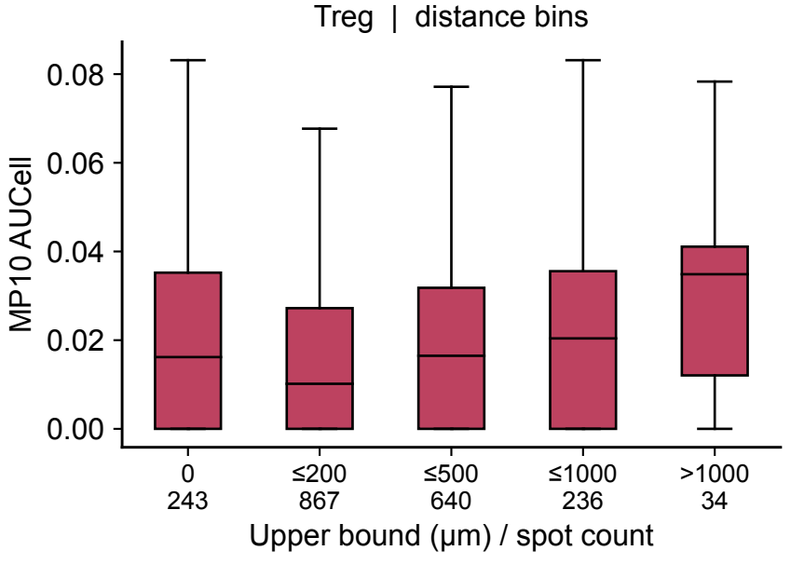
Orange = zero distance
Line = within-section linear fit



r +0.13 ρ +0.09
n 2020/2024 queries • 498 targets



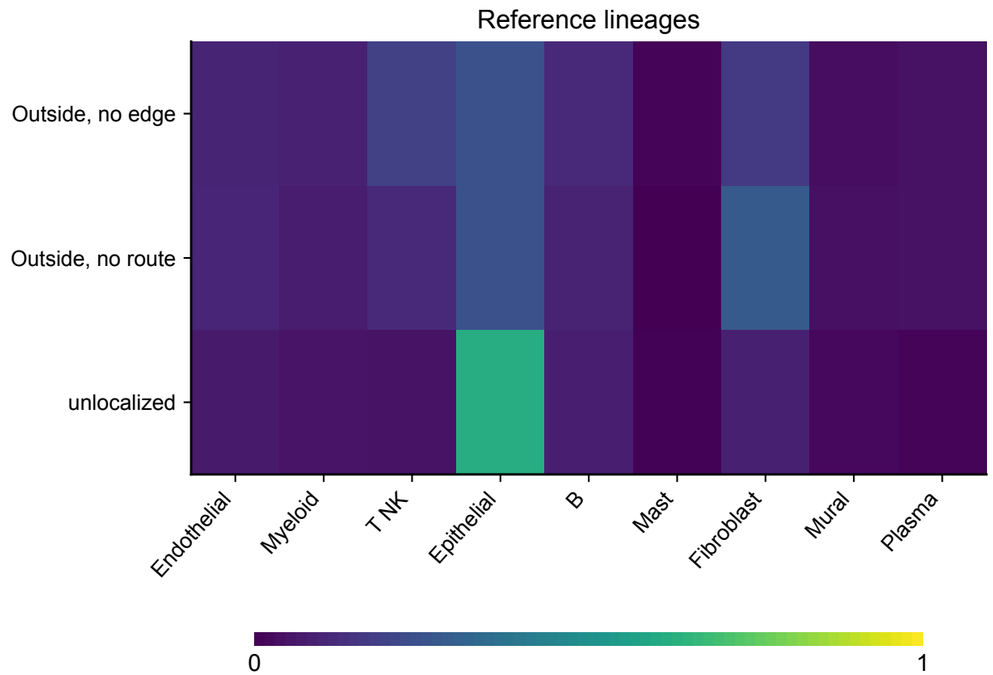
r +0.19 ρ +0.14
n 1777/2024 queries • 498 targets

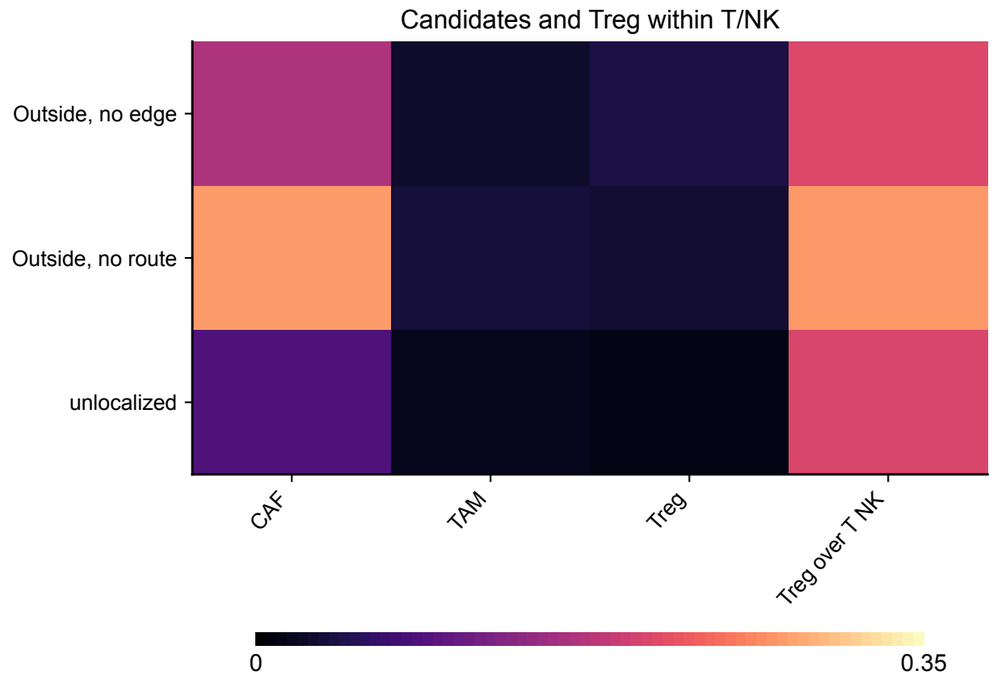


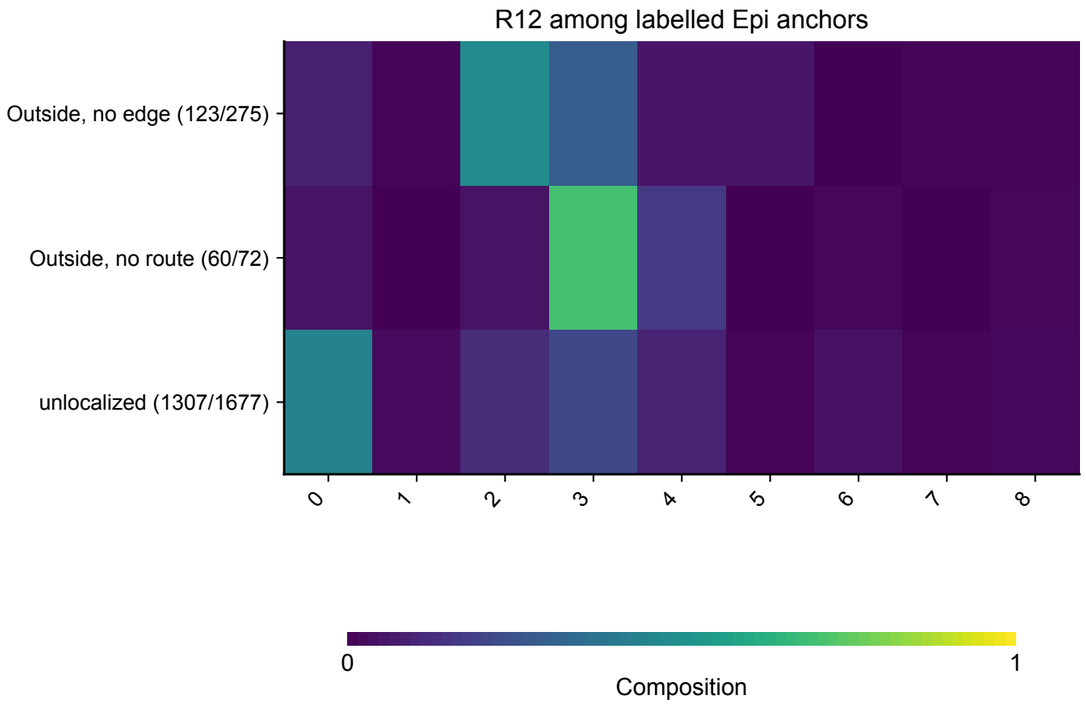
Orange = zero distance
Line = within-section linear fit

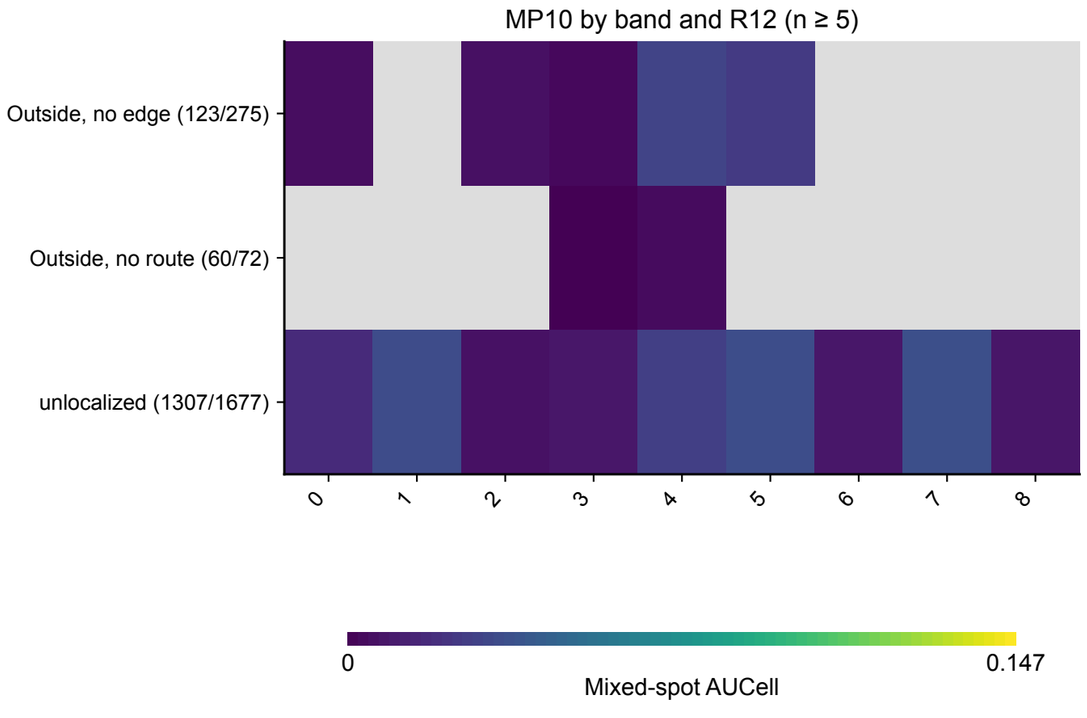
GSE285715_N3 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





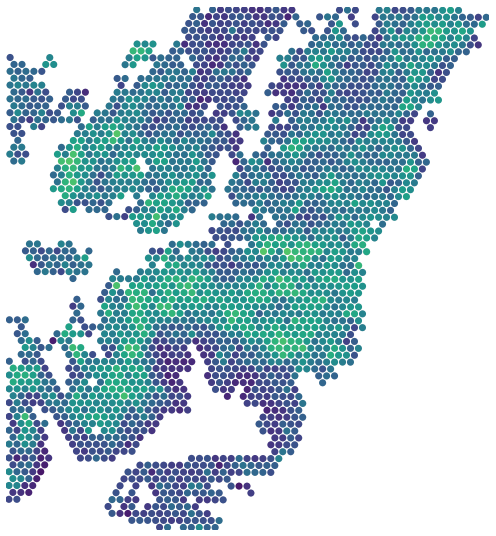




GSE285715_N3 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

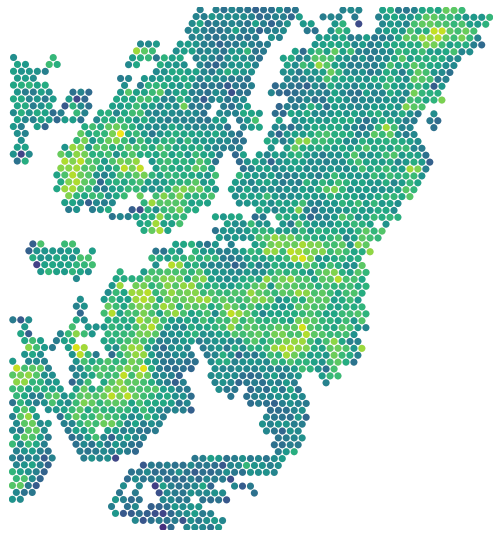
E2F Targets



0 0.152

UCell score

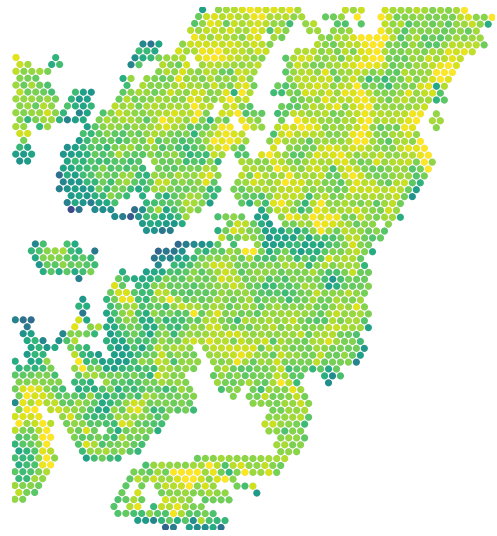
G2M Checkpoint



0 0.134

UCell score

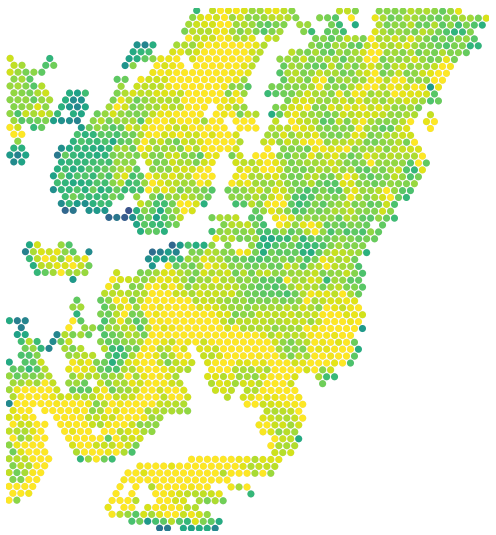
Hypoxia



0 0.211

UCell score

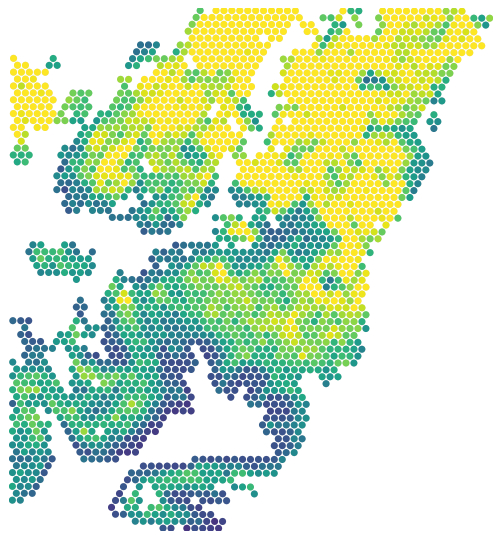
Tnfa Signaling Via Nfkb



0 0.275

UCell score

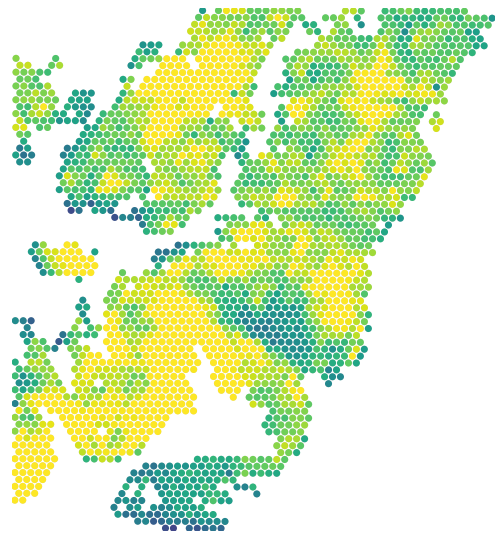
Emt



0 0.386

UCell score

Interferon Gamma Response



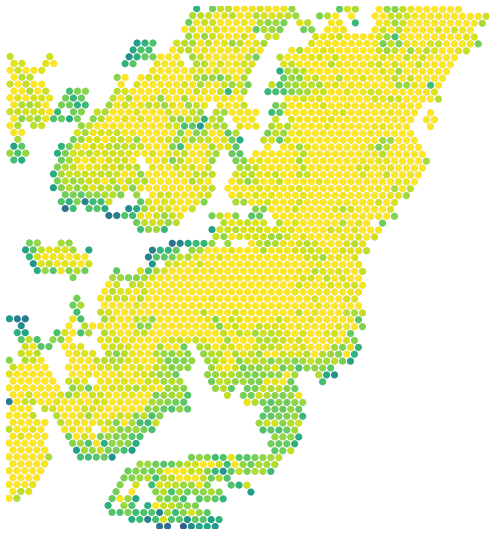
0 0.275

UCell score

GSE285715_N3 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

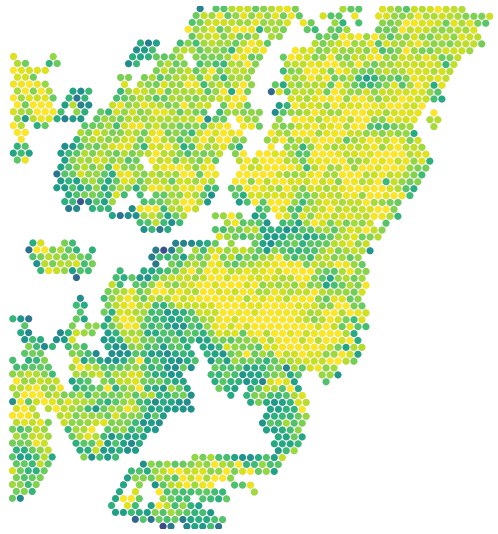
Il2 Stat5 Signaling



0 0.144

UCell score

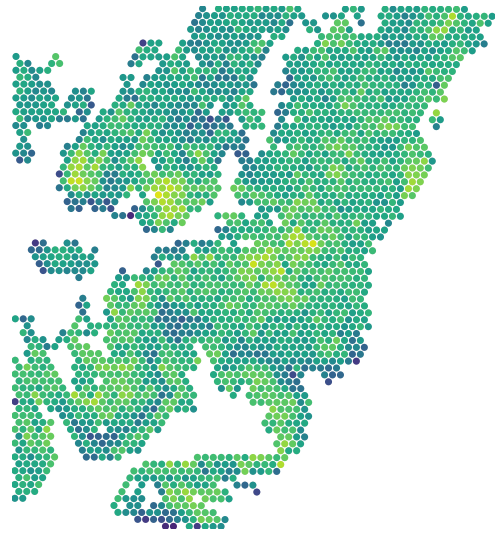
Tgf Beta Signaling



0 0.232

UCell score

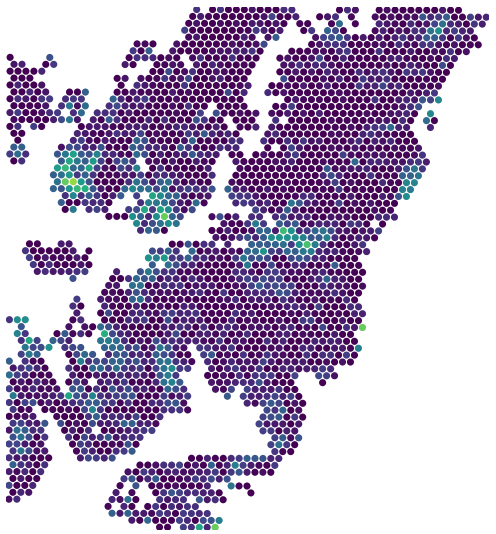
Cholesterol Homeostasis



0 0.21

UCell score

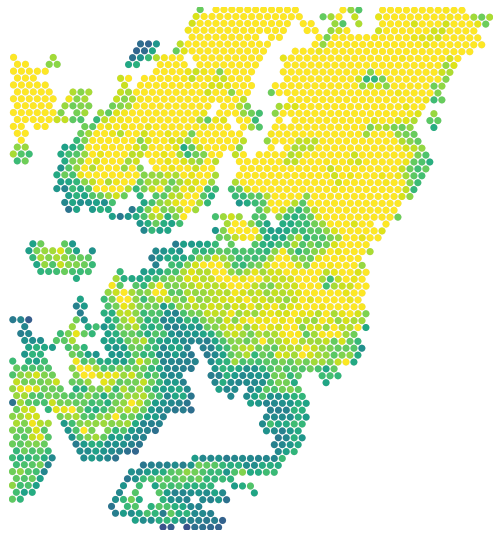
Cholesterol Biosynthesis



0 0.166

UCell score

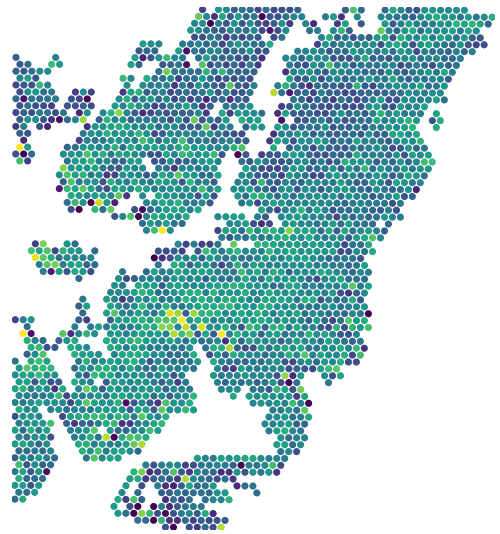
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



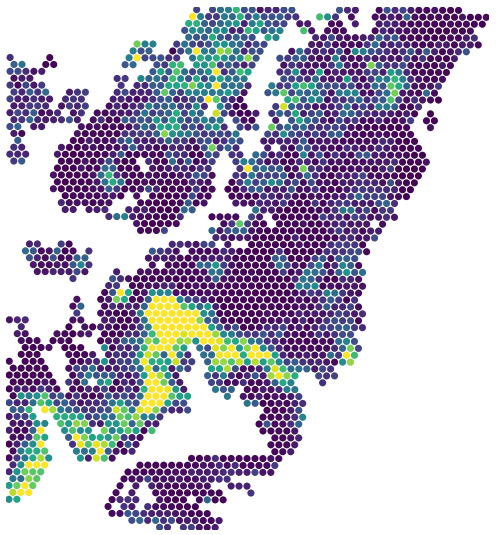
0 0.151

UCell score

GSE285715_N3 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

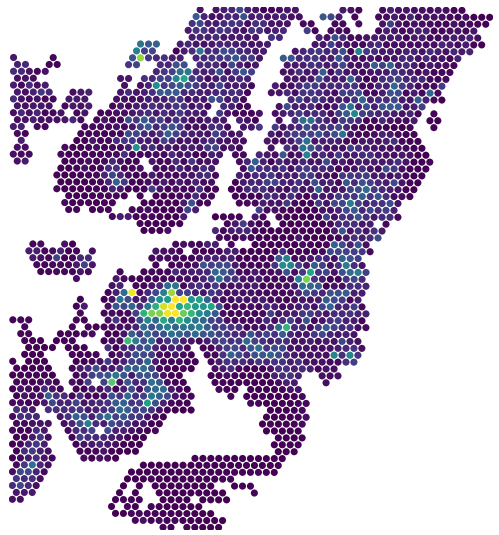
Treg composition



0 0.102

Fraction of all 30

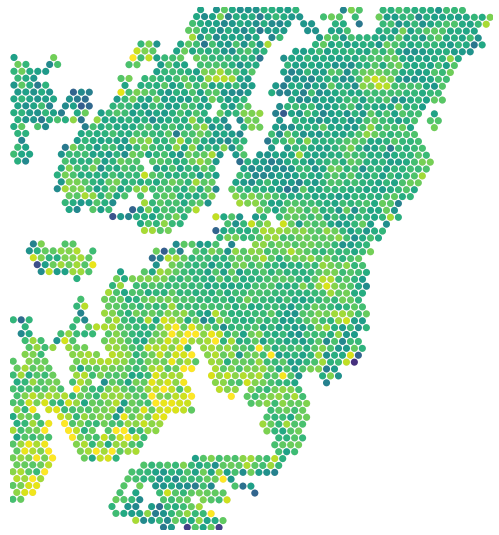
Treg estimated abundance



0 2.85

Posterior mean estimate

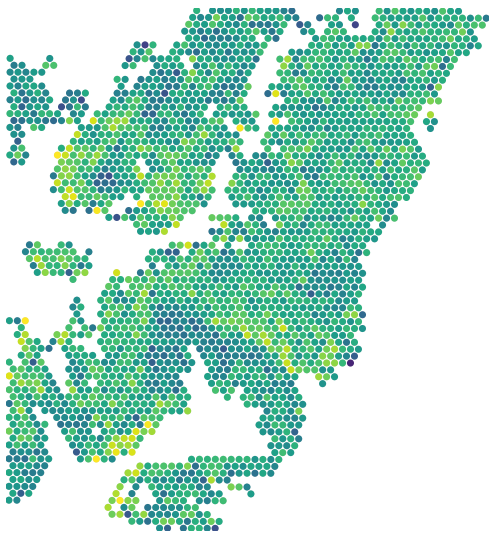
Treg Resting Vs Activated Dn



0 0.0778

UCell score

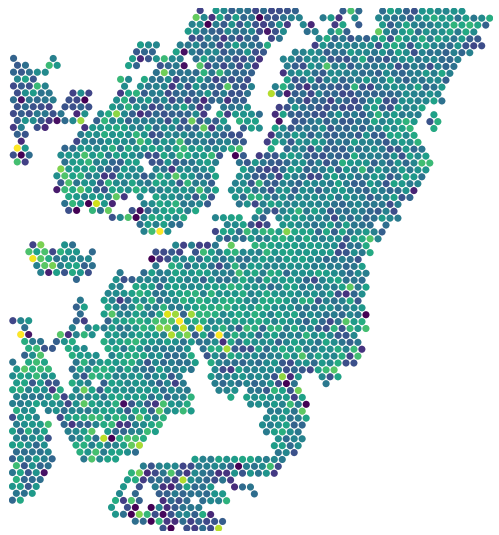
Treg Resting Vs Activated Up



0 0.0609

UCell score

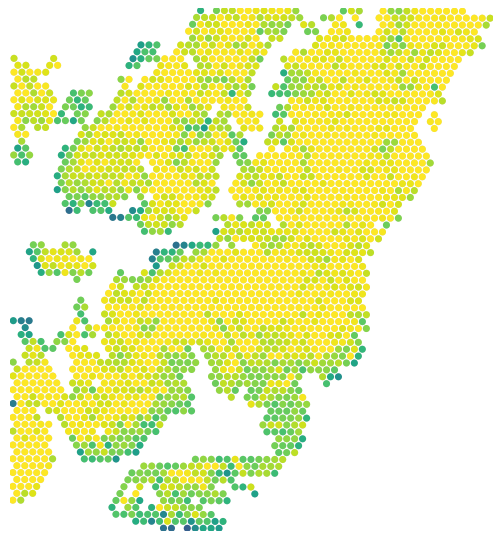
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

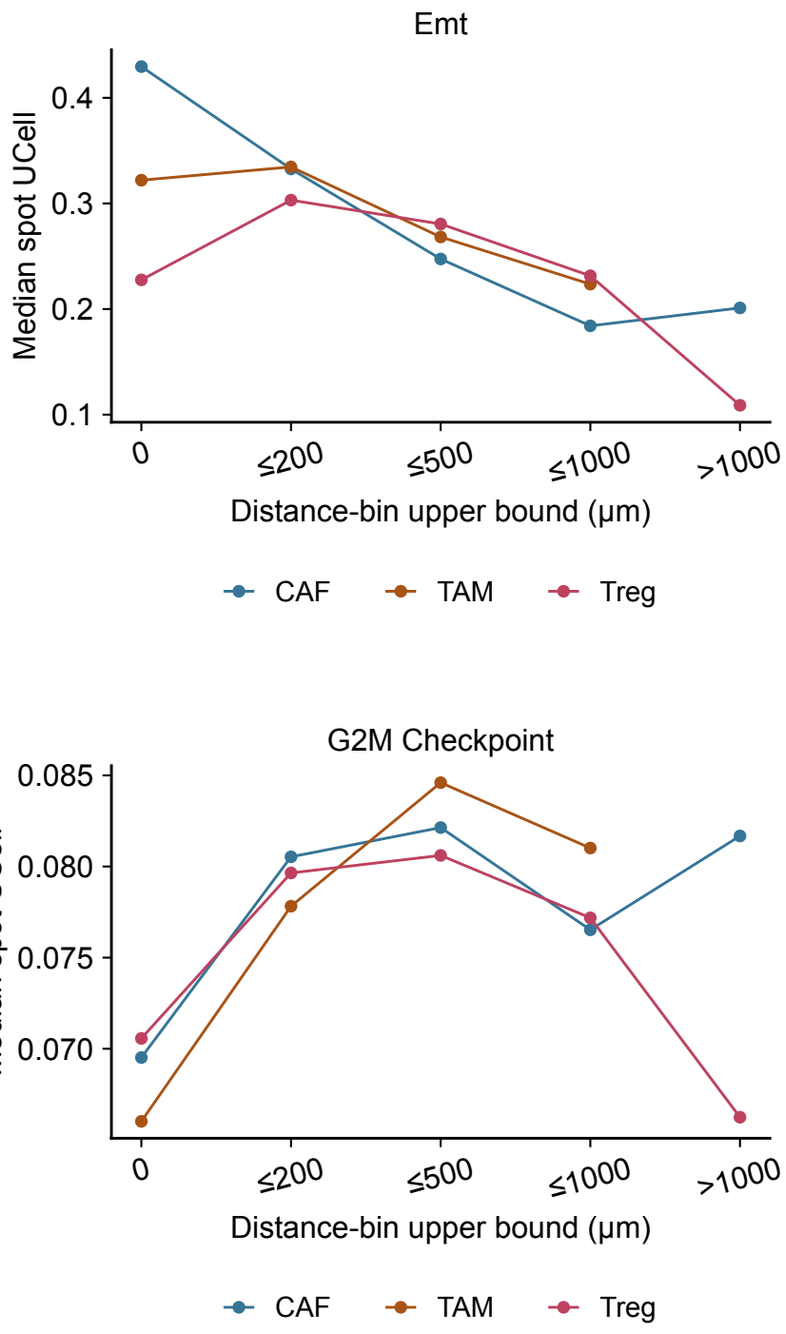


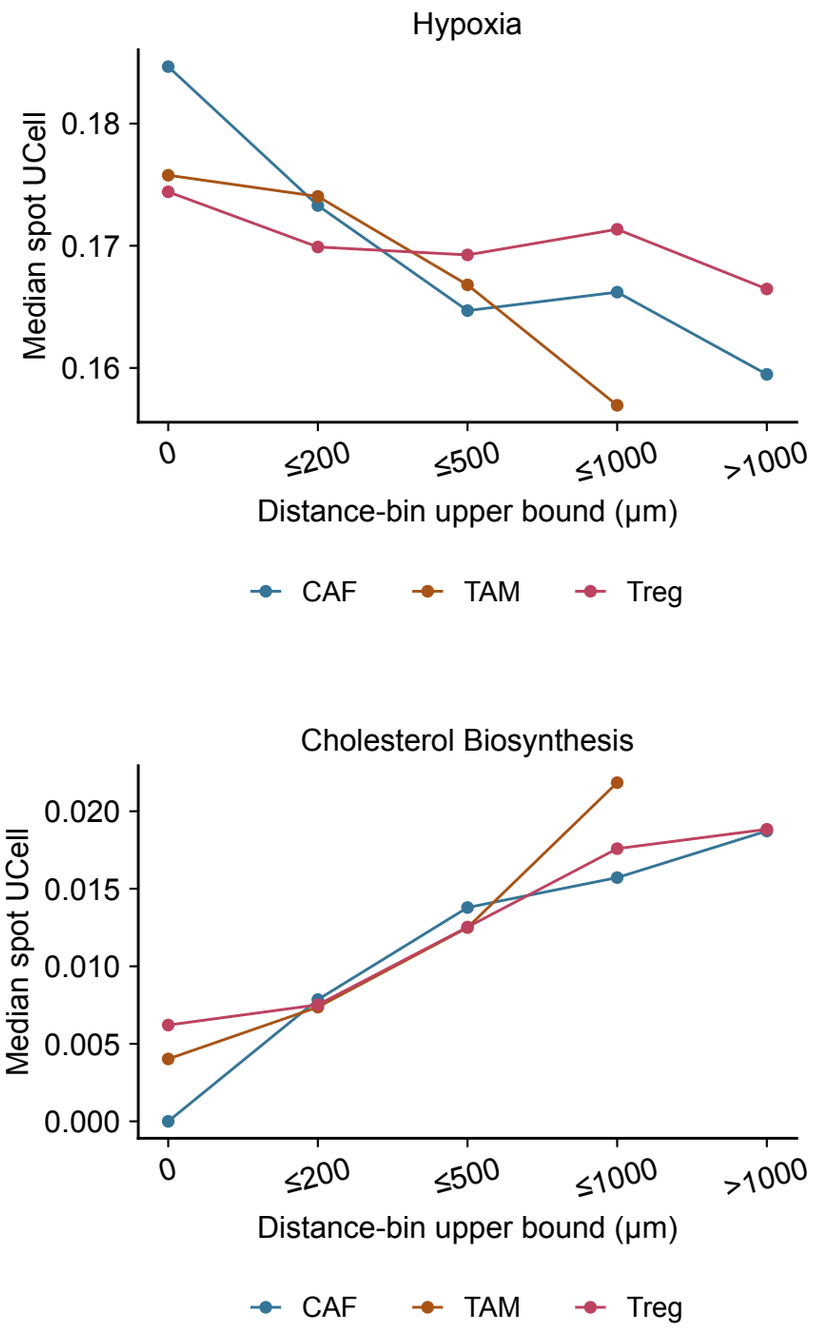
0 0.144

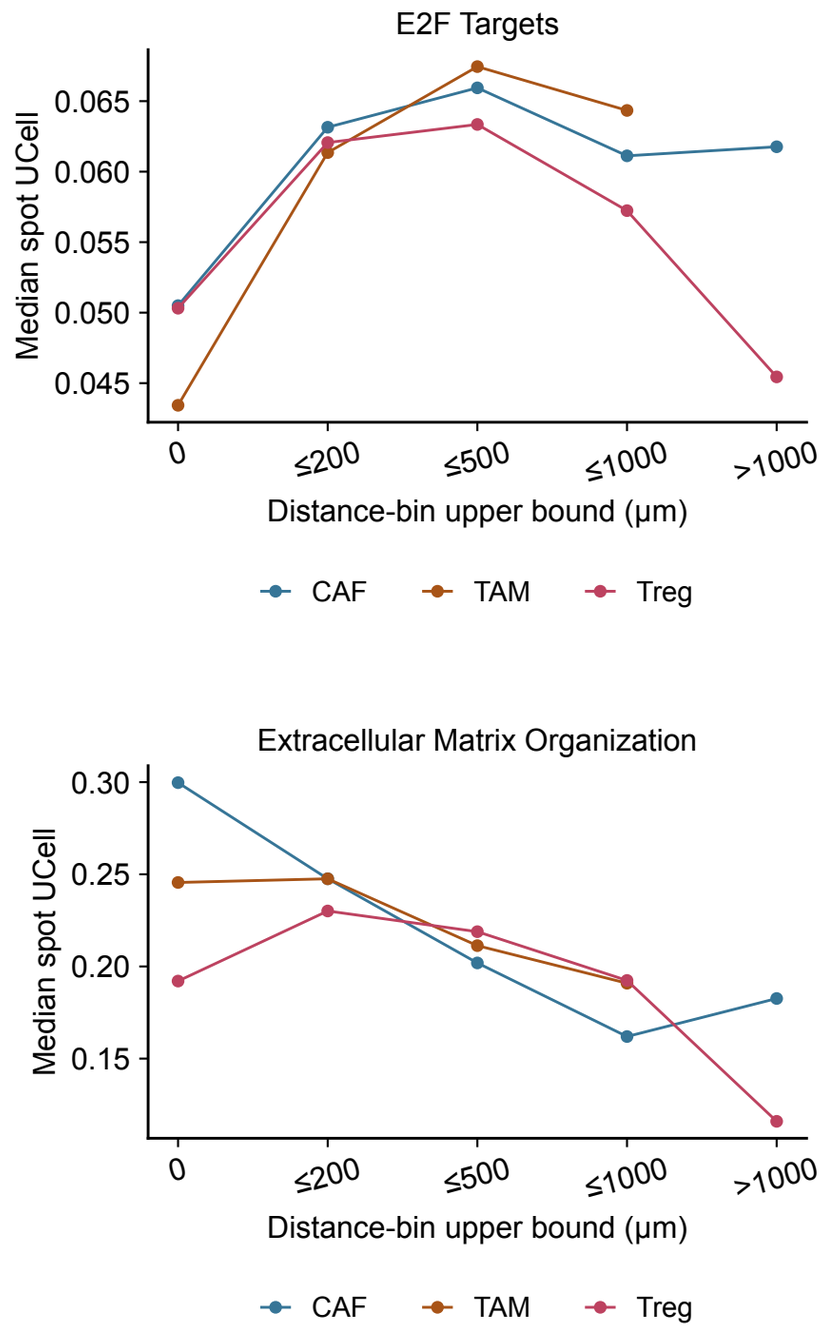
UCell score

GSE285715_N3 | Programs by target distance

Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

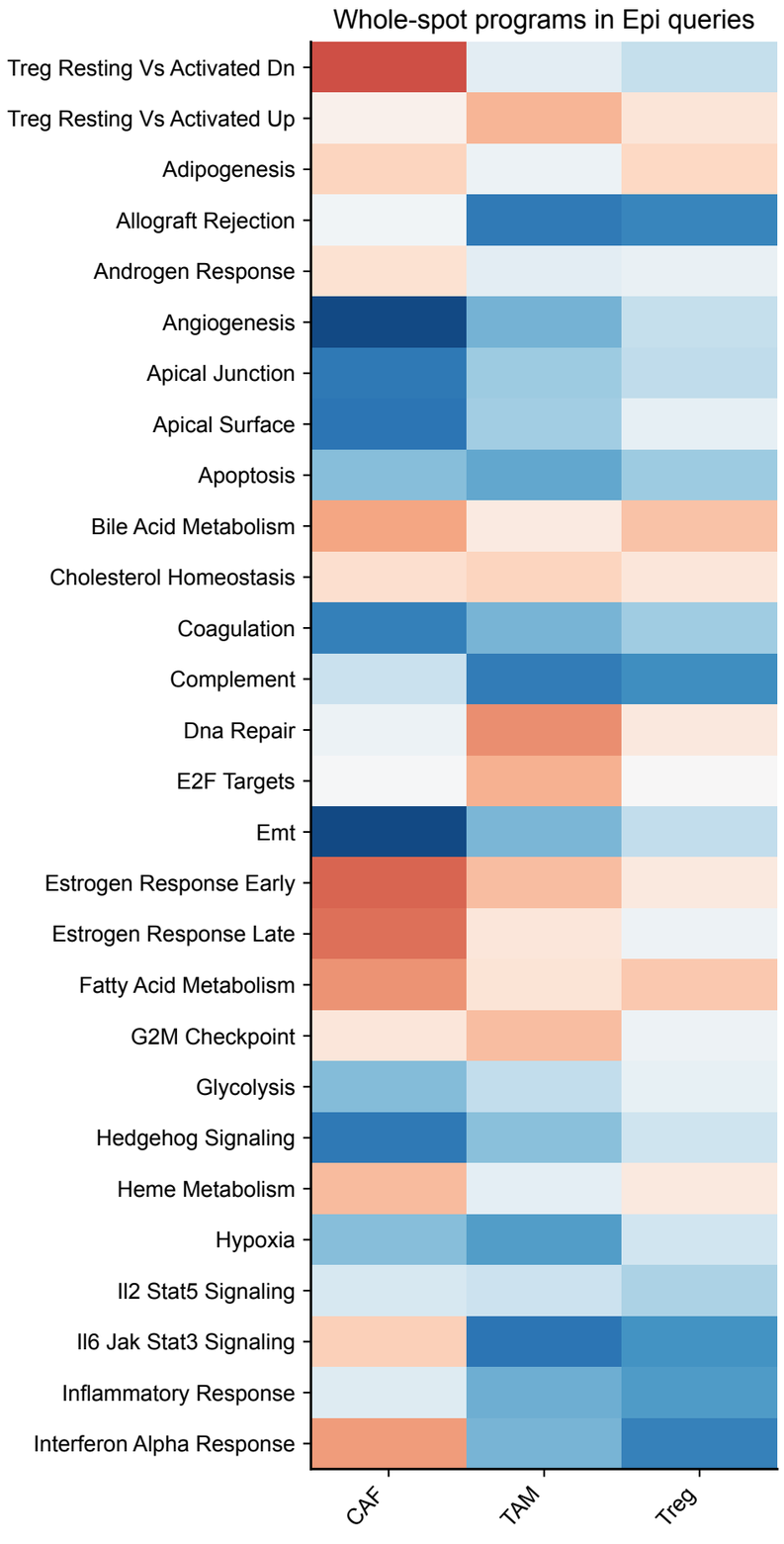


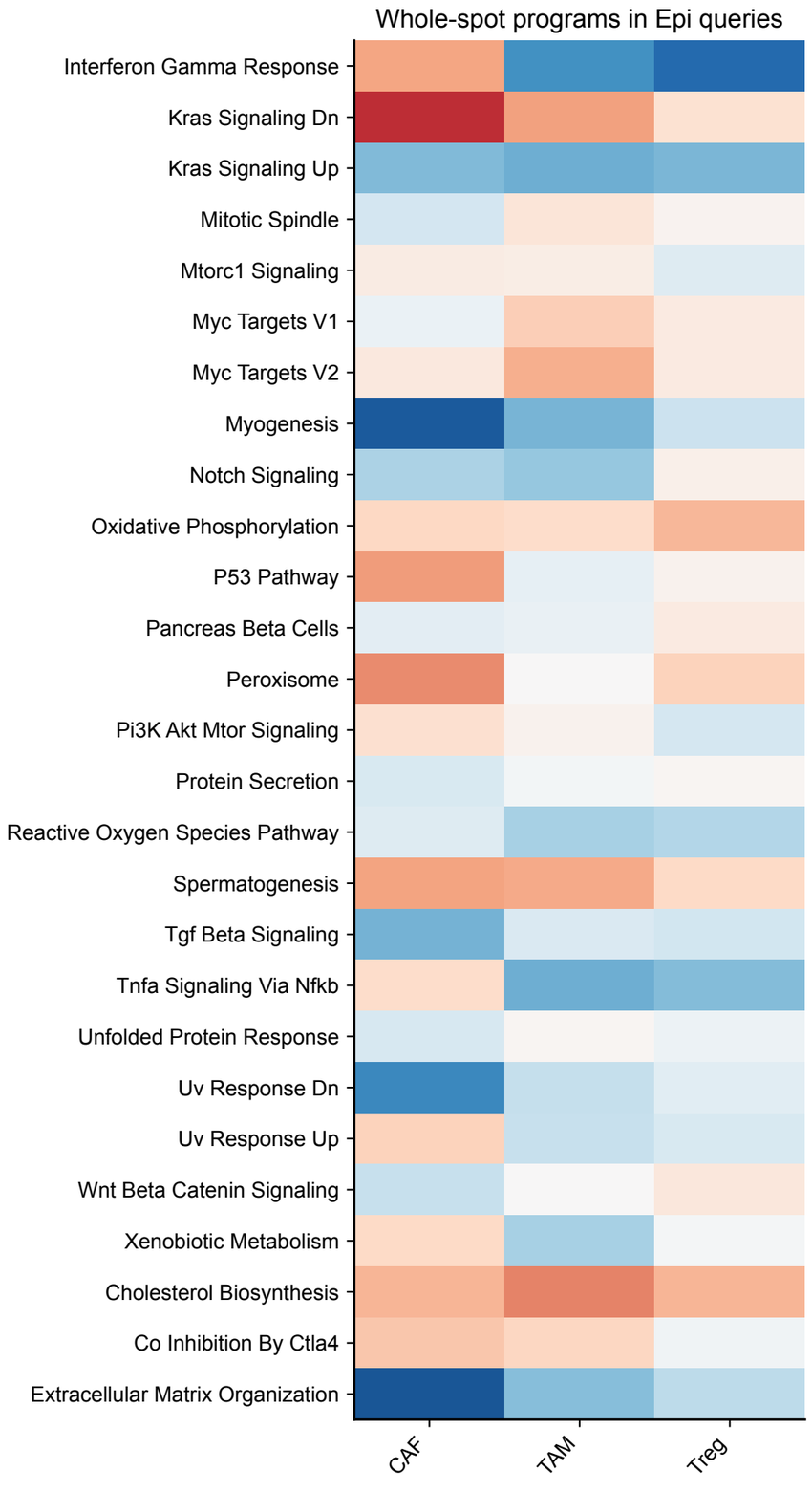




GSE285715_N3 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

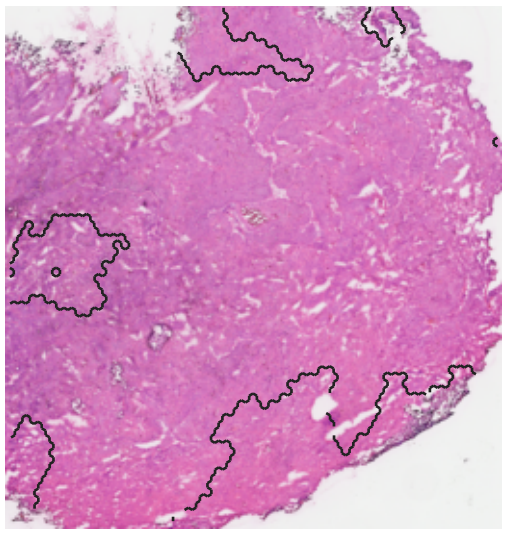




GSE285715_T1 | Tissue context

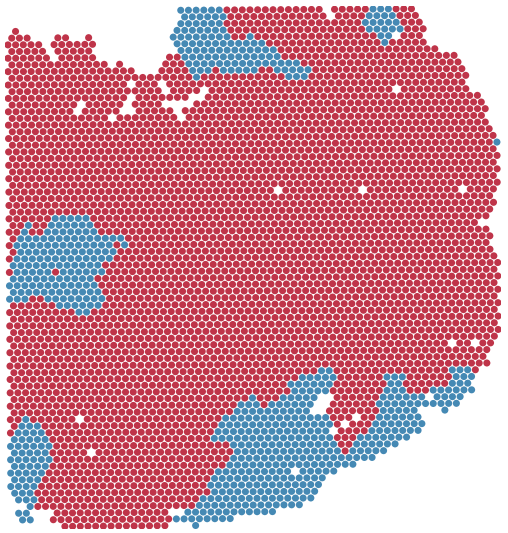
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



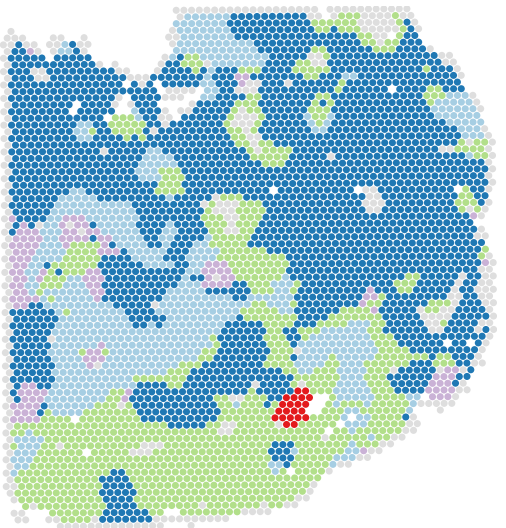
Black line: observed candidate outer edge

Final candidate location



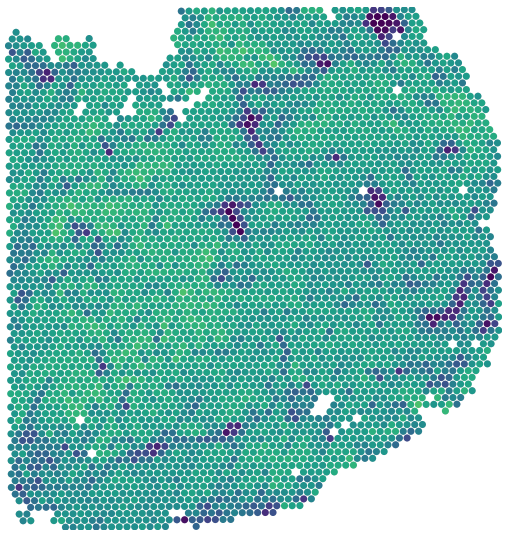
inside outside

Accepted R12 (original 0–8)



0 1 2 7 8

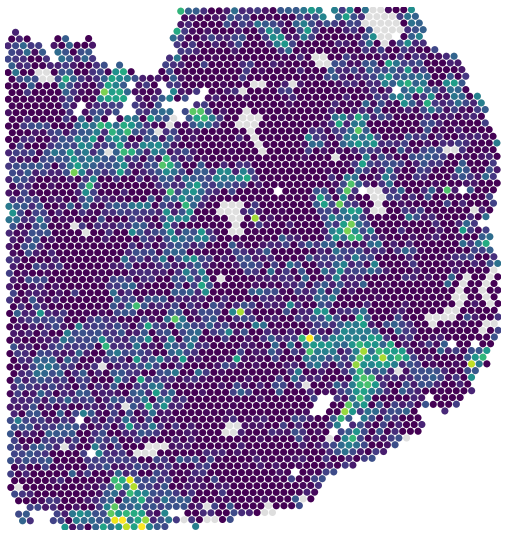
Epithelial support



0 1

Fraction of all 30

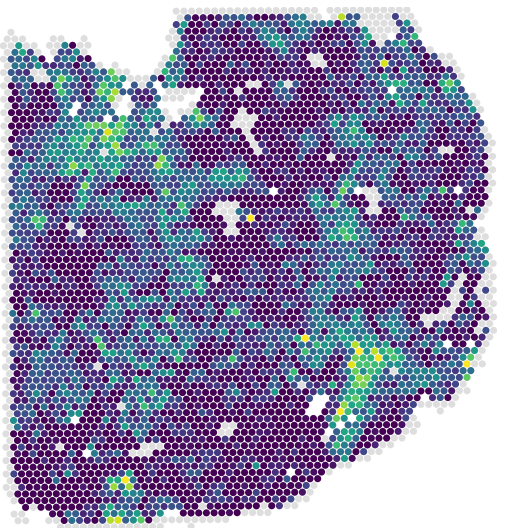
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

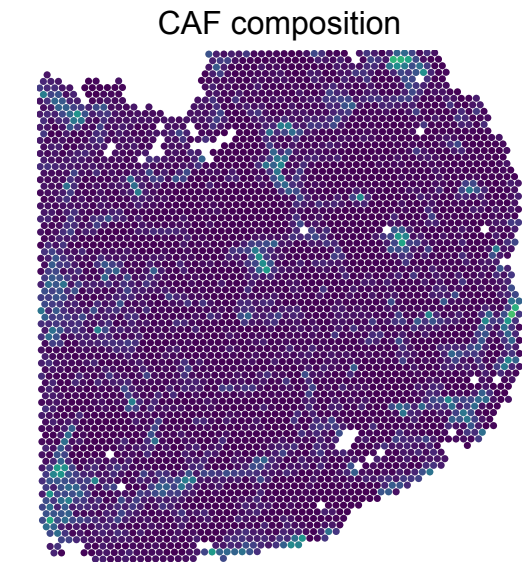


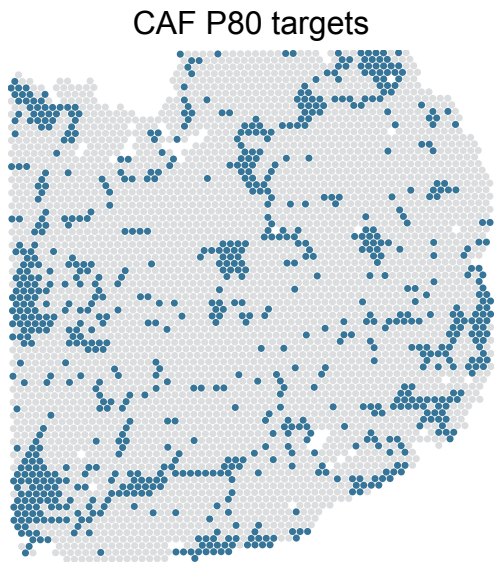
0 0.14

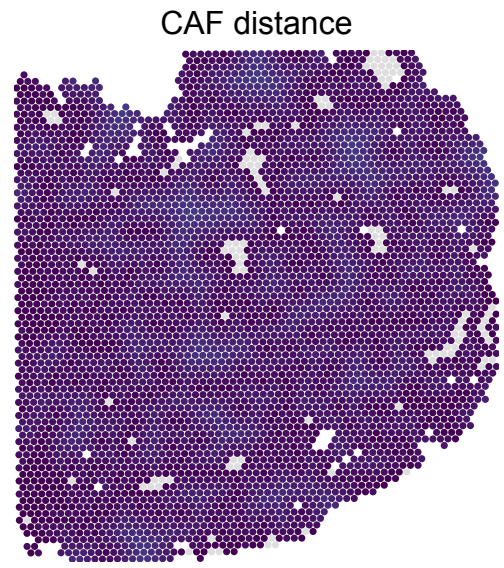
Matched-model AUCell

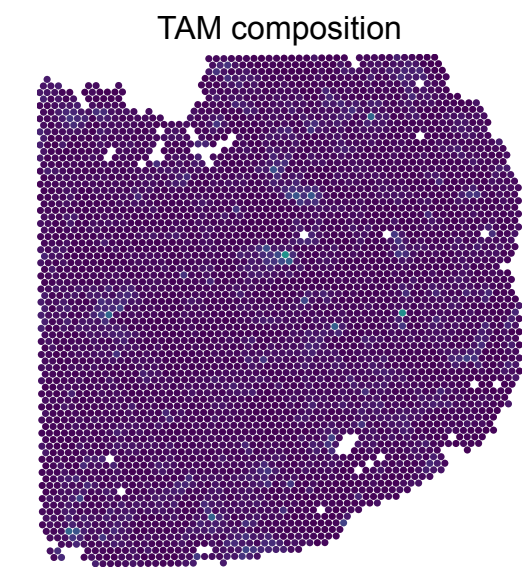
GSE285715_T1 | Candidate neighborhoods

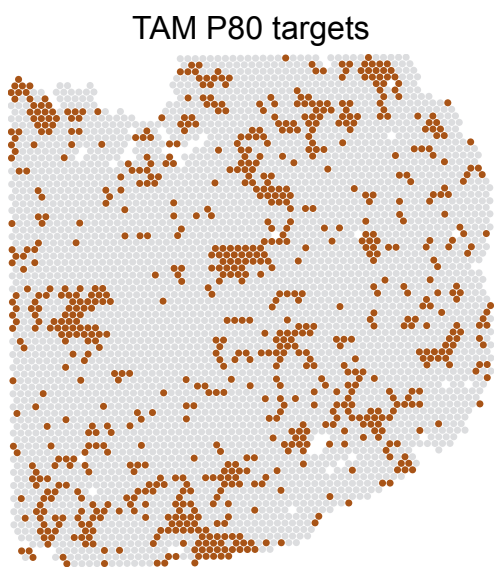
Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

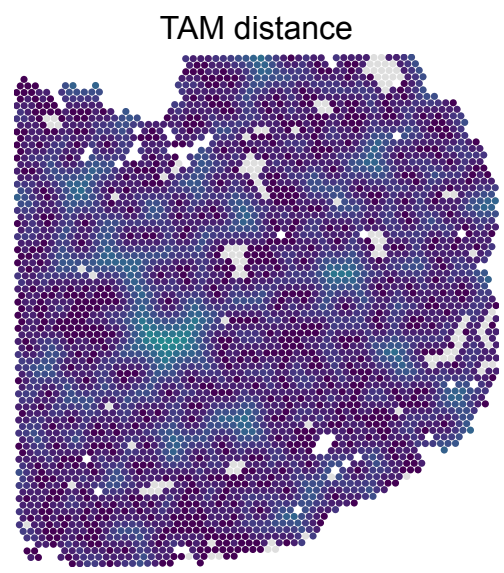


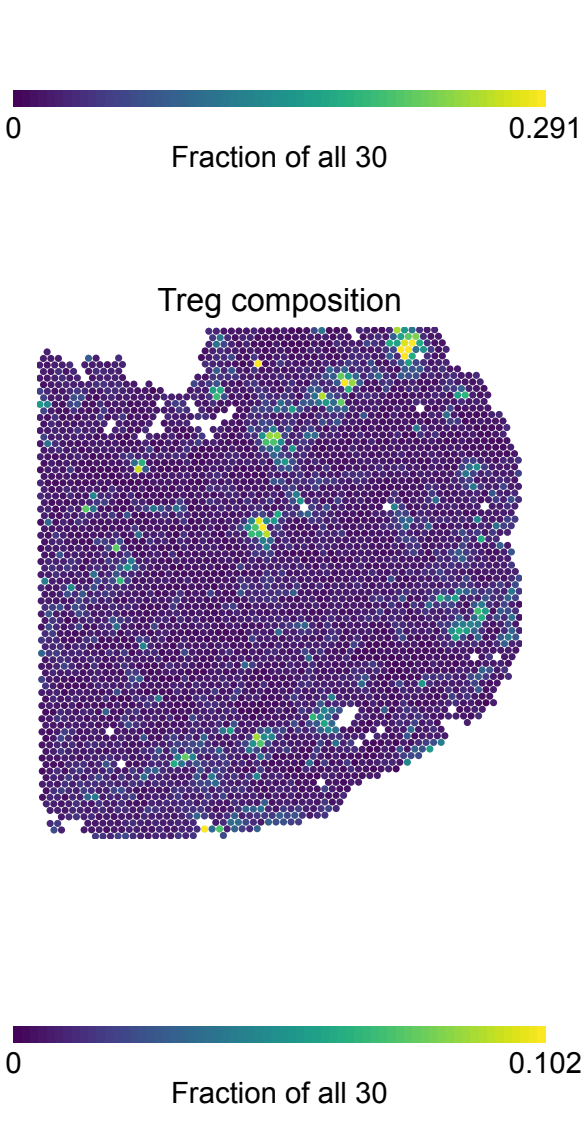


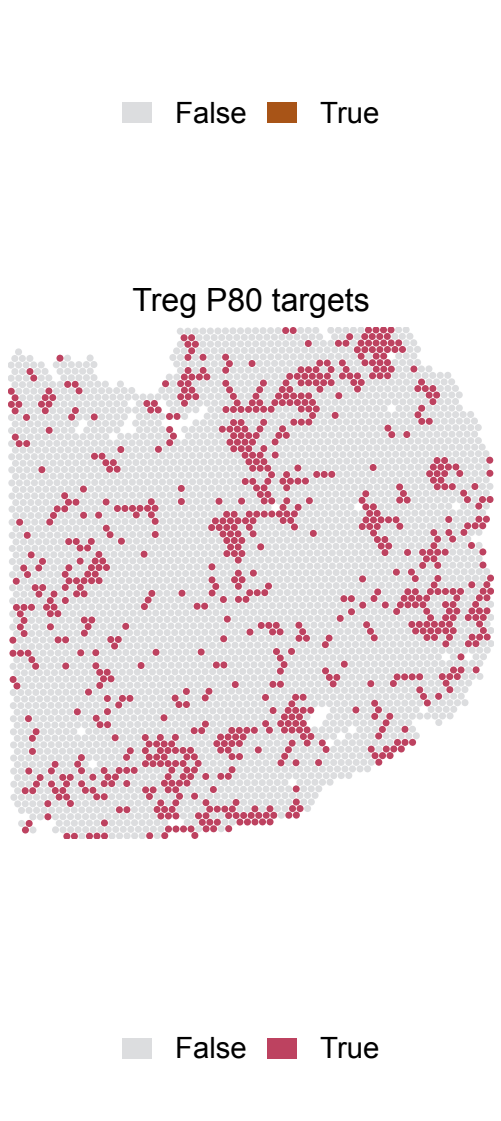


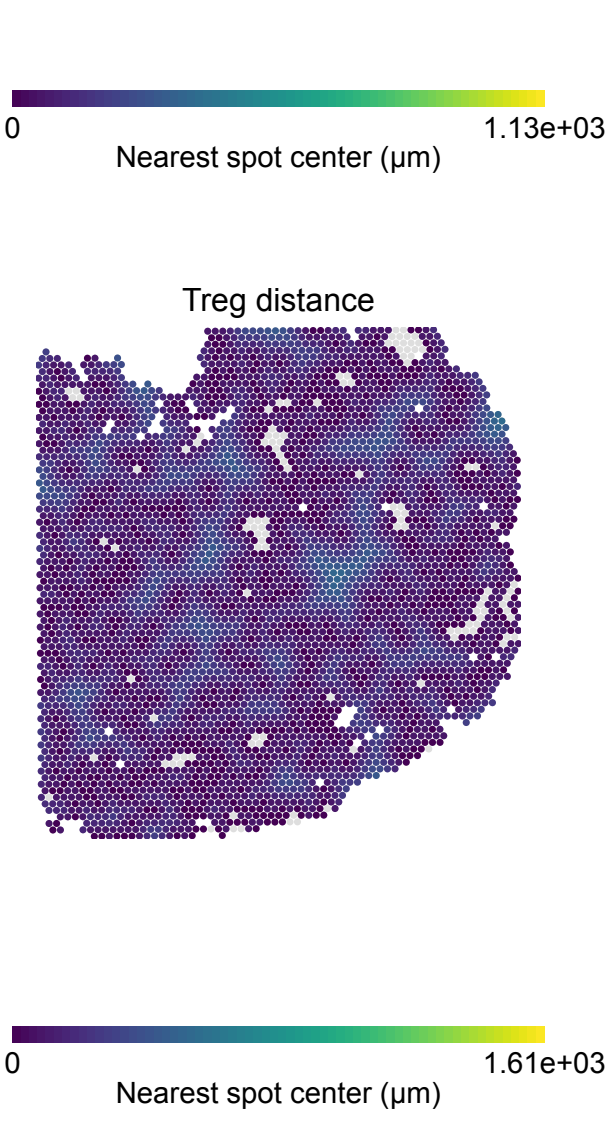






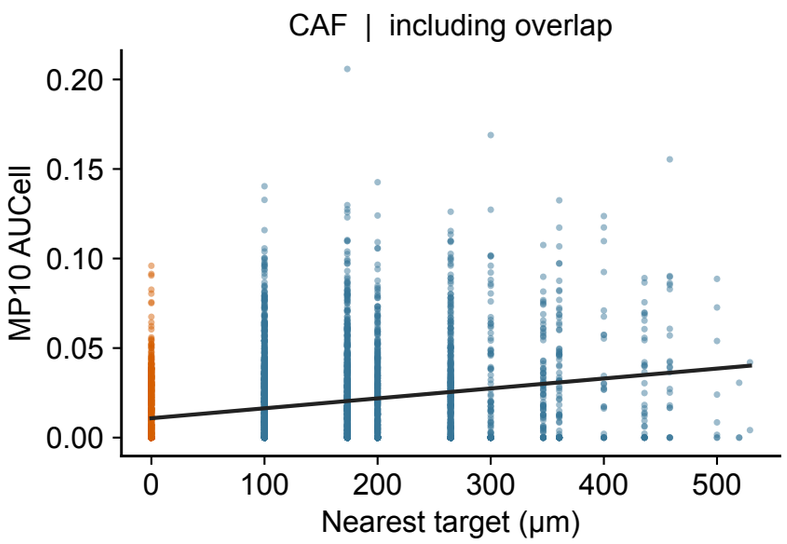




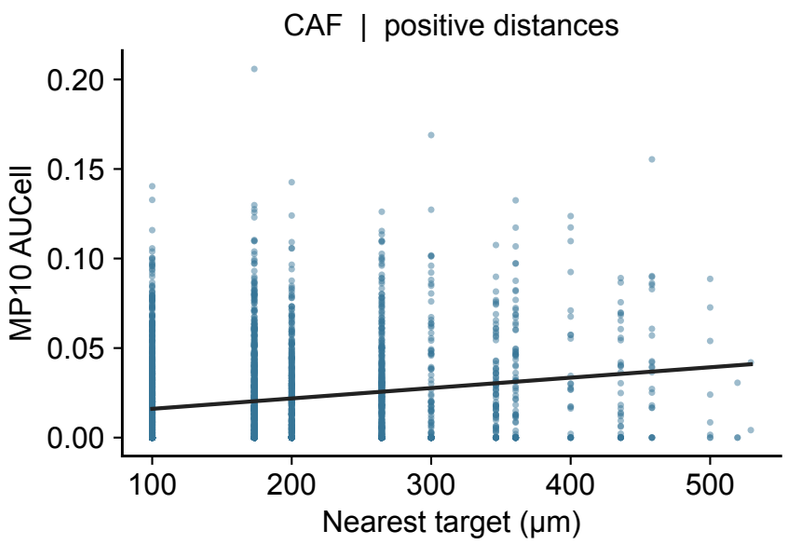


GSE285715_T1 | MP10 versus nearest target

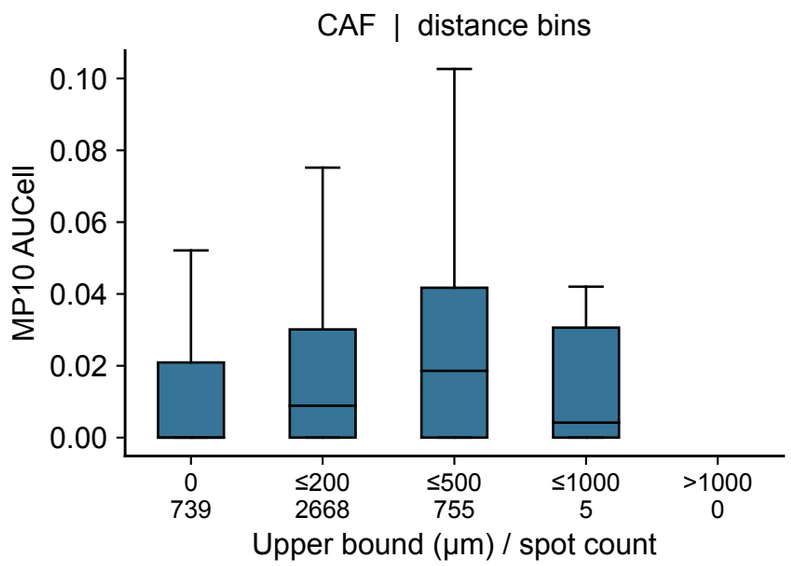
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



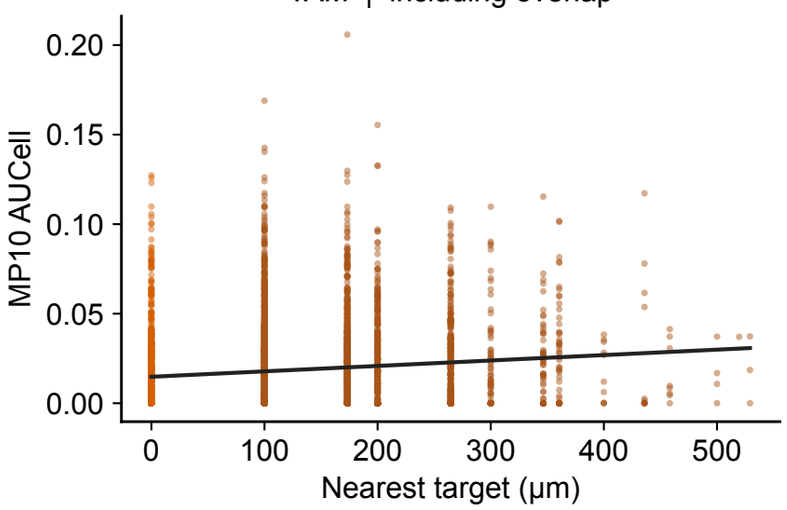
r +0.23 ρ +0.22
n 4167/4167 queries • 857 targets



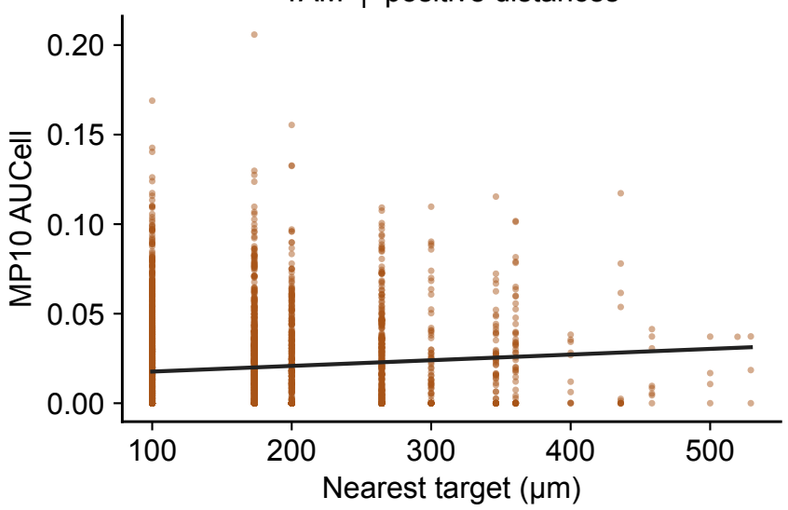
r +0.20 ρ +0.19
n 3428/4167 queries • 857 targets



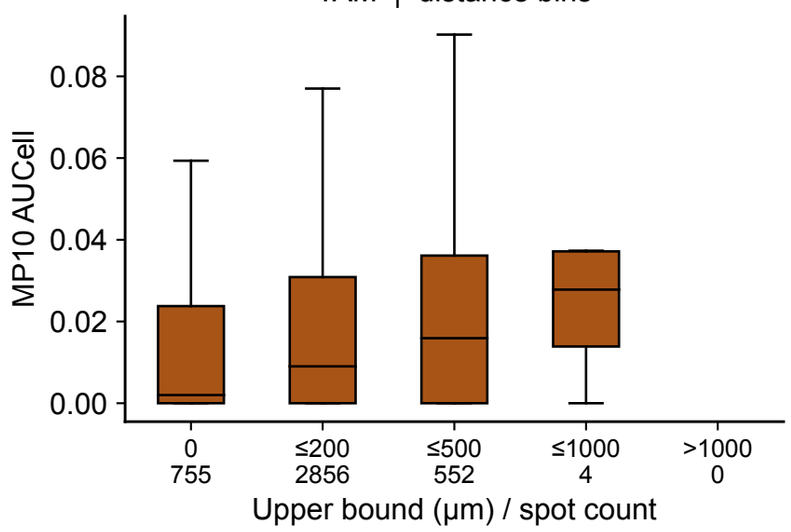
Orange = zero distance
Line = within-section linear fit



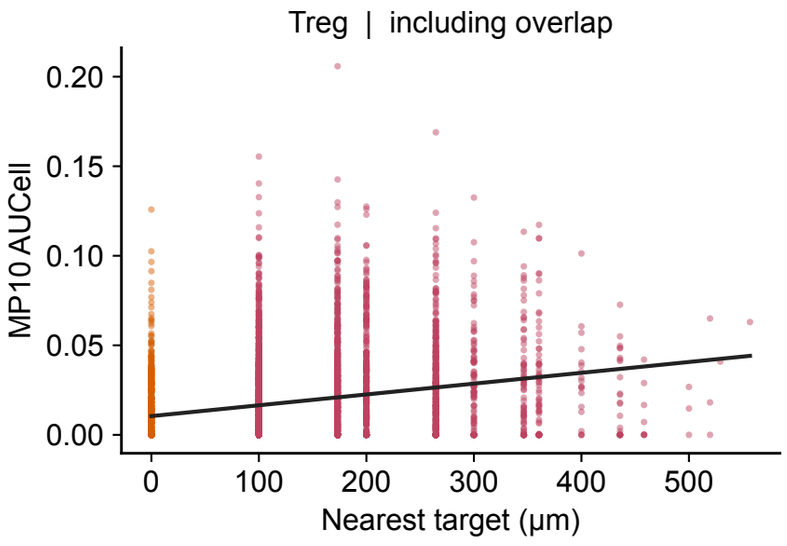
r +0.11 ρ +0.14
n 4167/4167 queries • 857 targets



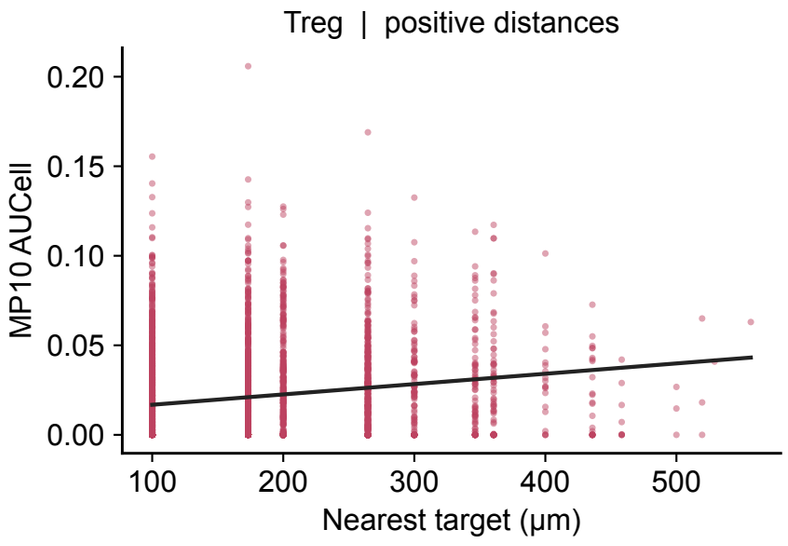
r +0.10 ρ +0.12
n 3412/4167 queries • 857 targets



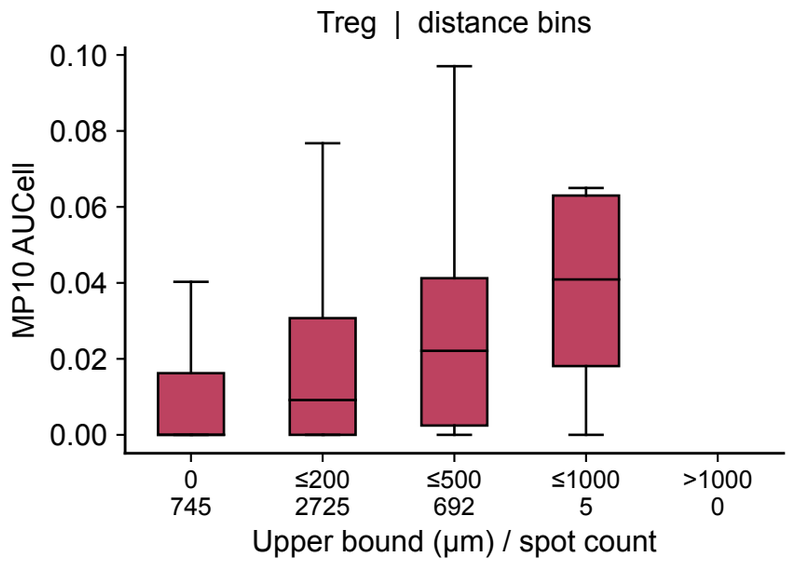
Orange = zero distance
Line = within-section linear fit



r +0.24 ρ +0.28
n 4167/4167 queries • 857 targets



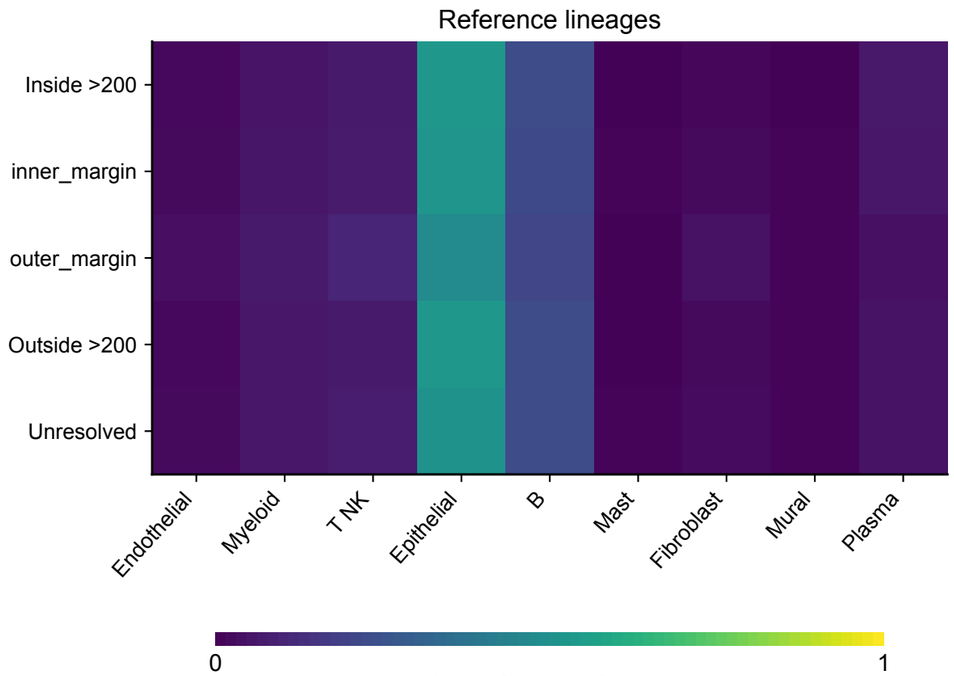
r +0.19 ρ +0.22
n 3422/4167 queries • 857 targets

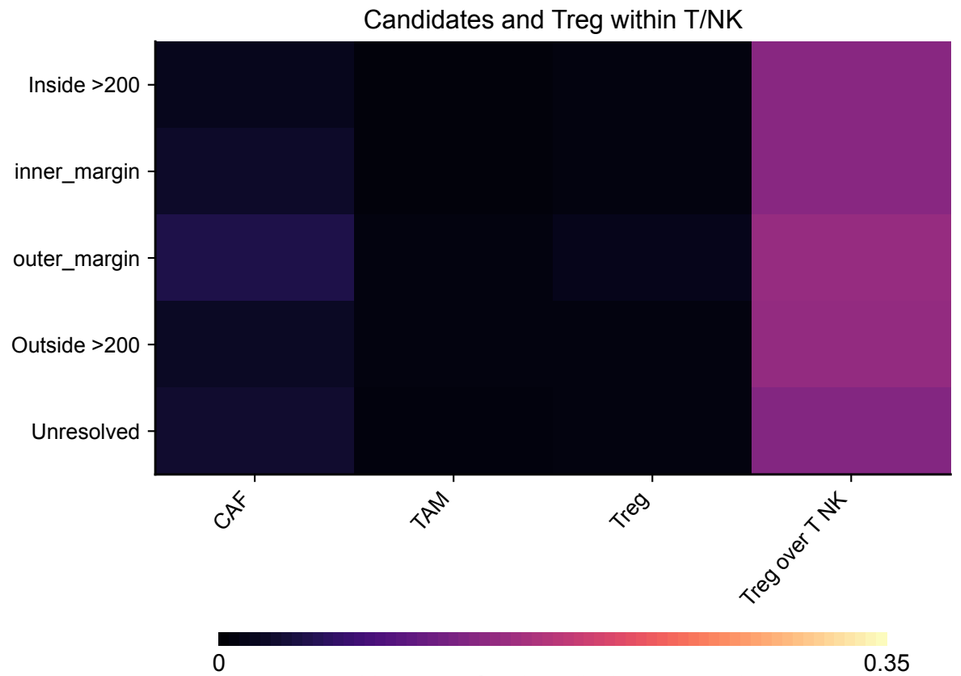


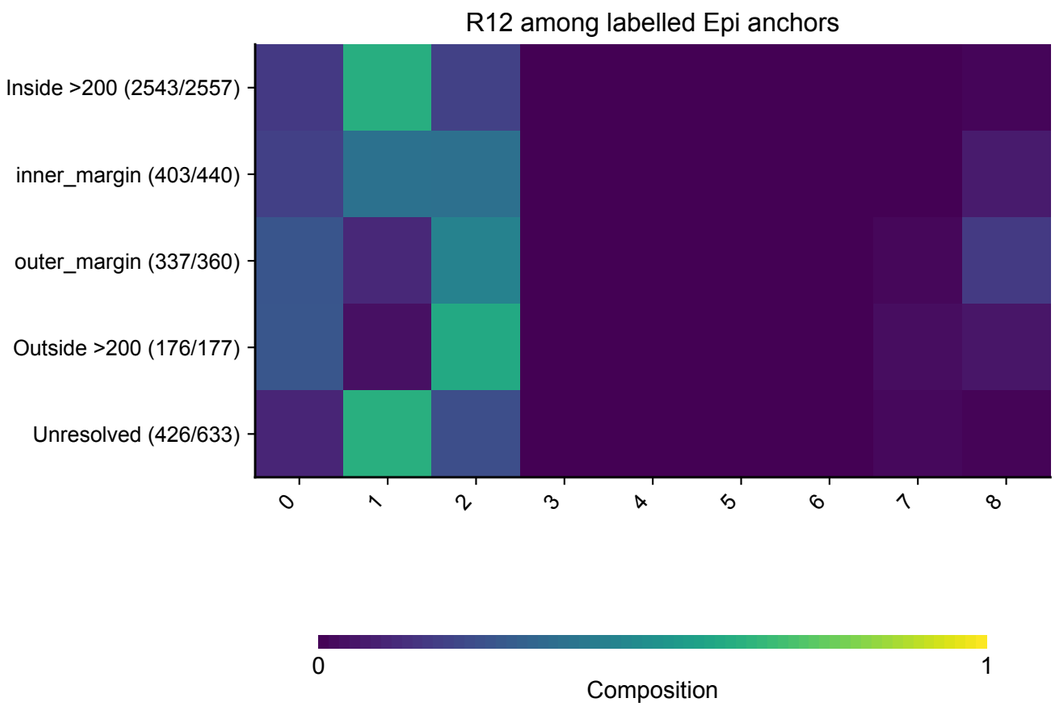
Orange = zero distance
Line = within-section linear fit

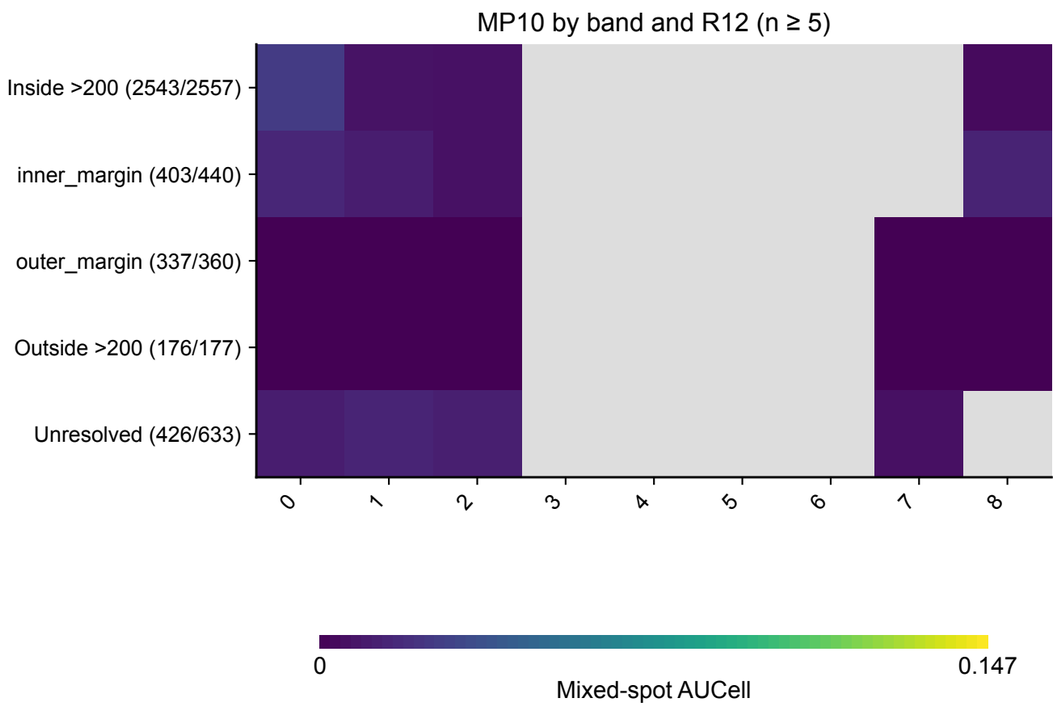
GSE285715_T1 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





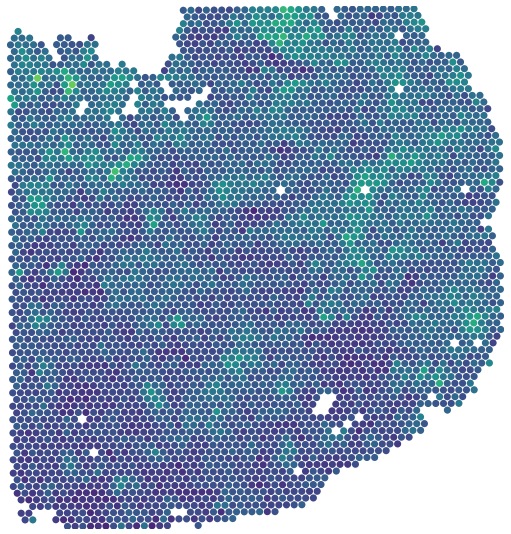




GSE285715_T1 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

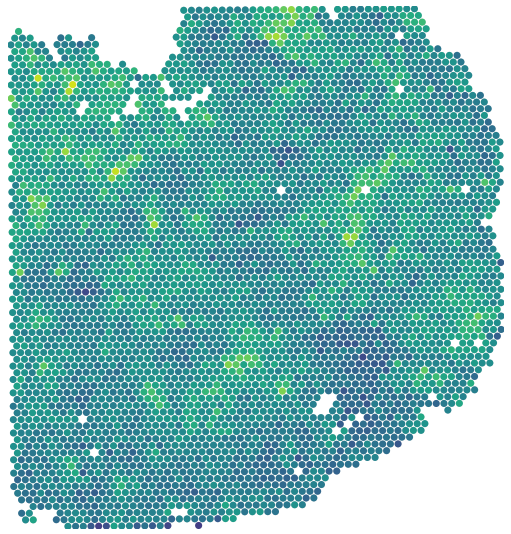
E2F Targets



0 0.152

UCell score

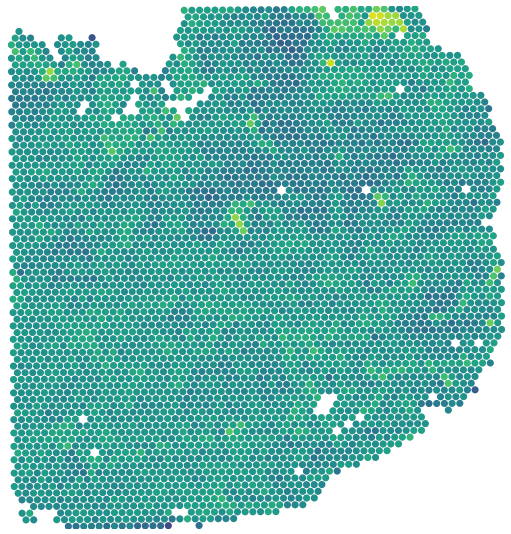
G2M Checkpoint



0 0.134

UCell score

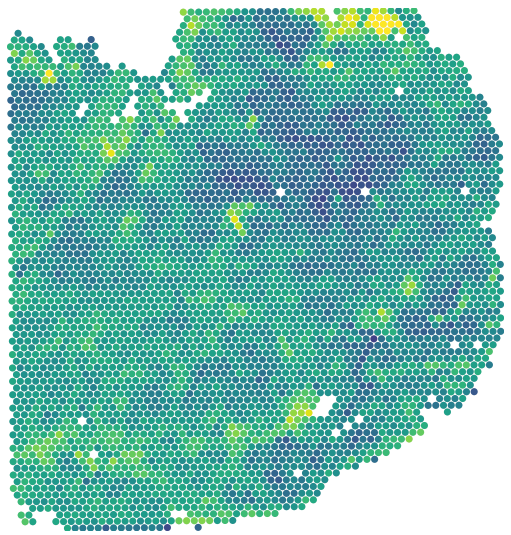
Hypoxia



0 0.211

UCell score

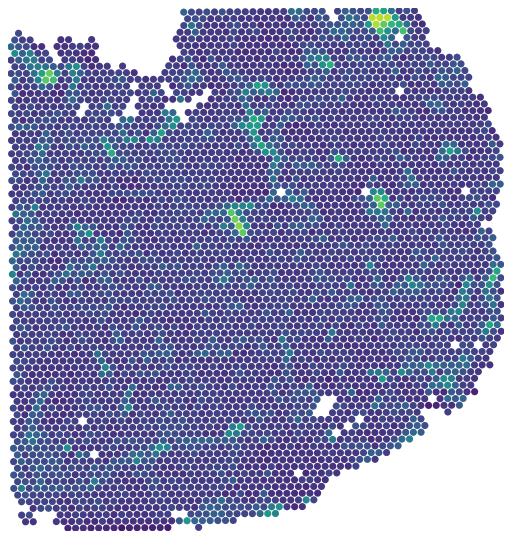
Tnfa Signaling Via Nfkb



0 0.275

UCell score

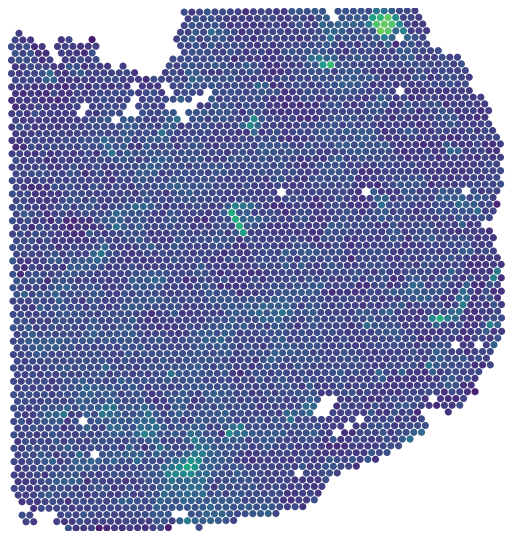
Emt



0 0.386

UCell score

Interferon Gamma Response



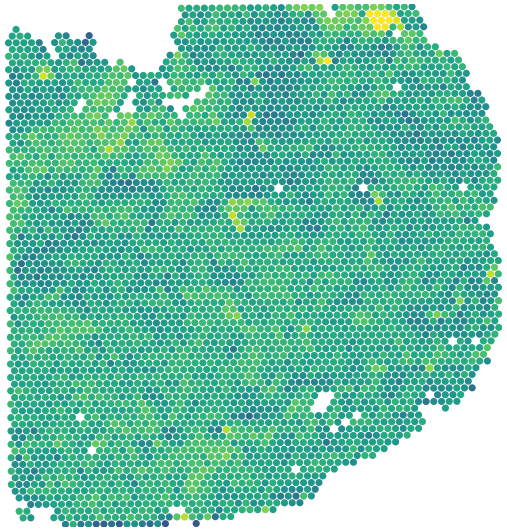
0 0.275

UCell score

GSE285715_T1 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

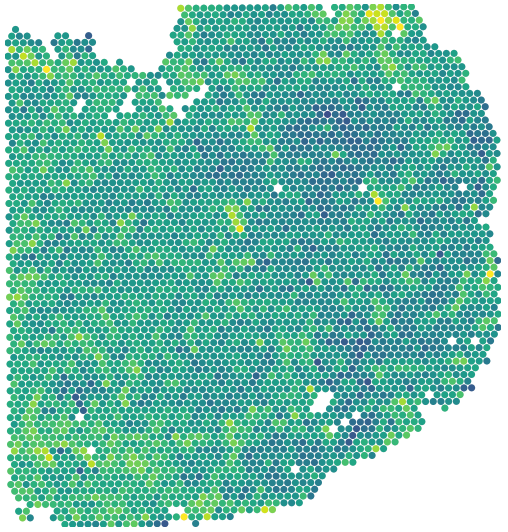
Il2 Stat5 Signaling



0 0.144

UCell score

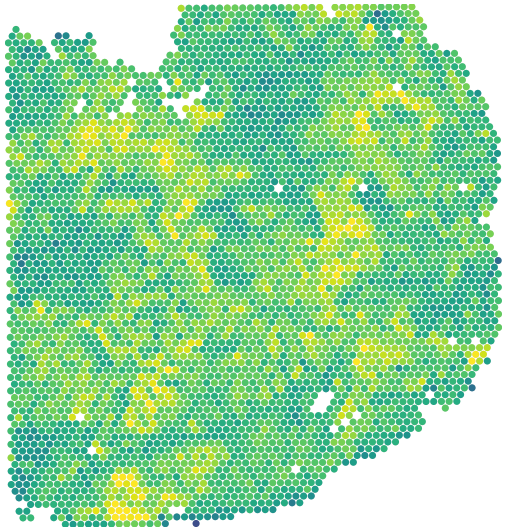
Tgf Beta Signaling



0 0.232

UCell score

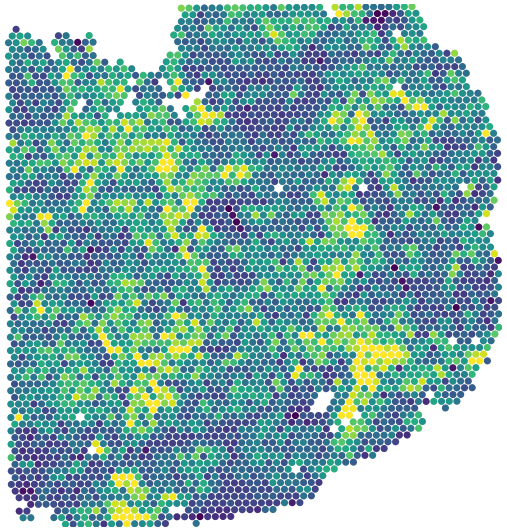
Cholesterol Homeostasis



0 0.21

UCell score

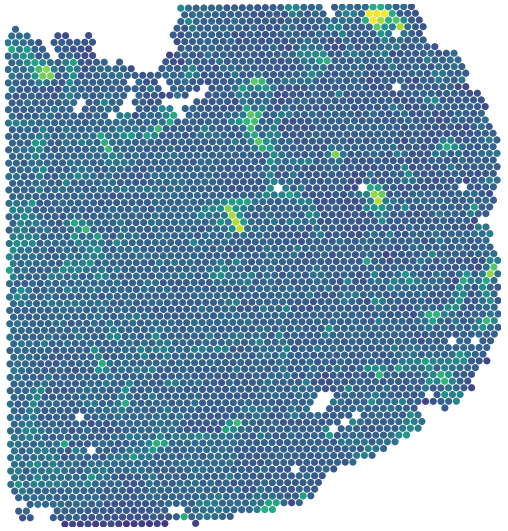
Cholesterol Biosynthesis



0 0.166

UCell score

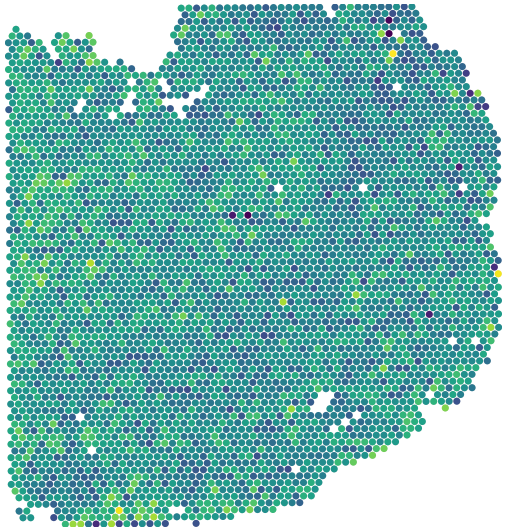
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



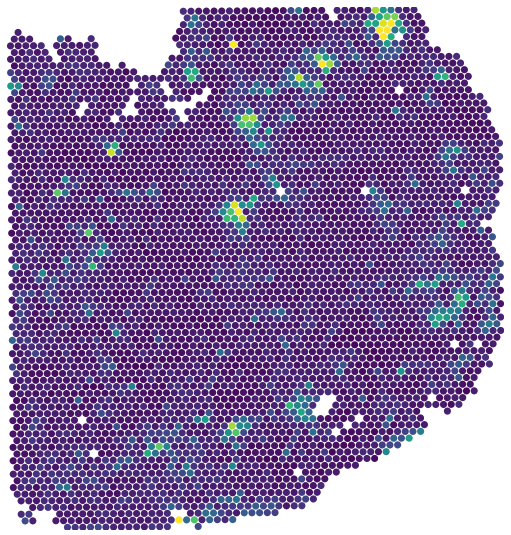
0 0.151

UCell score

GSE285715_T1 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

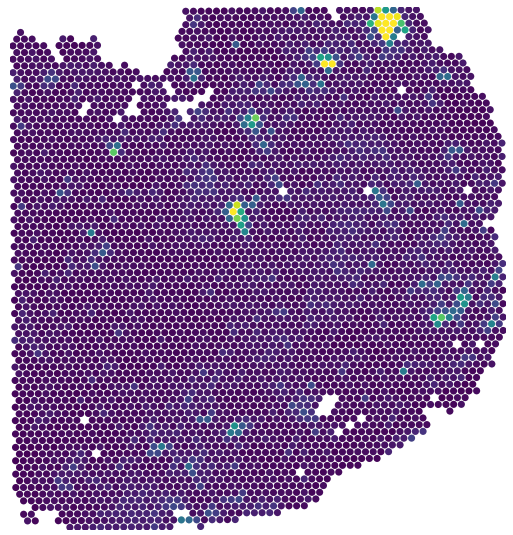
Treg composition



0 0.102

Fraction of all 30

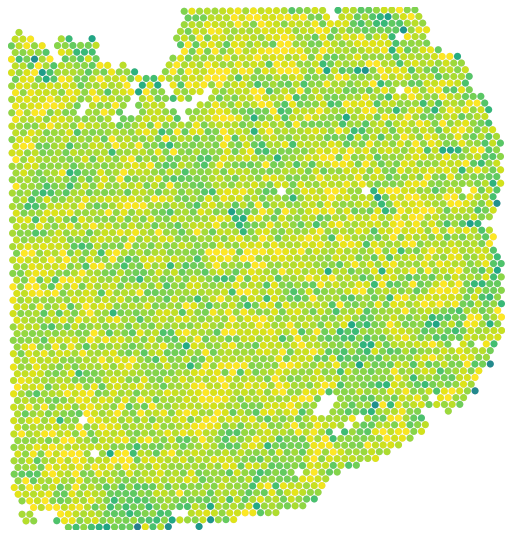
Treg estimated abundance



0 2.85

Posterior mean estimate

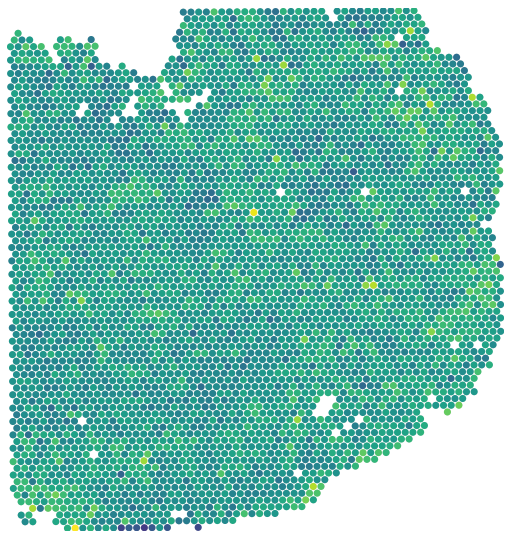
Treg Resting Vs Activated Dn



0 0.0778

UCell score

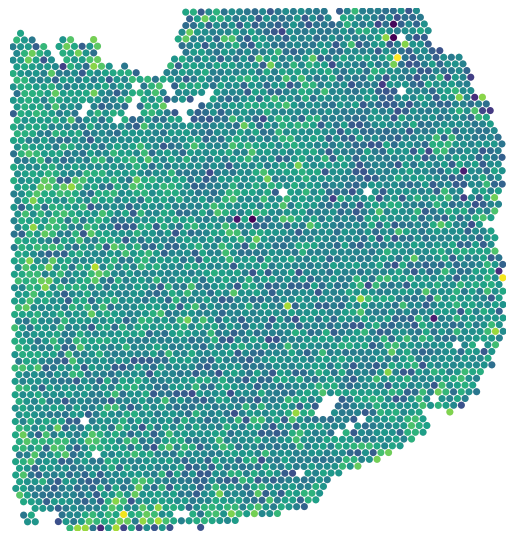
Treg Resting Vs Activated Up



0 0.0609

UCell score

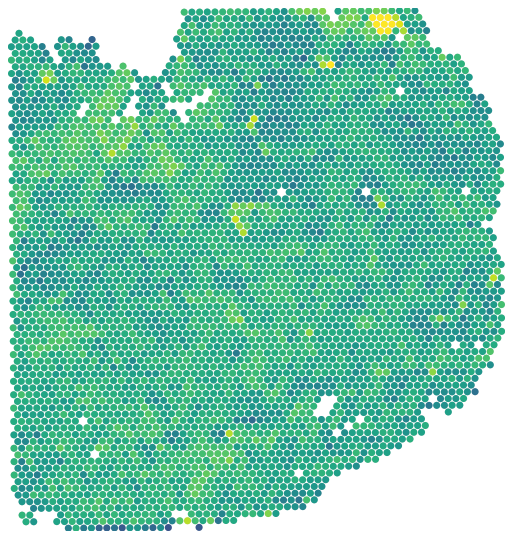
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

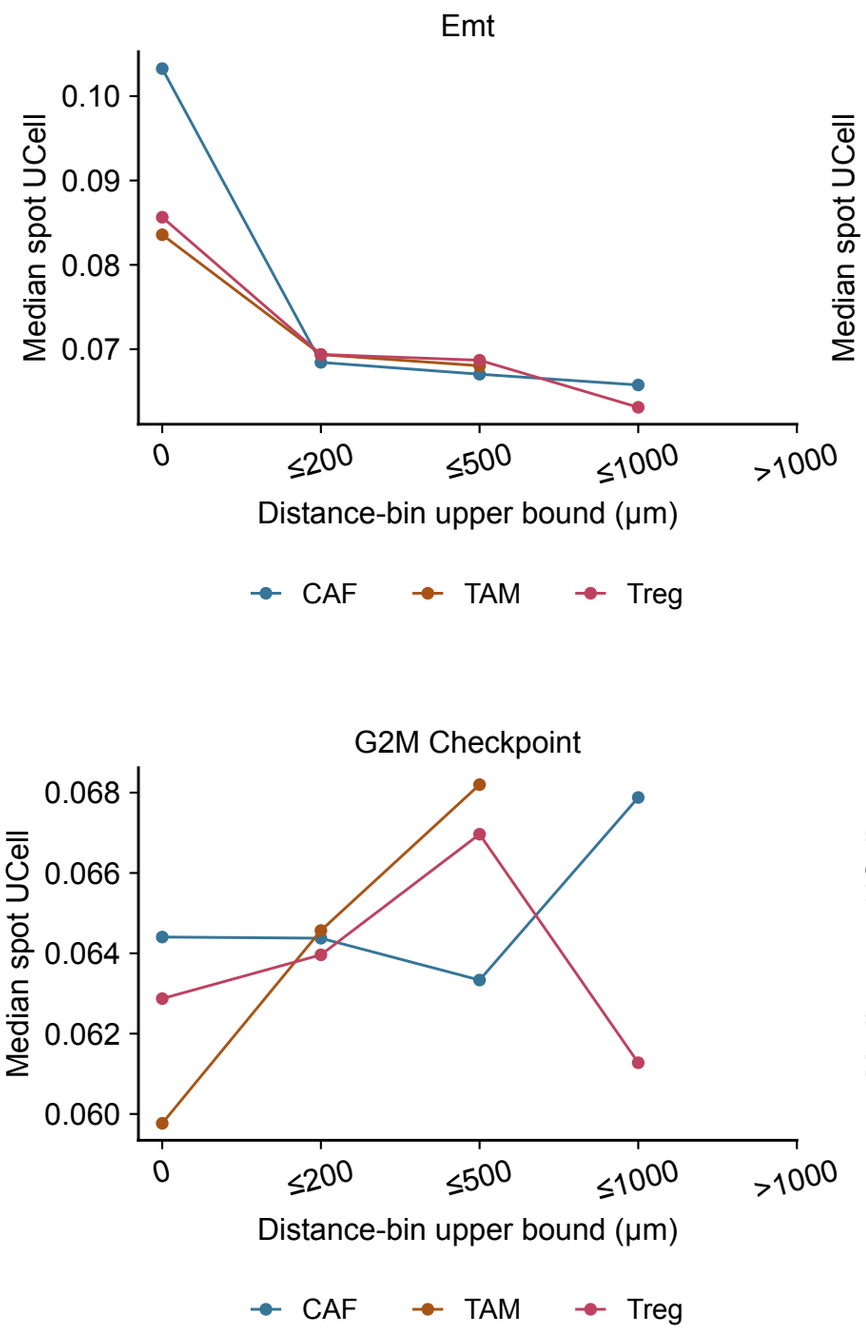


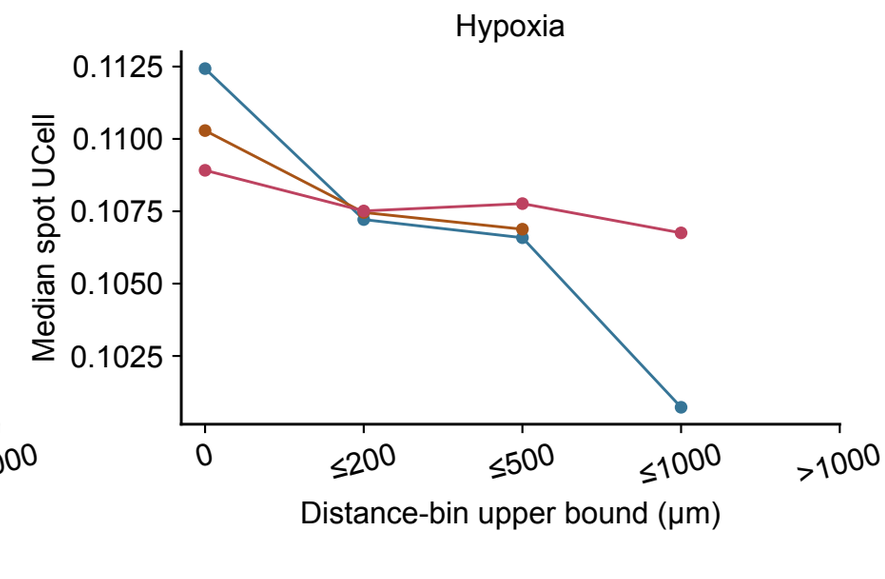
0 0.144

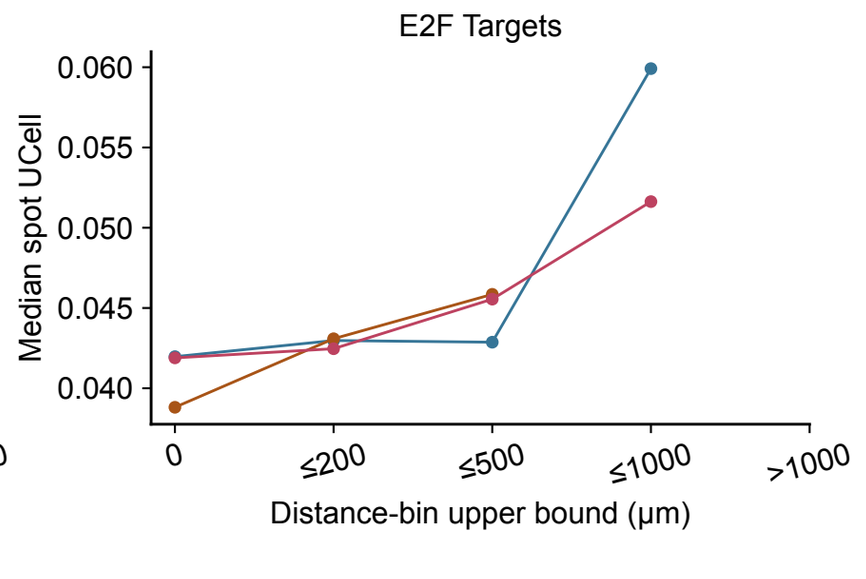
UCell score

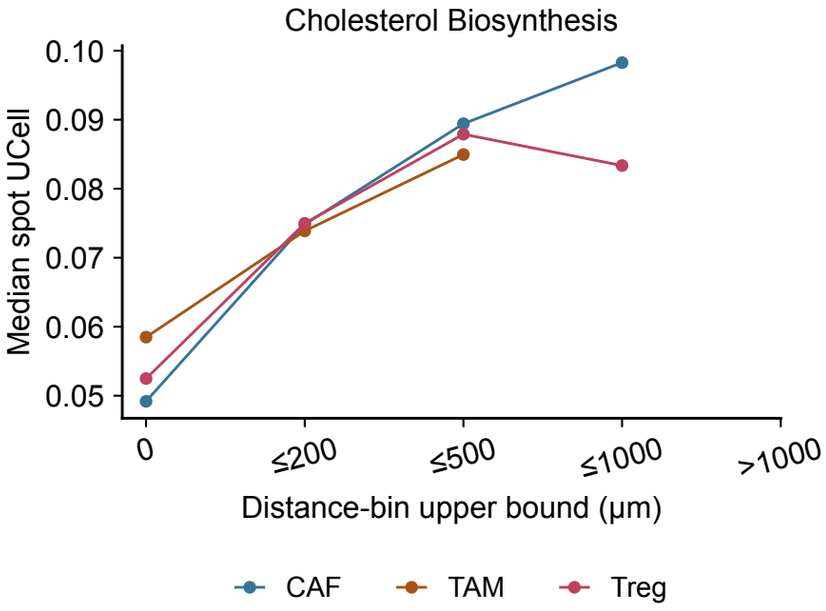
GSE285715_T1 | Programs by target distance

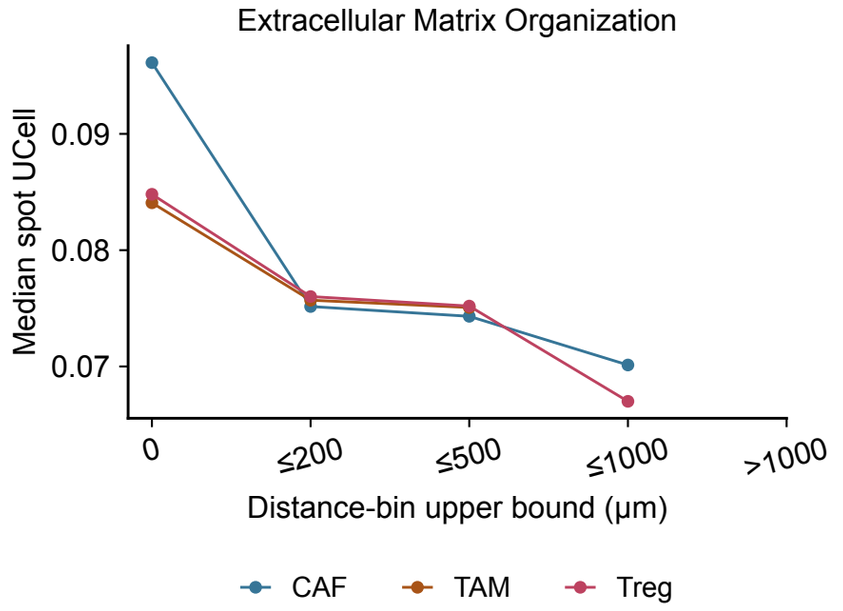
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots





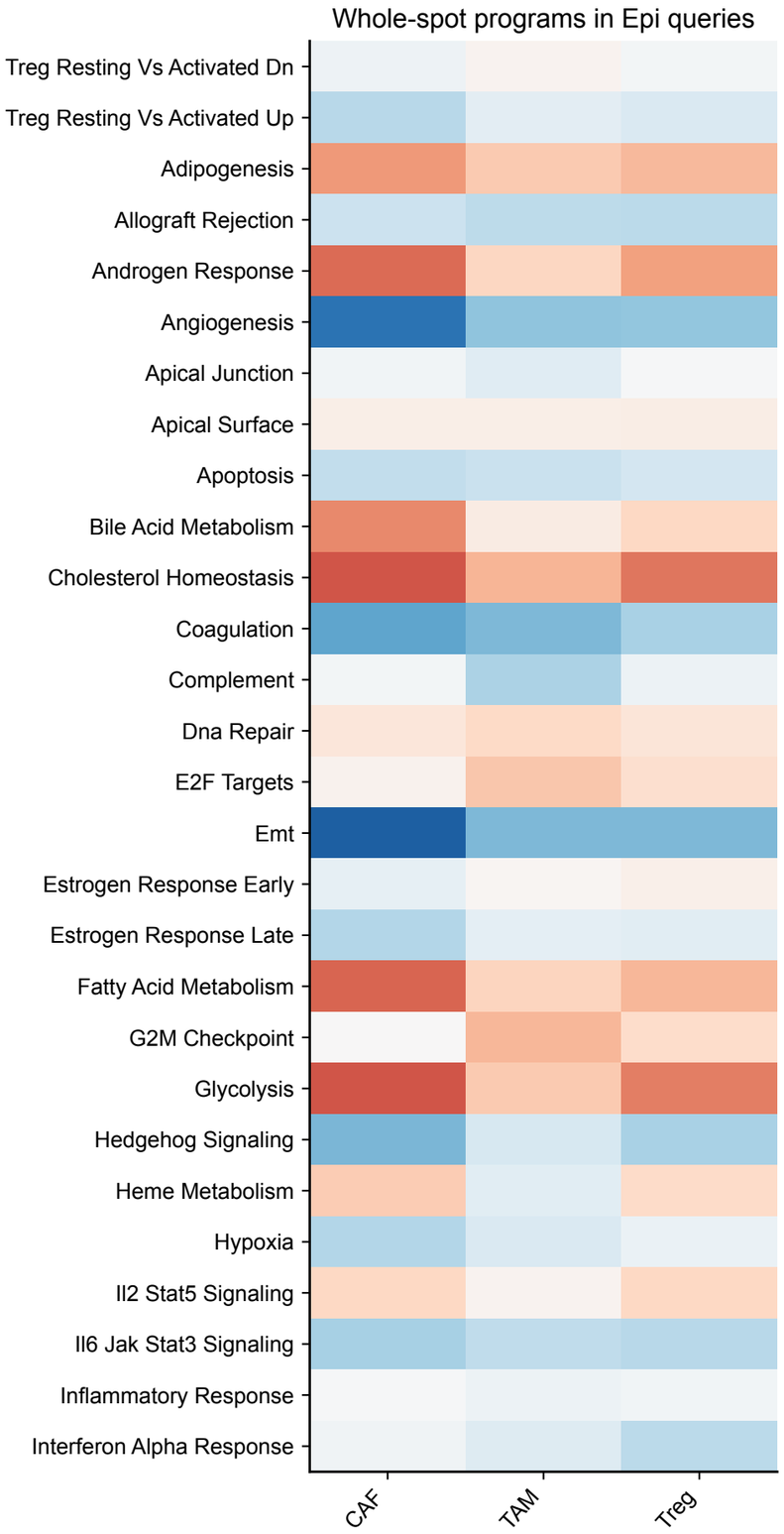


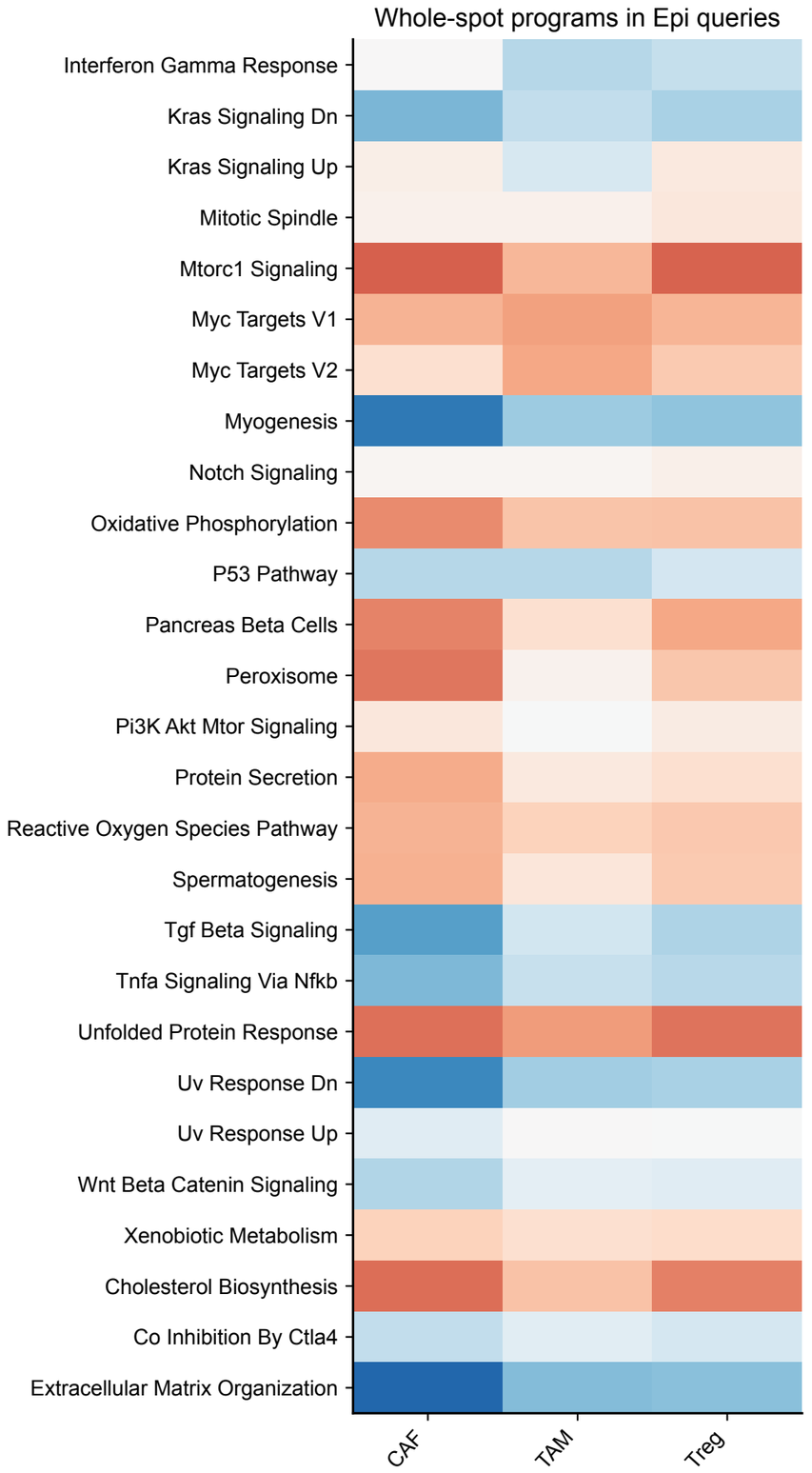




GSE285715_T1 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6





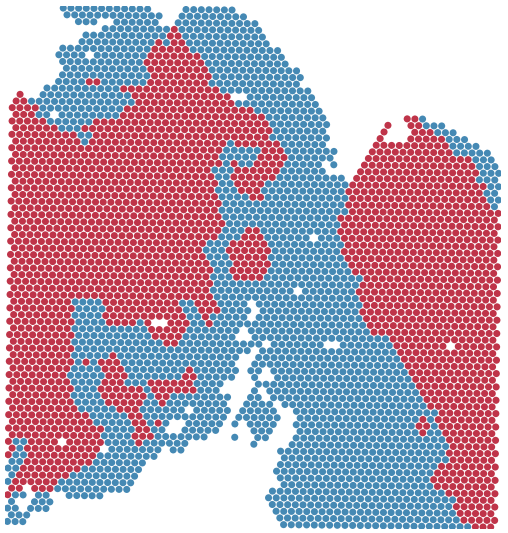
GSE285715_T2 | Tissue context

Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge

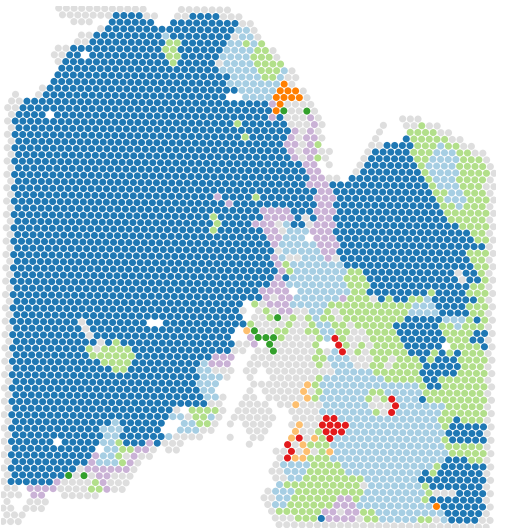
Black line: observed candidate outer edge

Final candidate location



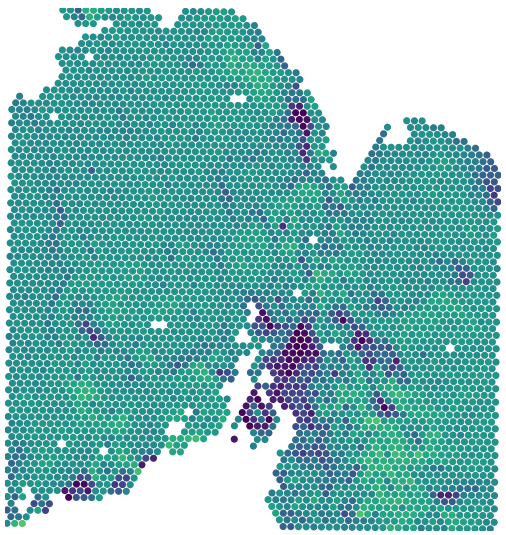
inside outside

Accepted R12 (original 0–8)



0 2 4 7 8
1 3 5

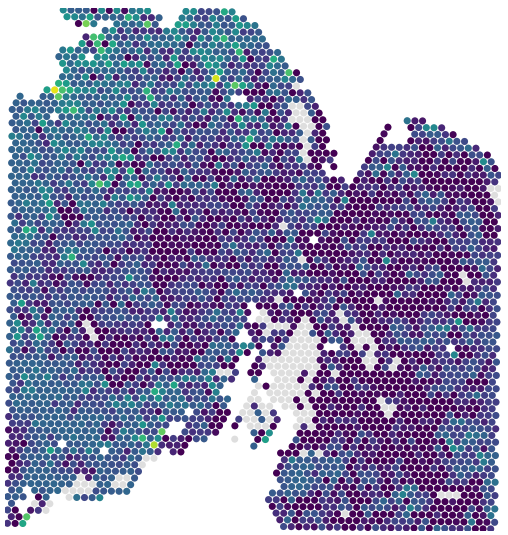
Epithelial support



0 1

Fraction of all 30

MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

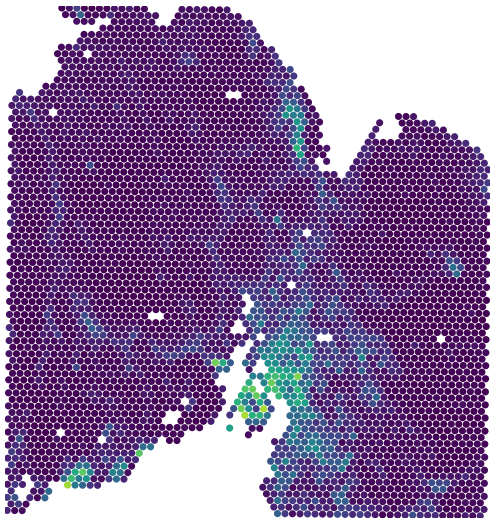
0 0.14

Matched-model AUCell

GSE285715_T2 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

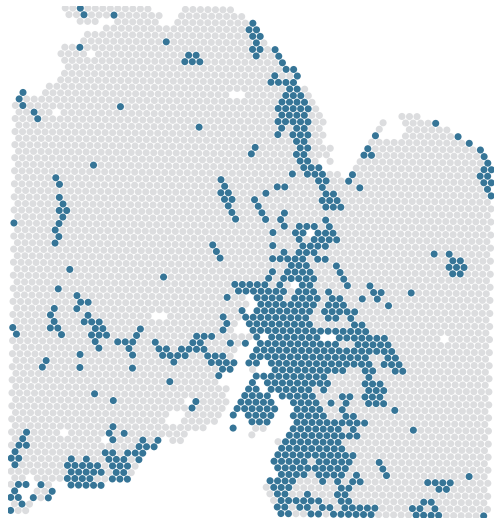
CAF composition



0 0.418

Fraction of all 30

CAF P80 targets



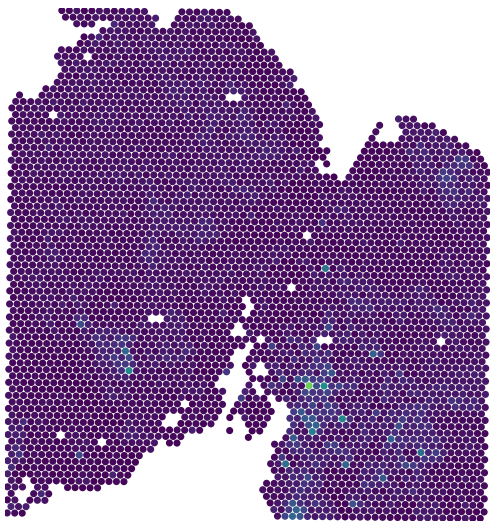
False True

CAF distance

0 3.1e+03

Nearest spot center (μm)

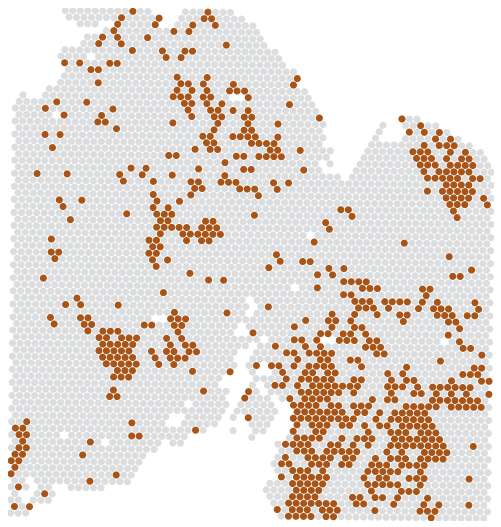
TAM composition



0 0.291

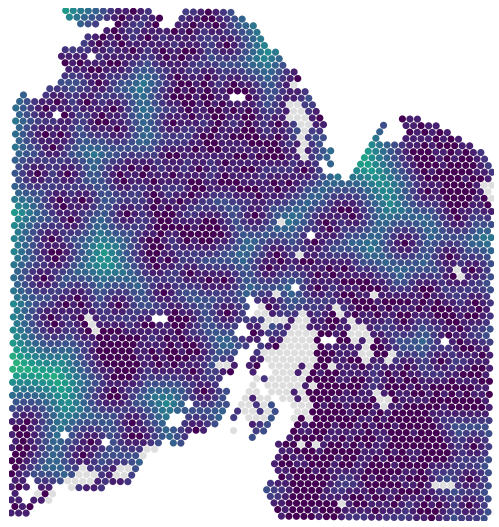
Fraction of all 30

TAM P80 targets



False True

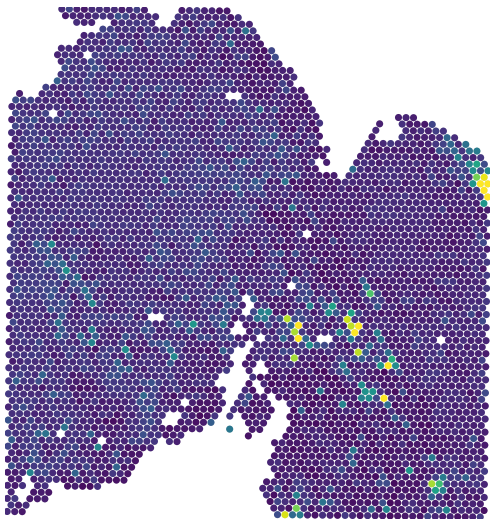
TAM distance



0 1.13e+03

Nearest spot center (μm)

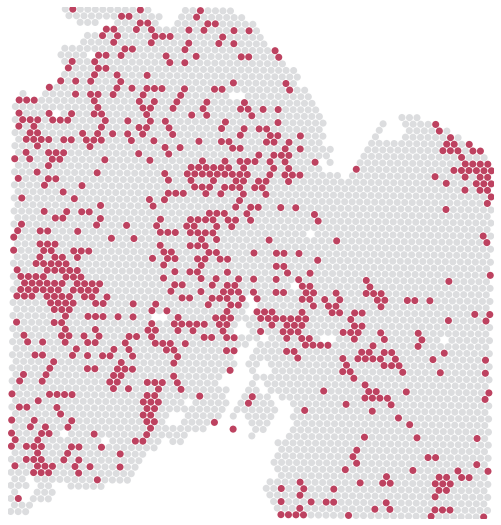
Treg composition



0 0.102

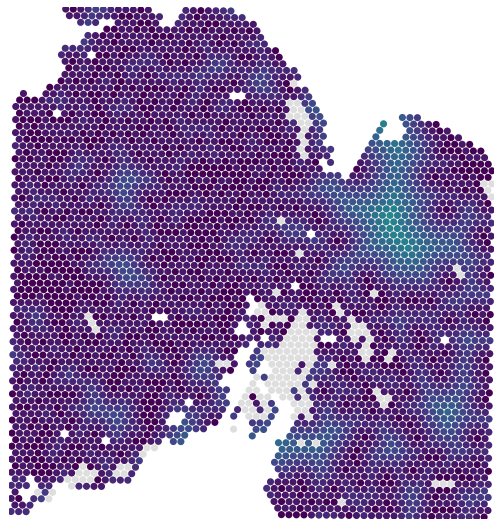
Fraction of all 30

Treg P80 targets



False True

Treg distance

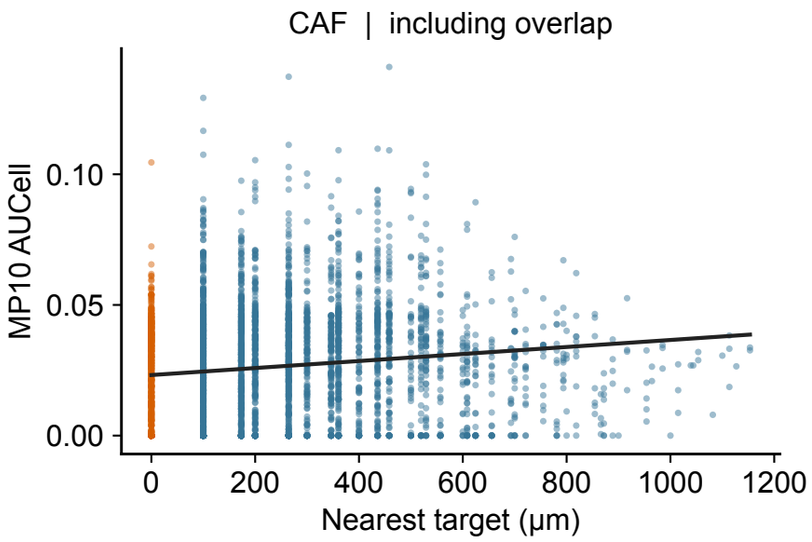


0 1.61e+03

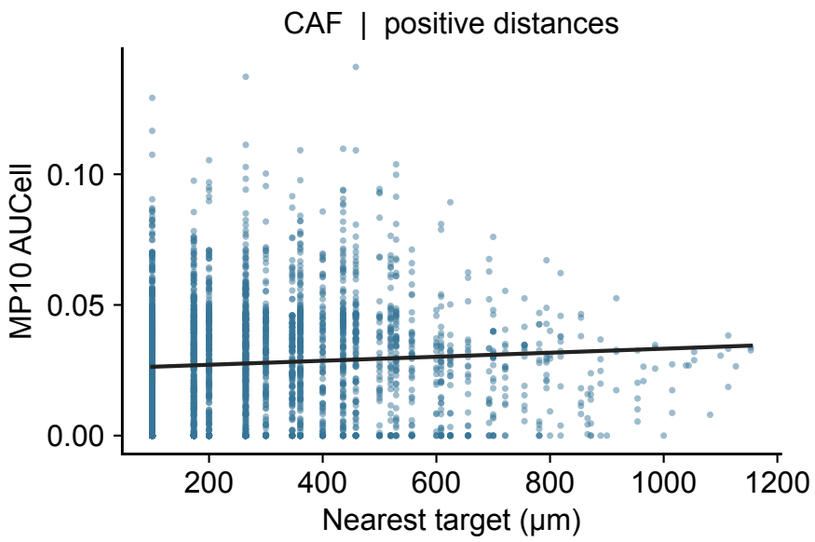
Nearest spot center (μm)

GSE285715_T2 | MP10 versus nearest target

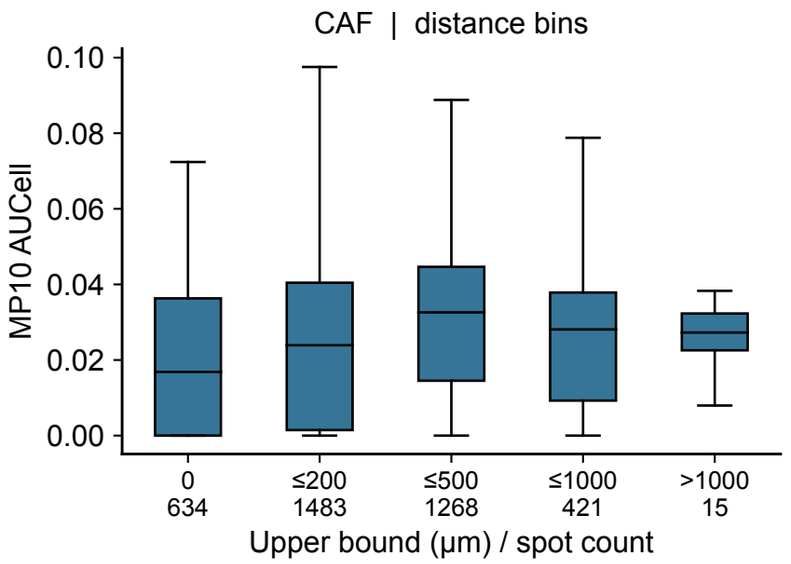
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



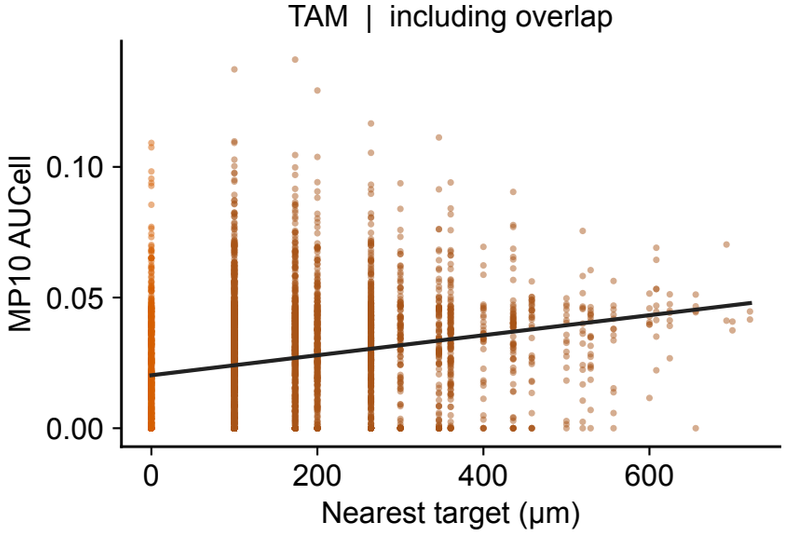
r +0.13 ρ +0.18
n 3821/3821 queries • 798 targets



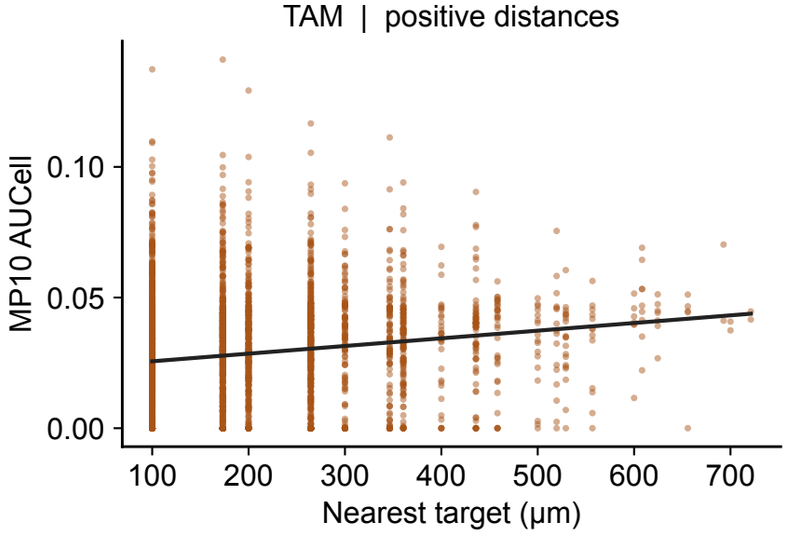
r +0.07 ρ +0.13
n 3187/3821 queries • 798 targets



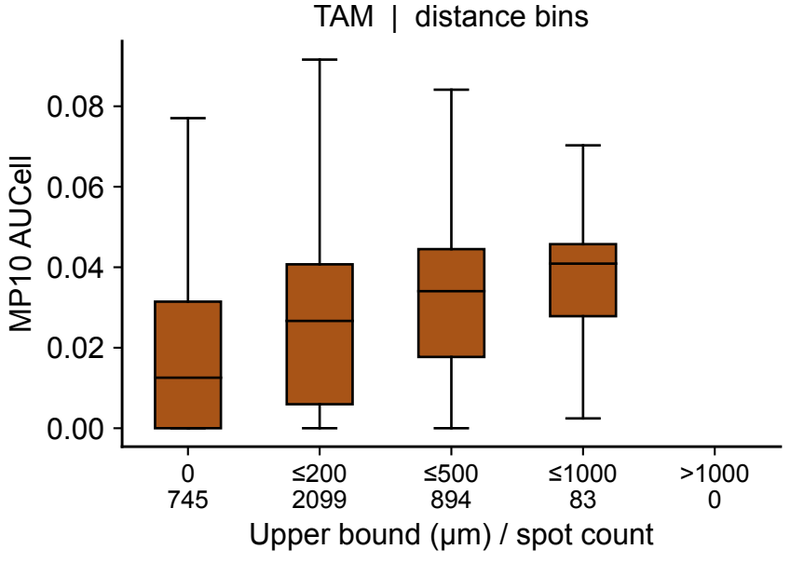
Orange = zero distance
Line = within-section linear fit



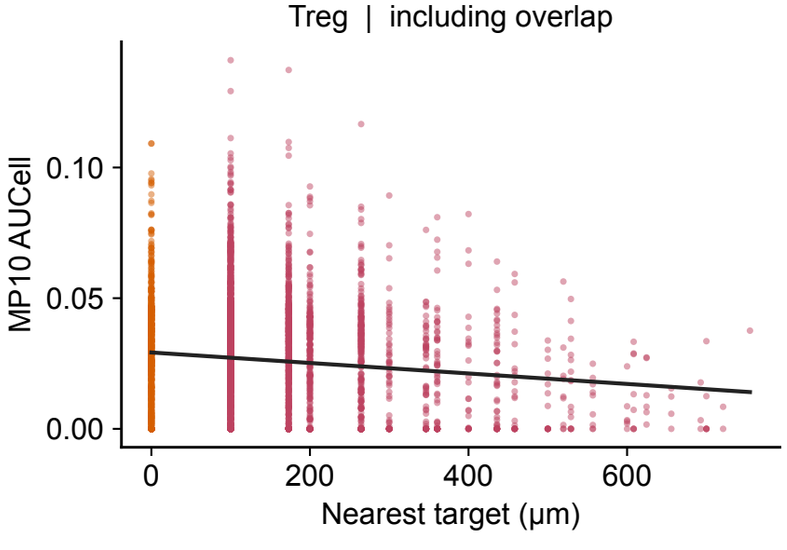
r +0.23 ρ +0.27
n 3821/3821 queries • 798 targets



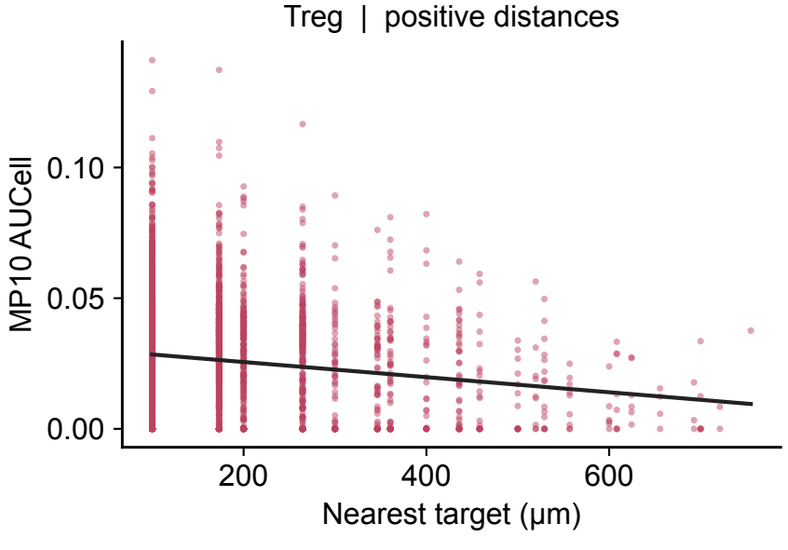
r +0.16 ρ +0.19
n 3076/3821 queries • 798 targets



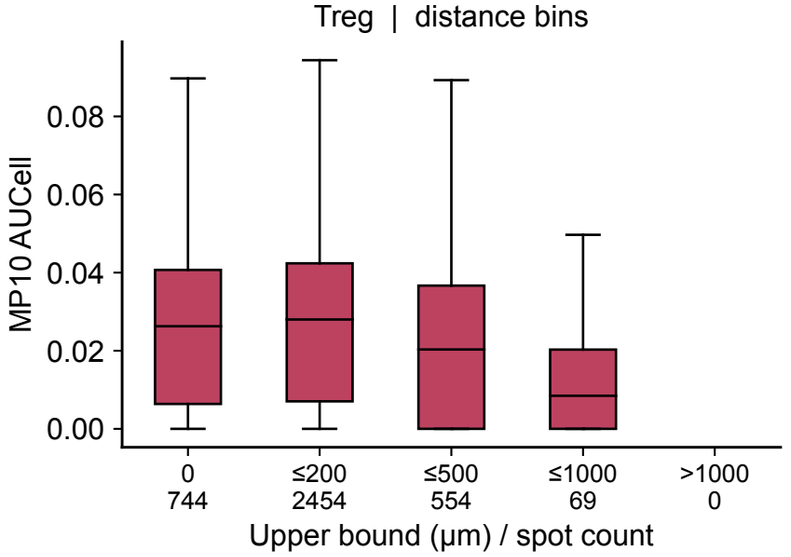
Orange = zero distance
Line = within-section linear fit



r -0.11 ρ -0.08
n 3821/3821 queries • 798 targets



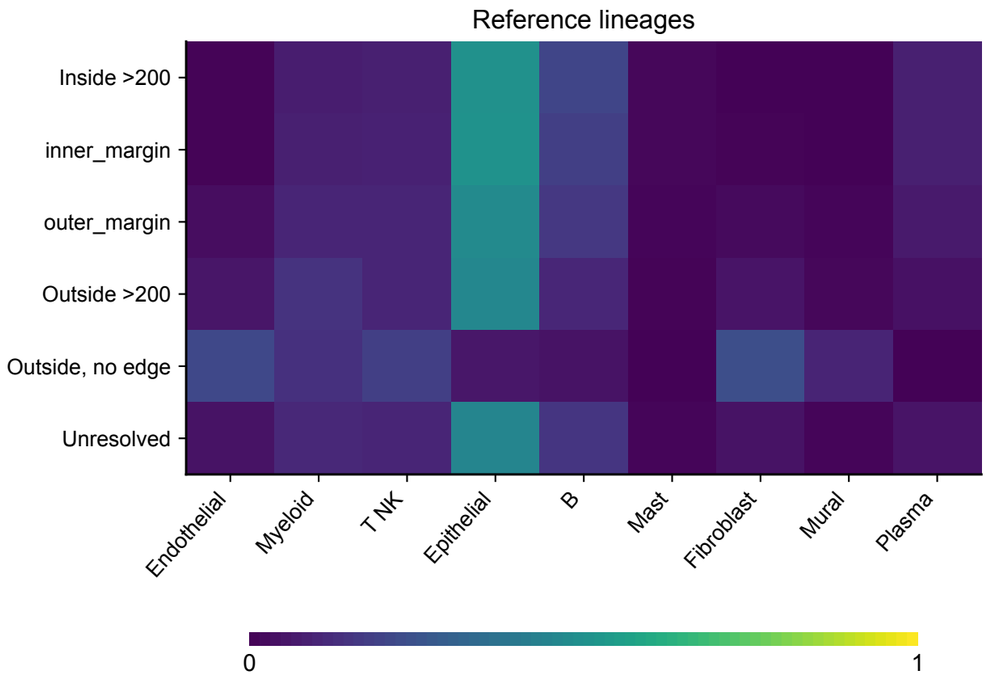
r -0.14 ρ -0.11
n 3077/3821 queries • 798 targets

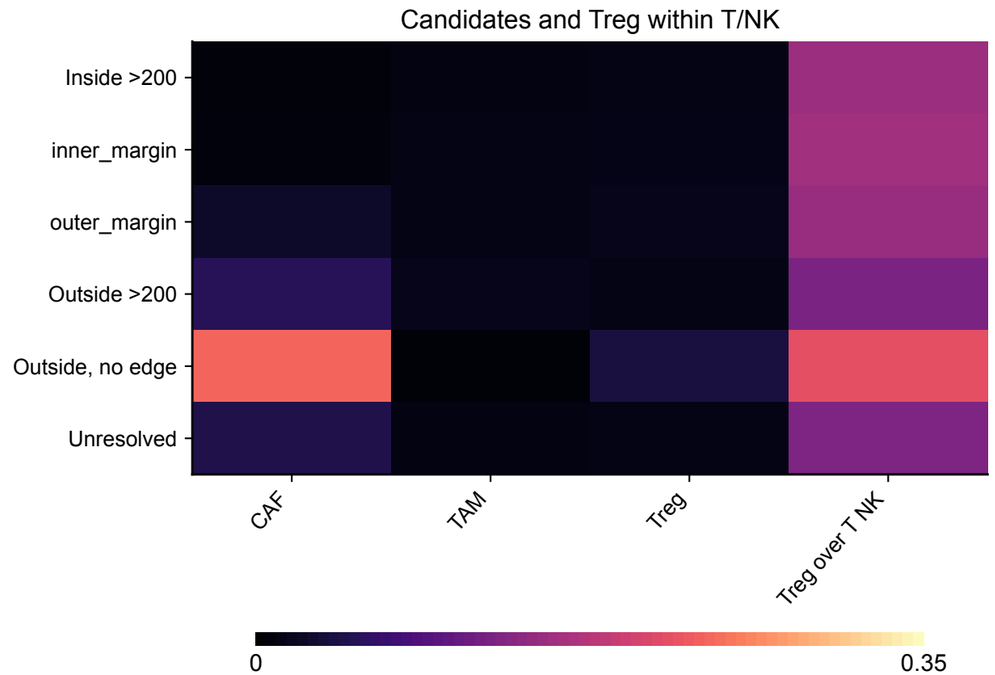


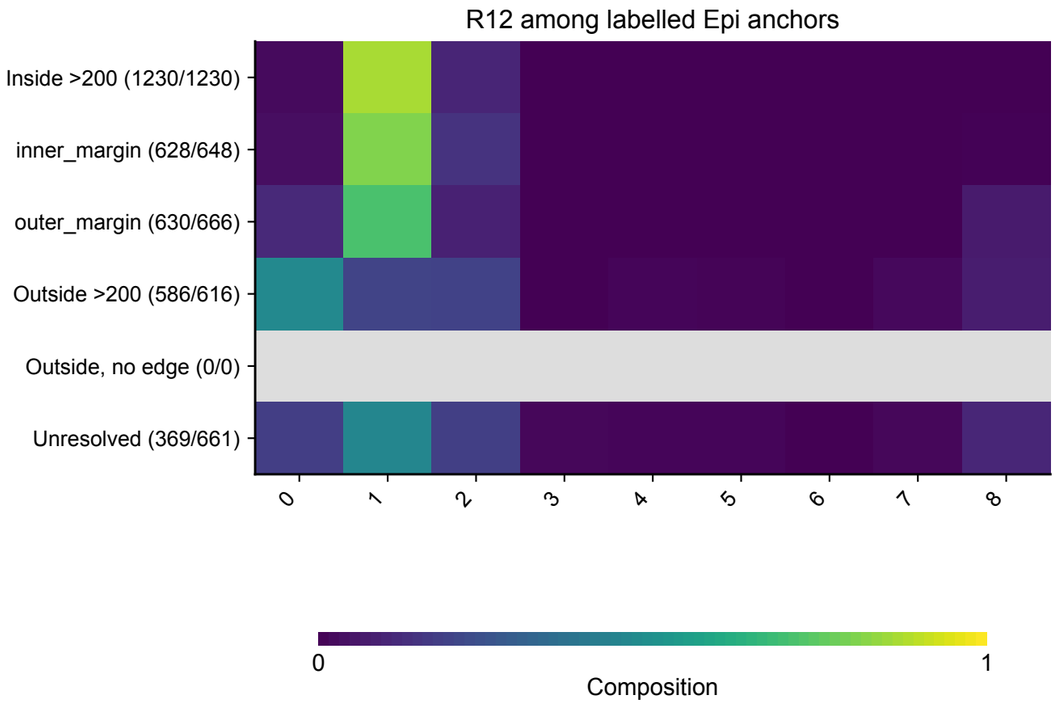
Orange = zero distance
Line = within-section linear fit

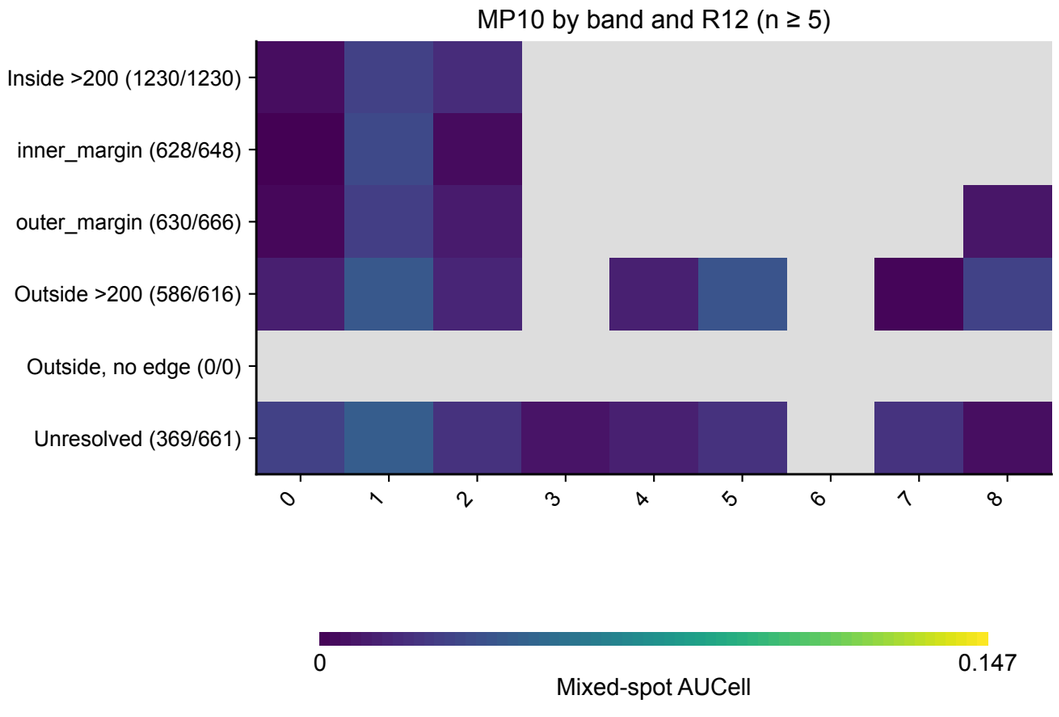
GSE285715_T2 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





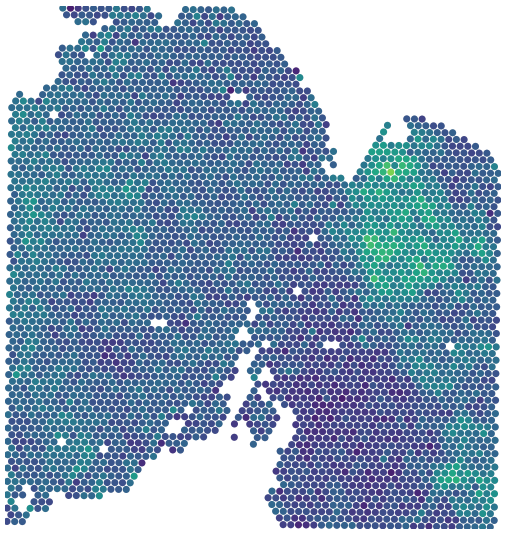




GSE285715_T2 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

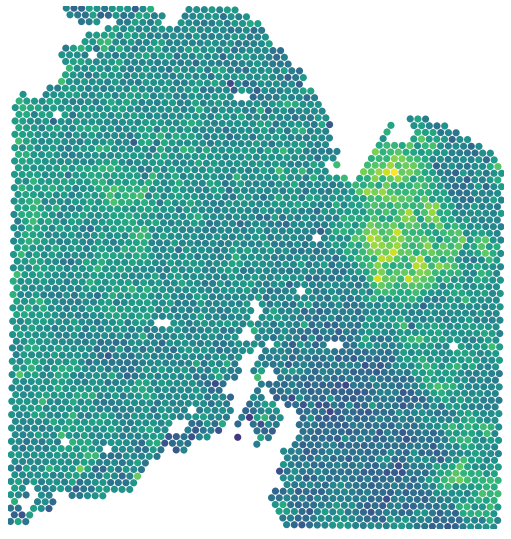
E2F Targets



0 0.152

UCell score

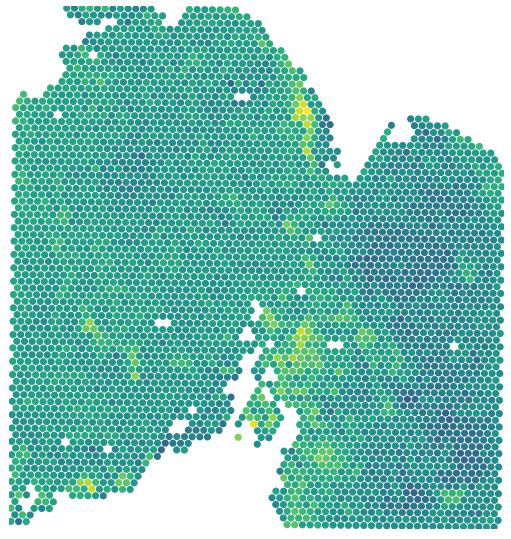
G2M Checkpoint



0 0.134

UCell score

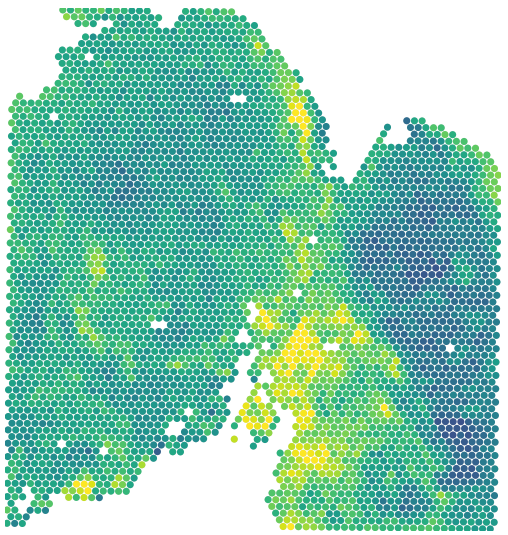
Hypoxia



0 0.211

UCell score

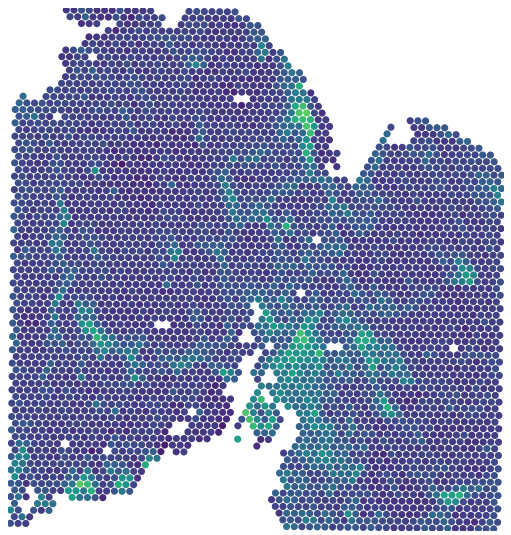
Tnfa Signaling Via Nfkb



0 0.275

UCell score

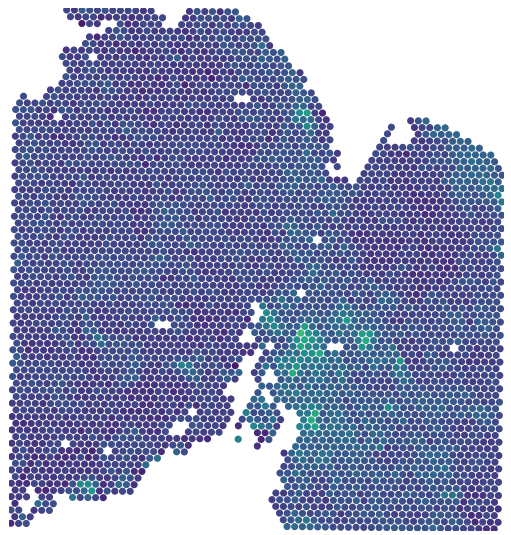
Emt



0 0.386

UCell score

Interferon Gamma Response



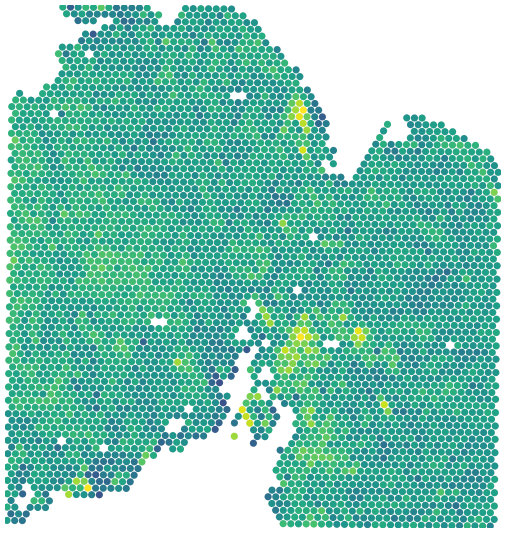
0 0.275

UCell score

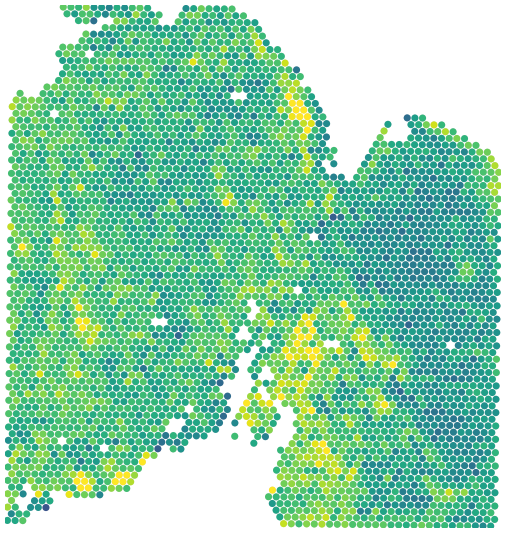
GSE285715_T2 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Il2 Stat5 Signaling

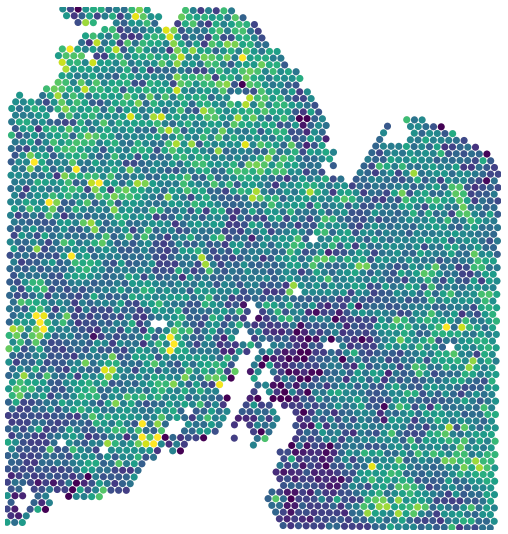


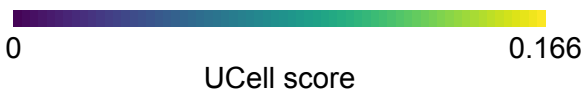
Tgf Beta Signaling



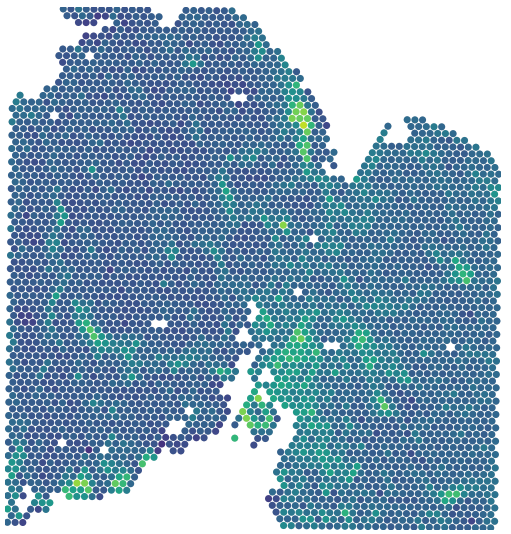
Cholesterol Homeostasis

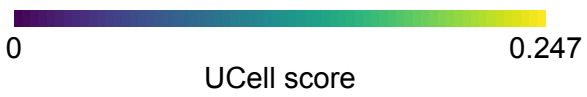
Cholesterol Biosynthesis



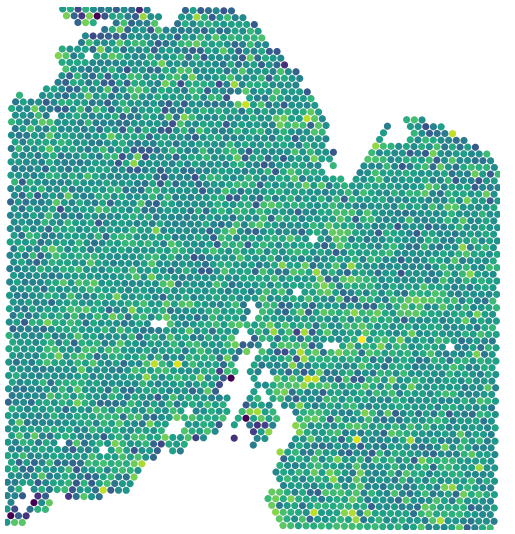


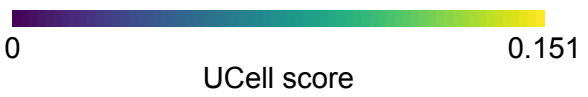
Extracellular Matrix Organization





Co Inhibition By Ctla4

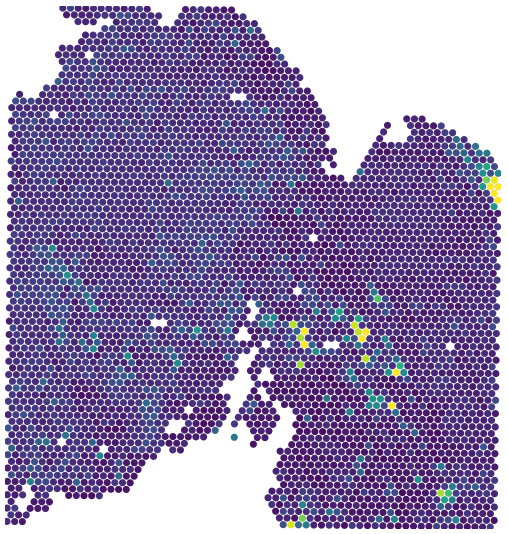




GSE285715_T2 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Treg composition



0 0.102

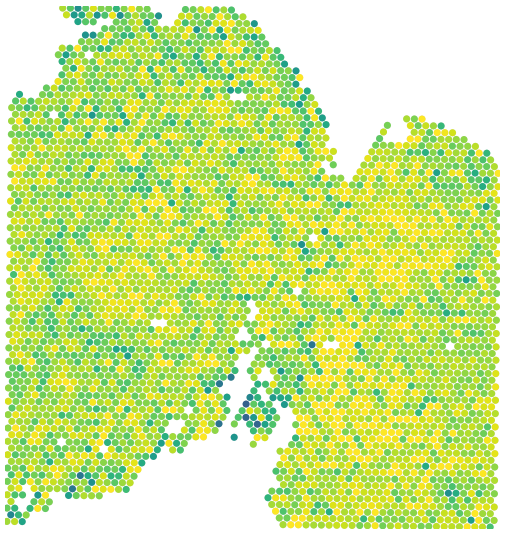
Fraction of all 30

Treg estimated abundance

0 2.85

Posterior mean estimate

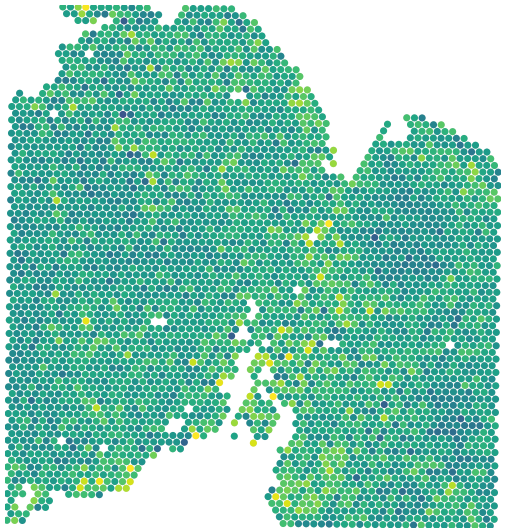
Treg Resting Vs Activated Dn



0 0.0778

UCell score

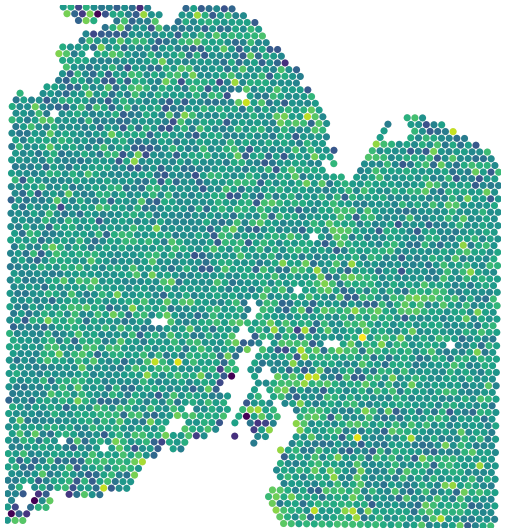
Treg Resting Vs Activated Up



0 0.0609

UCell score

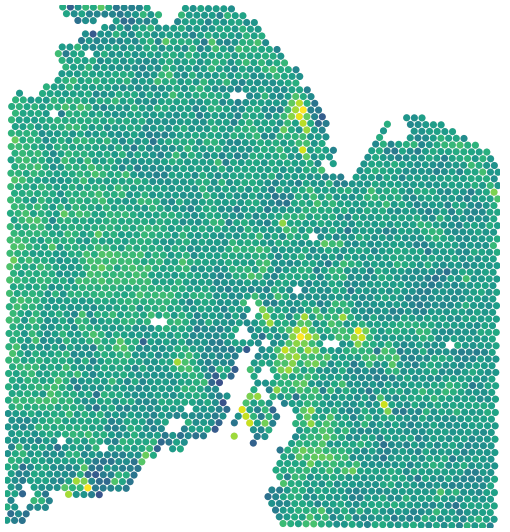
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

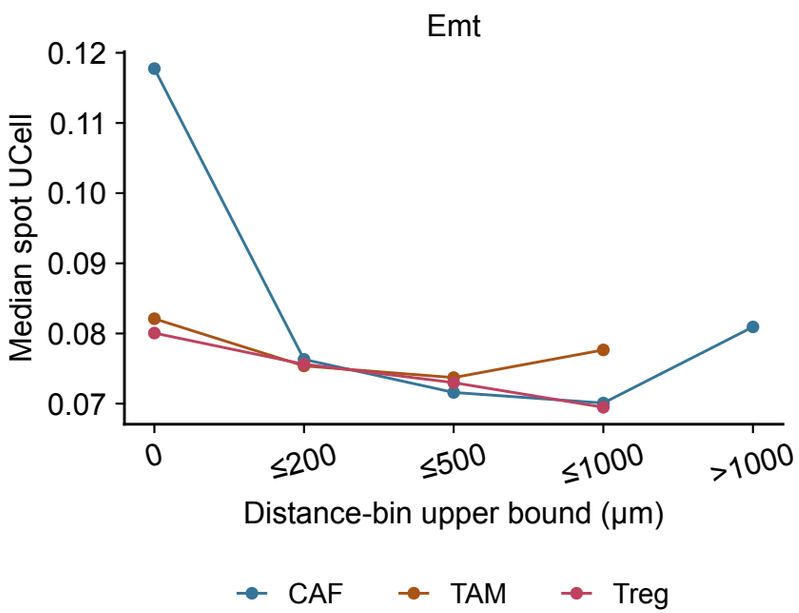


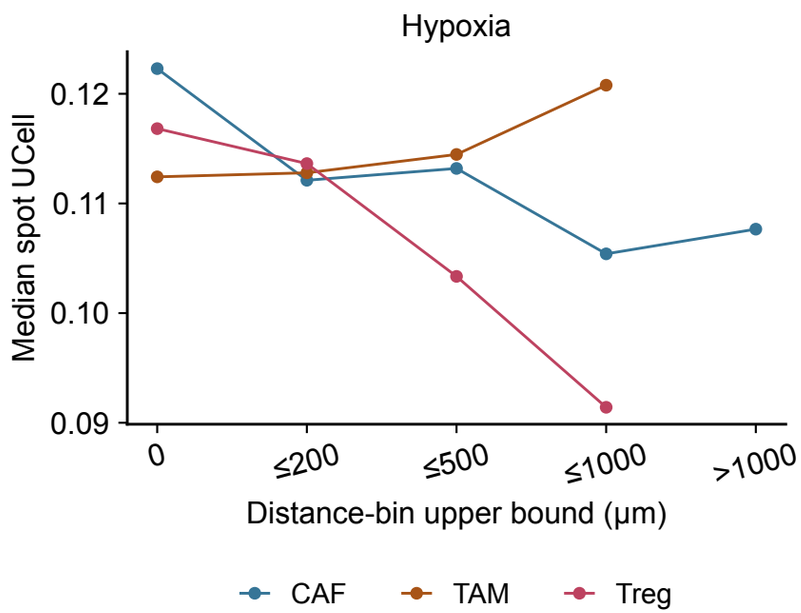
0 0.144

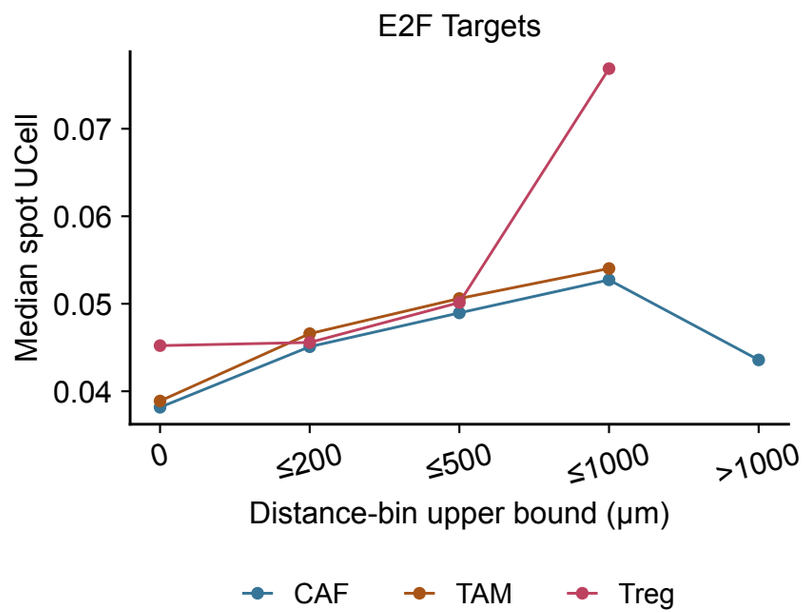
UCell score

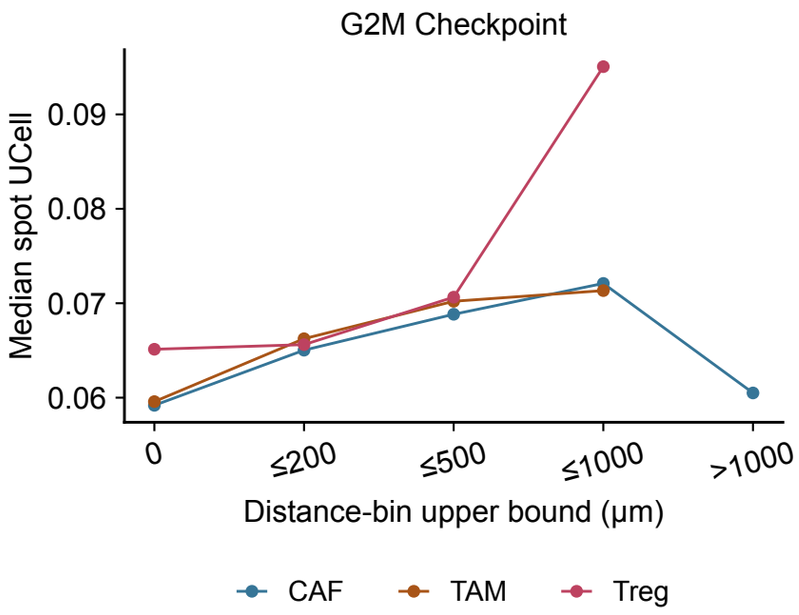
GSE285715_T2 | Programs by target distance

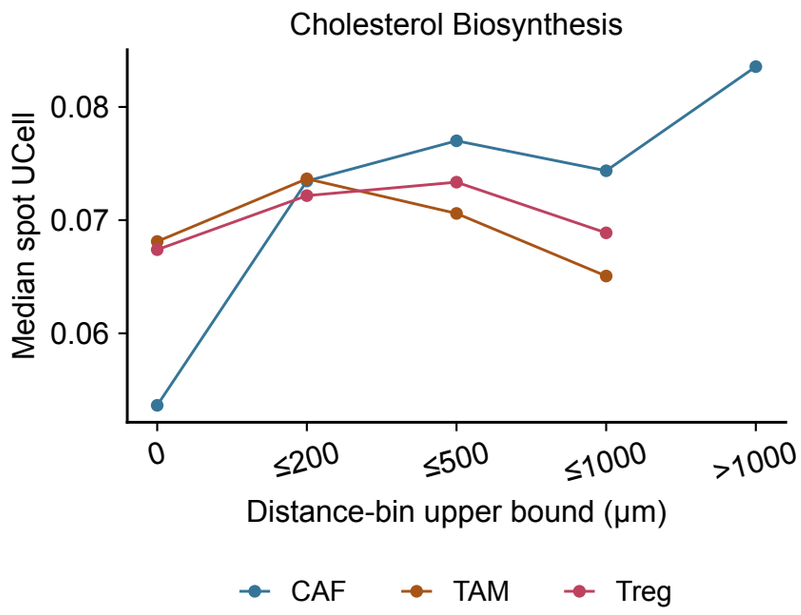
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

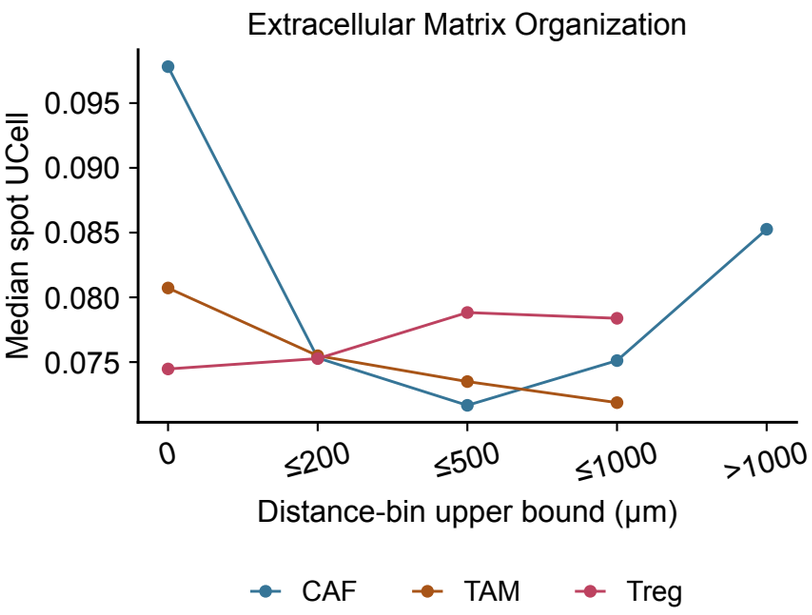






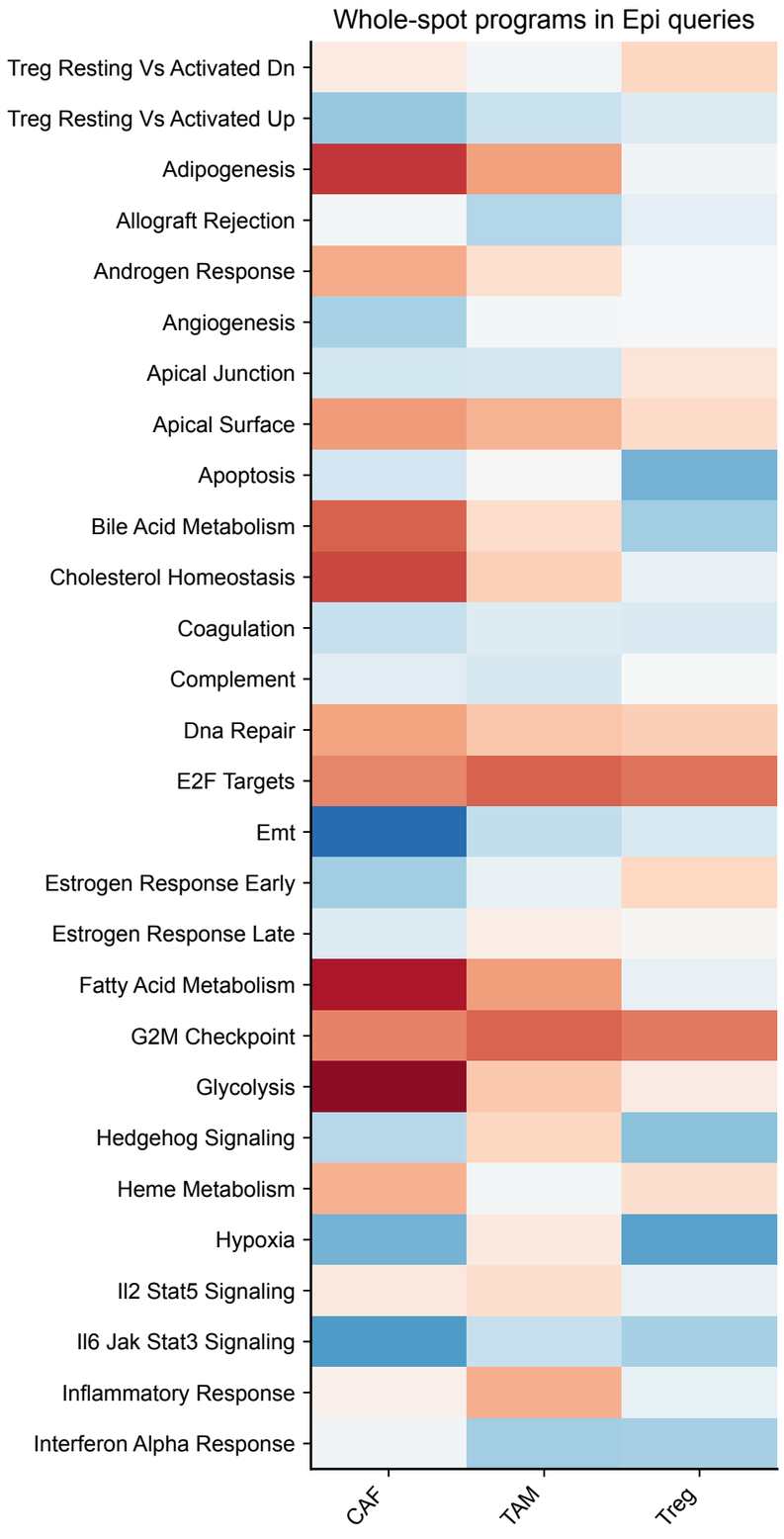


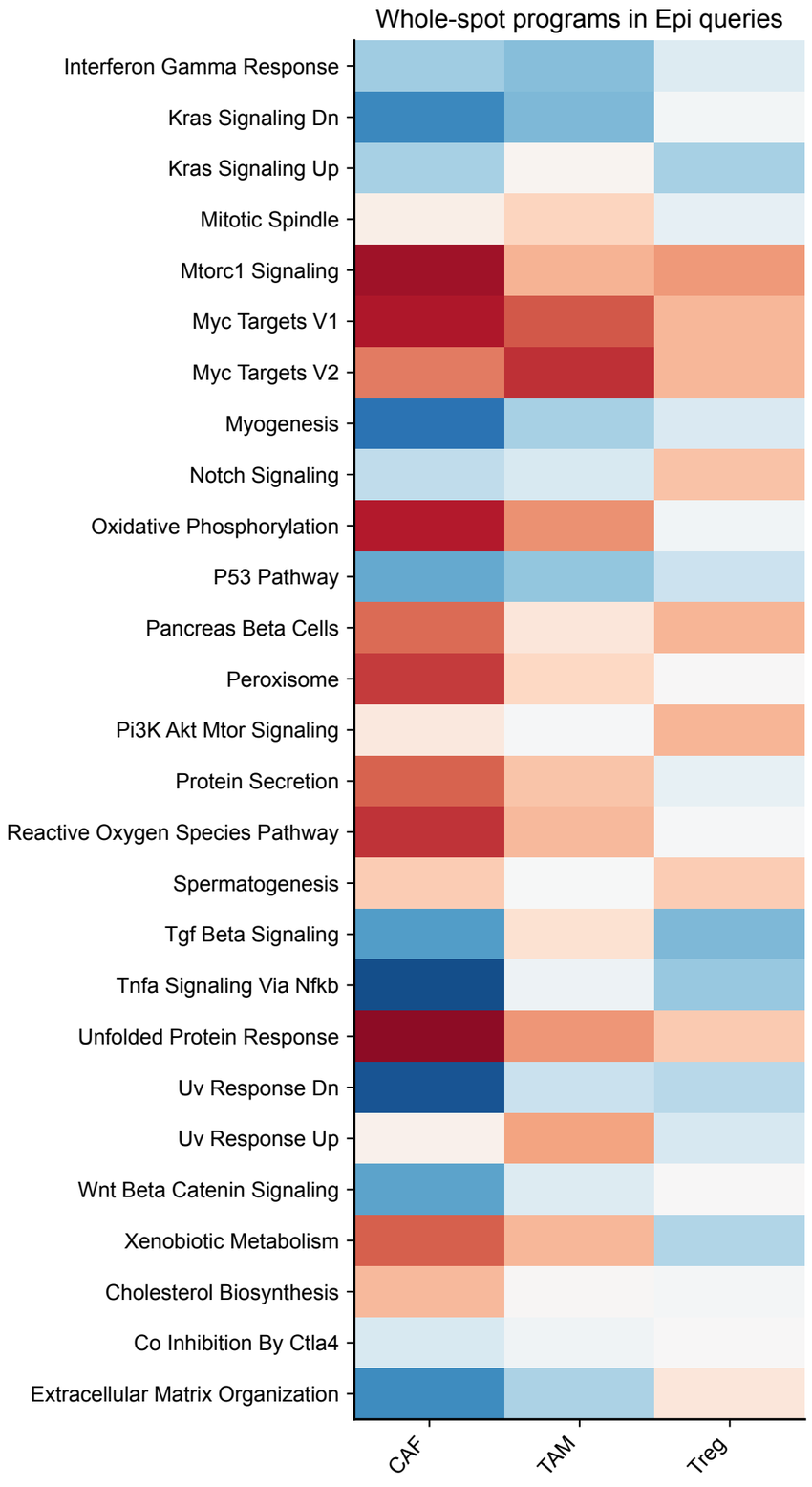




GSE285715_T2 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

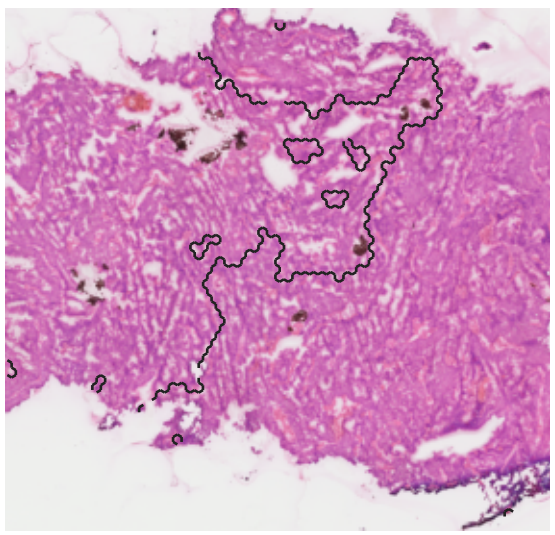




GSE285715_T3 | Tissue context

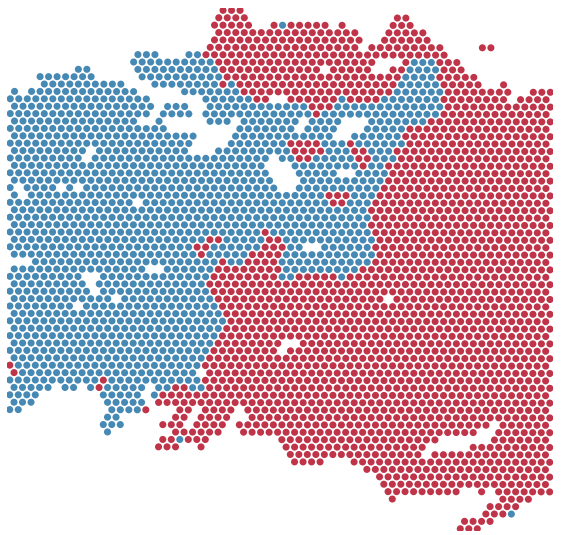
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



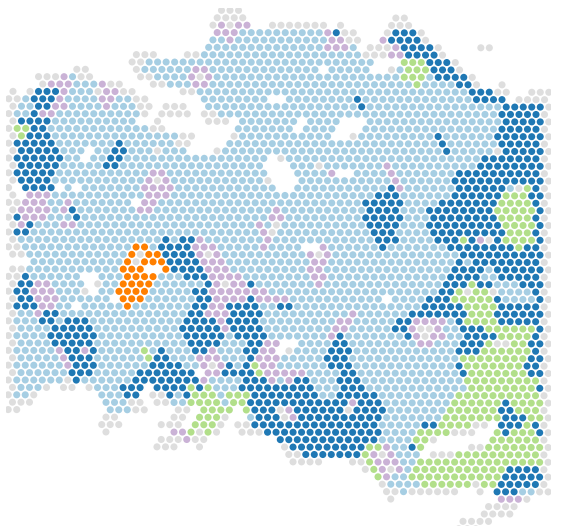
Black line: observed candidate outer edge

Final candidate location



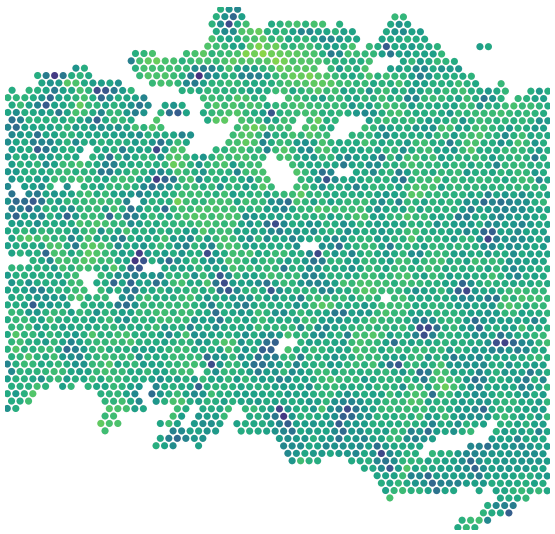
inside outside

Accepted R12 (original 0–8)



0 1 2 5 8

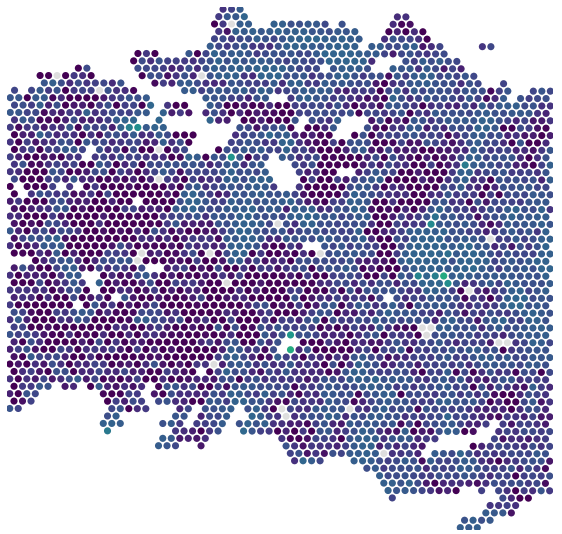
Epithelial support



0 1

Fraction of all 30

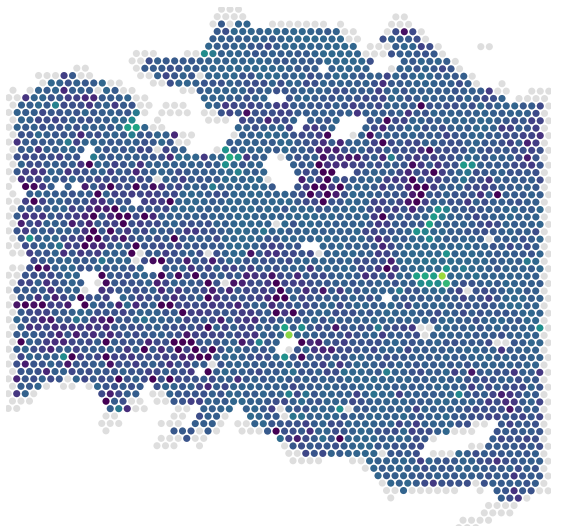
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

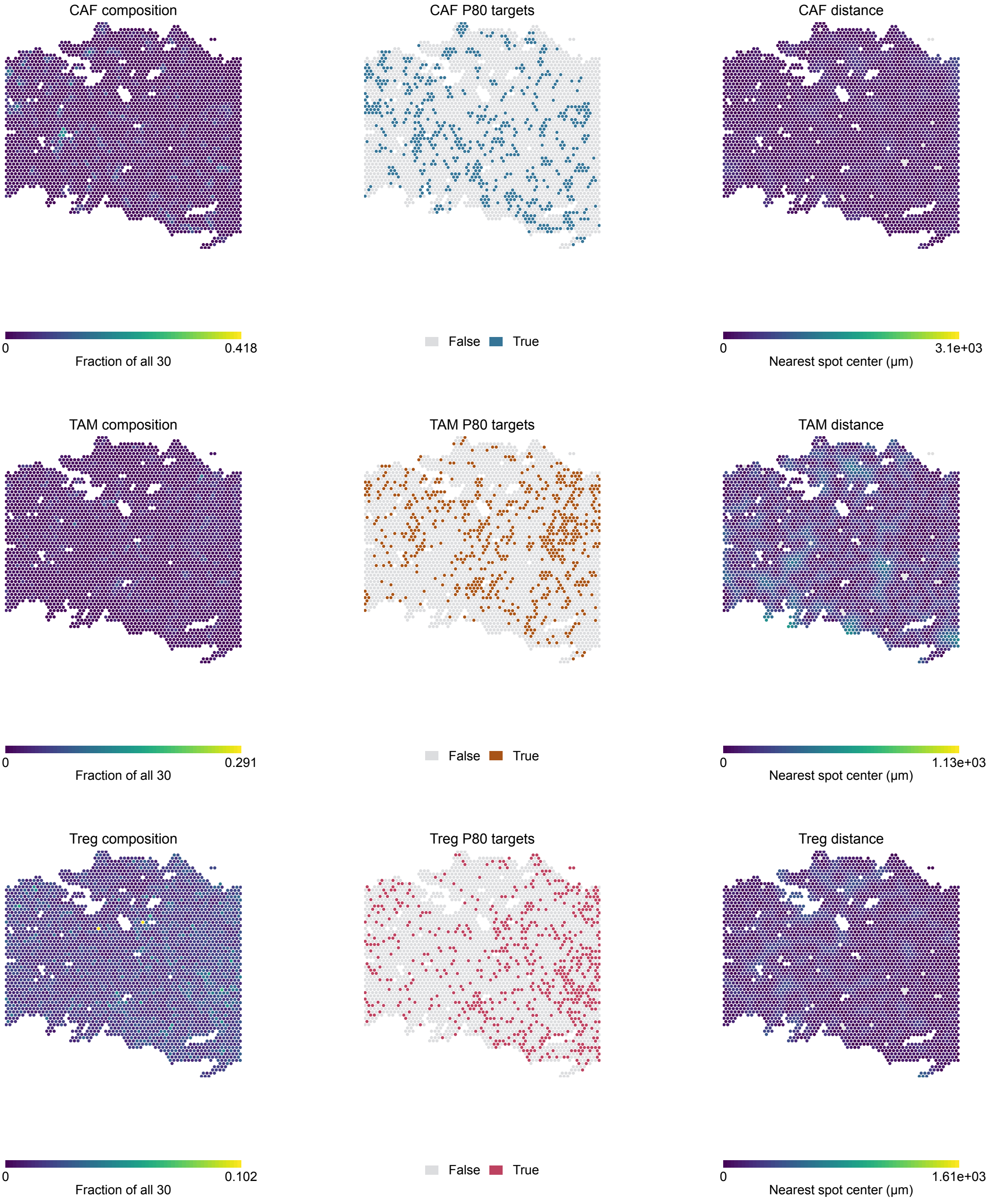


0 0.14

Matched-model AUCell

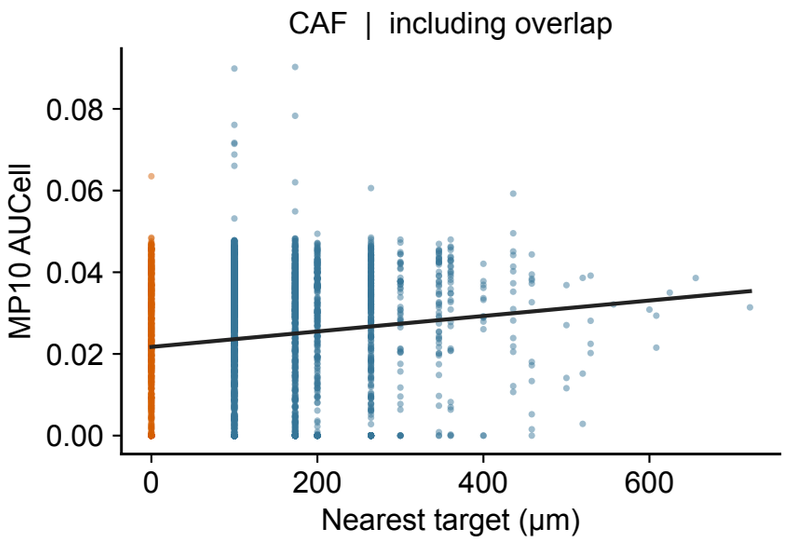
GSE285715_T3 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

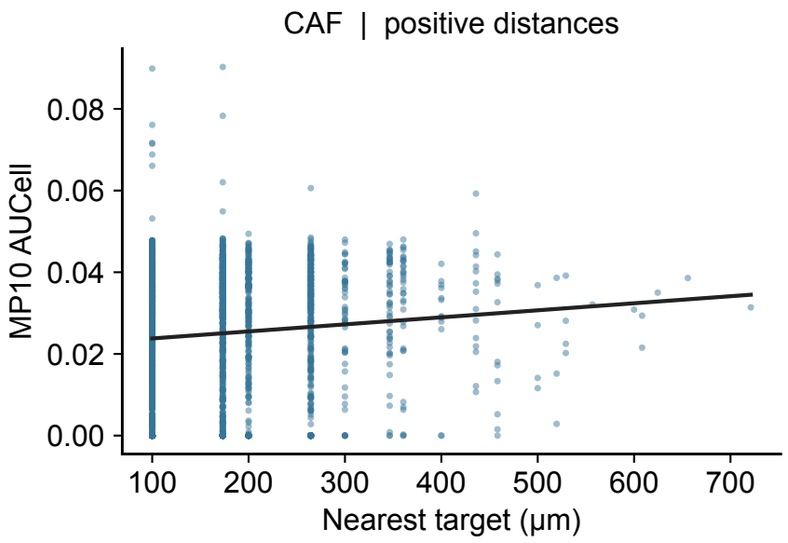


GSE285715_T3 | MP10 versus nearest target

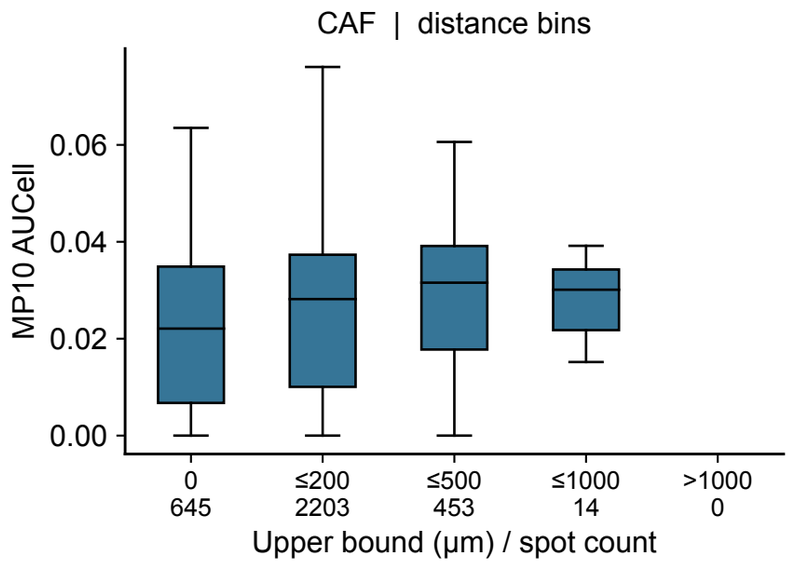
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



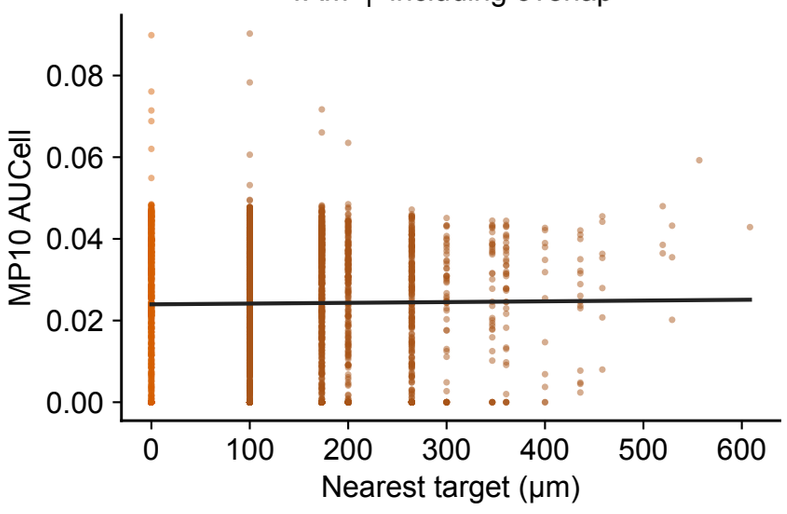
r +0.12 ρ +0.12
n 3315/3317 queries • 669 targets



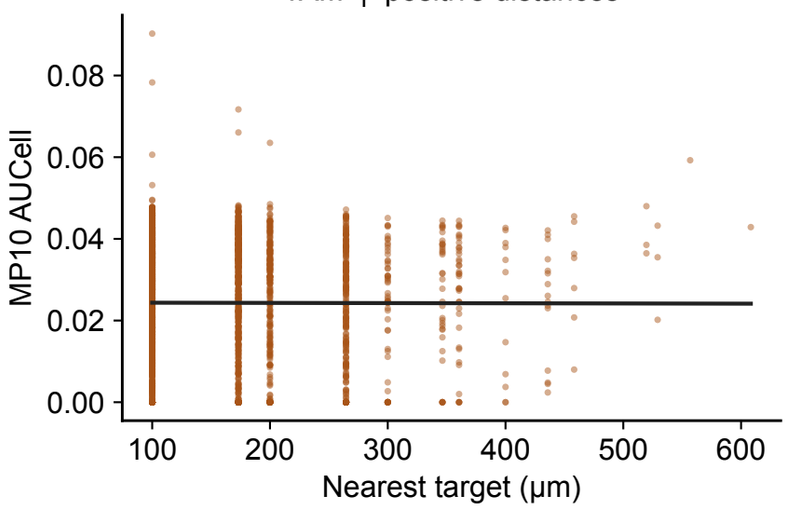
r +0.09 ρ +0.09
n 2670/3317 queries • 669 targets



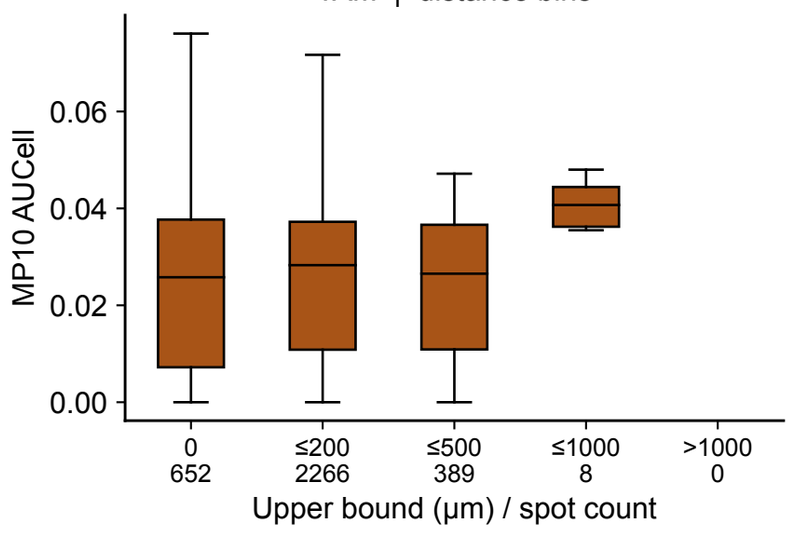
Orange = zero distance
Line = within-section linear fit



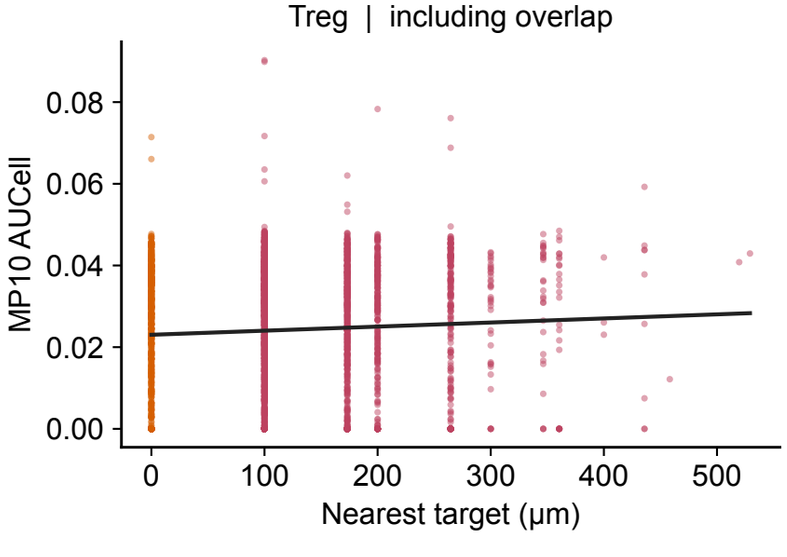
r +0.01 ρ -0.00
n 3315/3317 queries • 669 targets



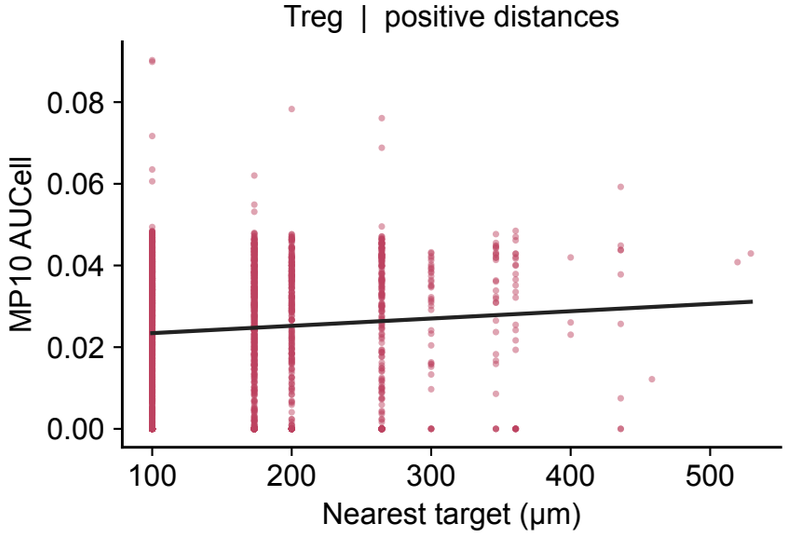
r -0.00 ρ -0.02
n 2663/3317 queries • 669 targets



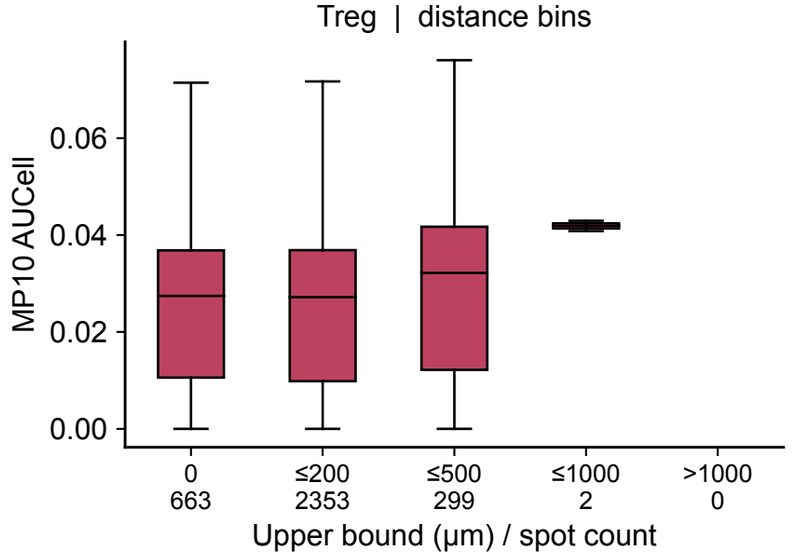
Orange = zero distance
Line = within-section linear fit



r +0.05 ρ +0.04
n 3317/3317 queries • 669 targets



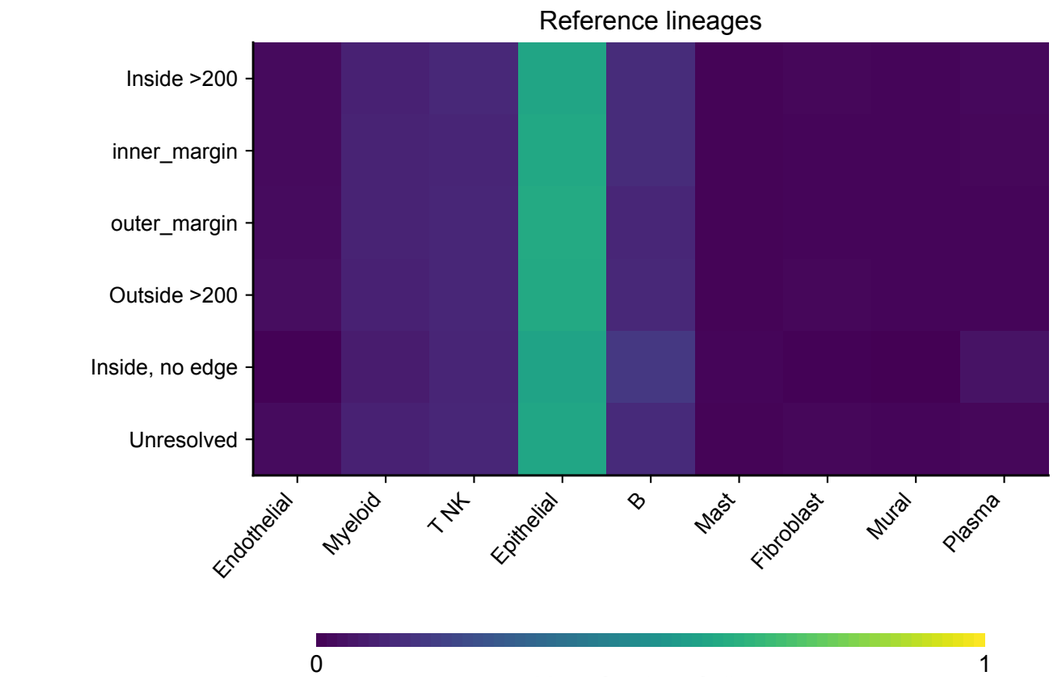
r +0.08 ρ +0.05
n 2654/3317 queries • 669 targets

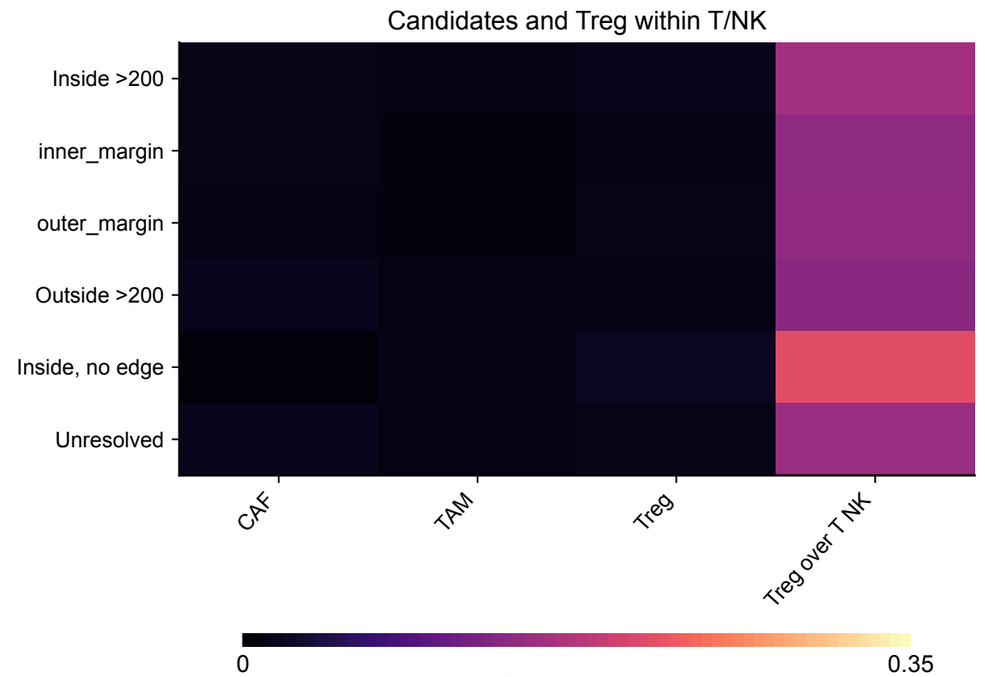


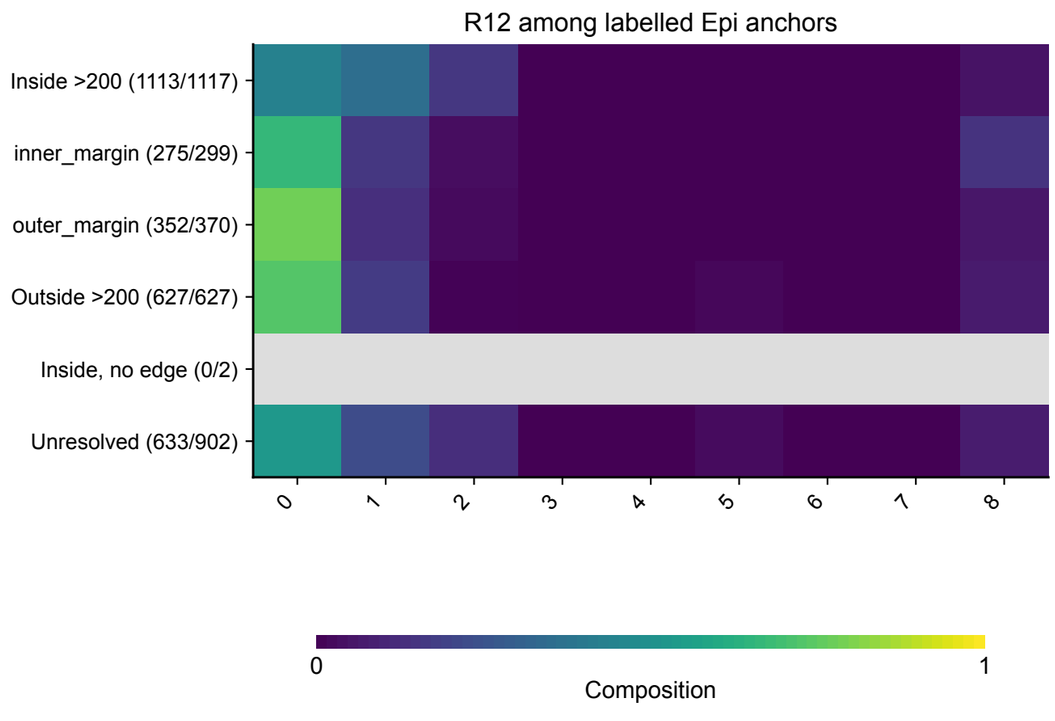
Orange = zero distance
Line = within-section linear fit

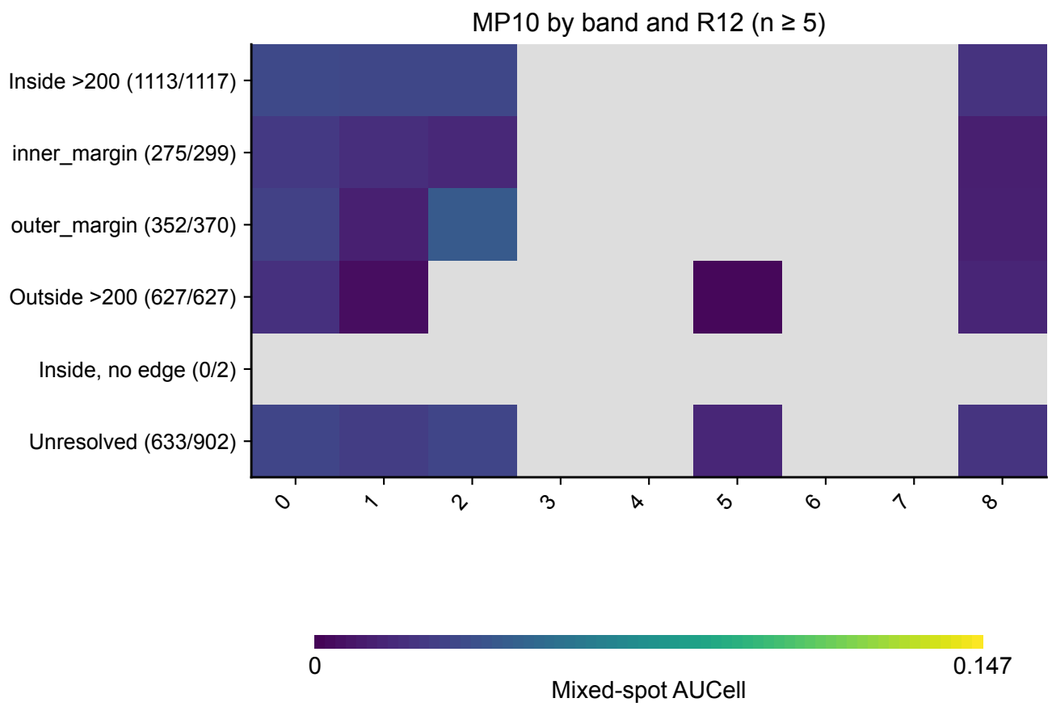
GSE285715_T3 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





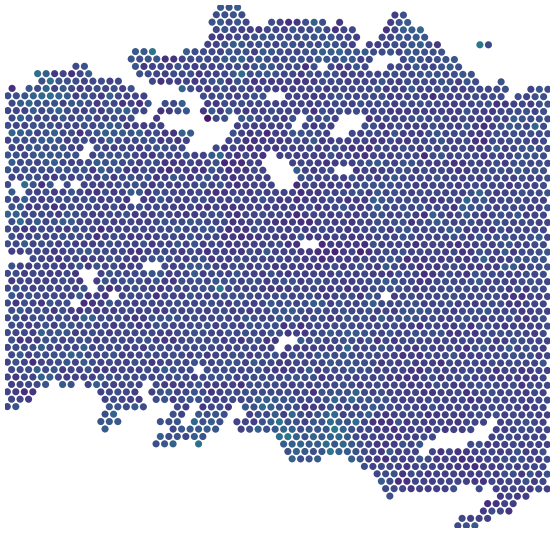




GSE285715_T3 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

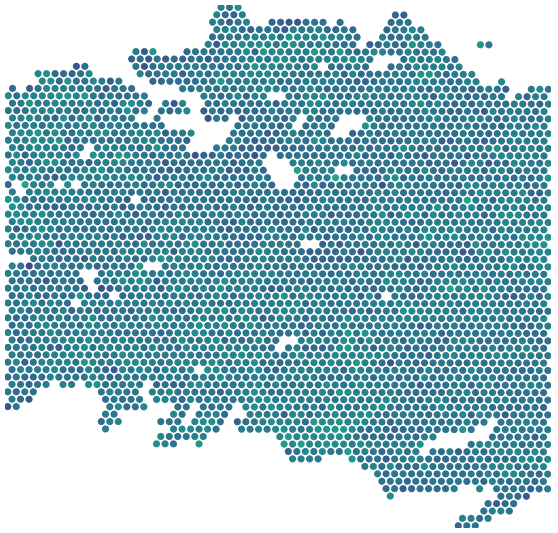
E2F Targets



0 0.152

UCell score

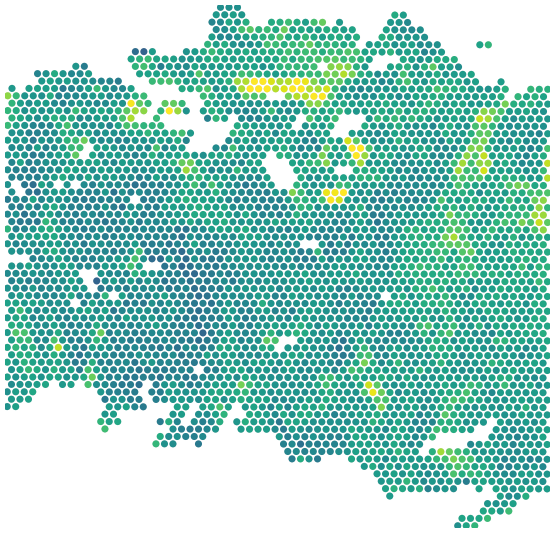
G2M Checkpoint



0 0.134

UCell score

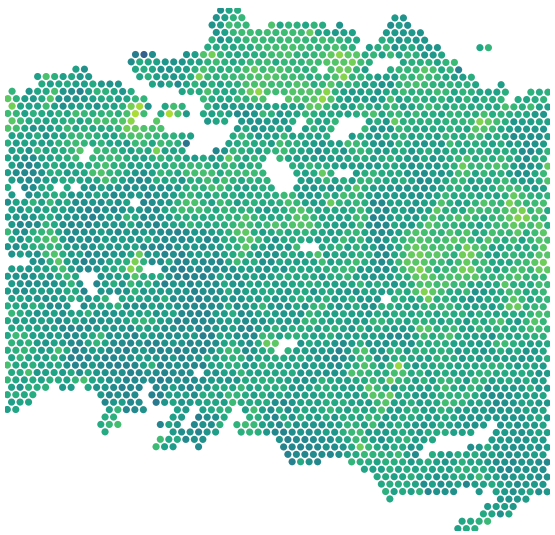
Hypoxia



0 0.211

UCell score

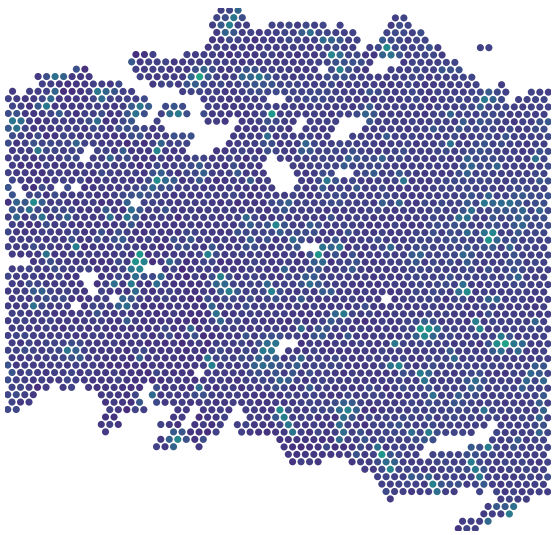
Tnfa Signaling Via Nfkb



0 0.275

UCell score

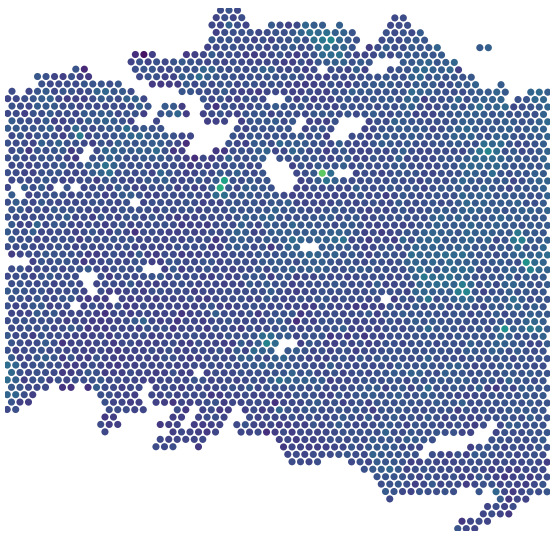
Emt



0 0.386

UCell score

Interferon Gamma Response



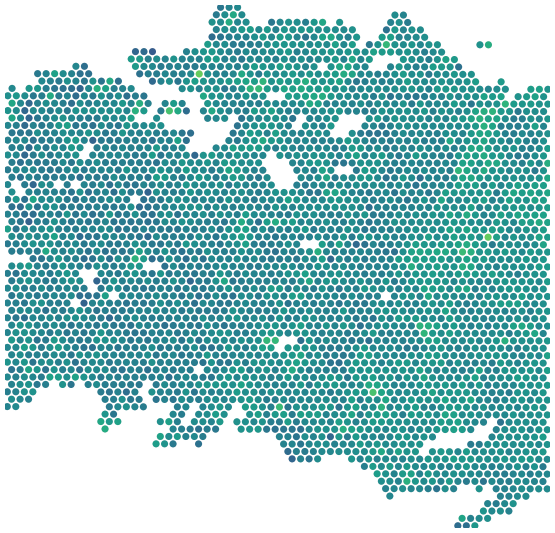
0 0.275

UCell score

GSE285715_T3 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

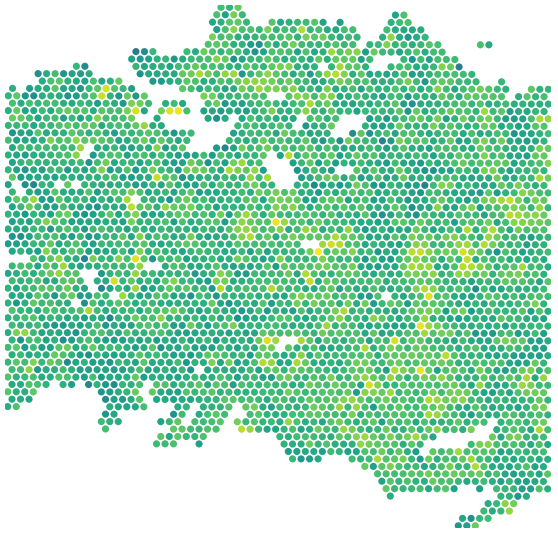
Il2 Stat5 Signaling



0 0.144

UCell score

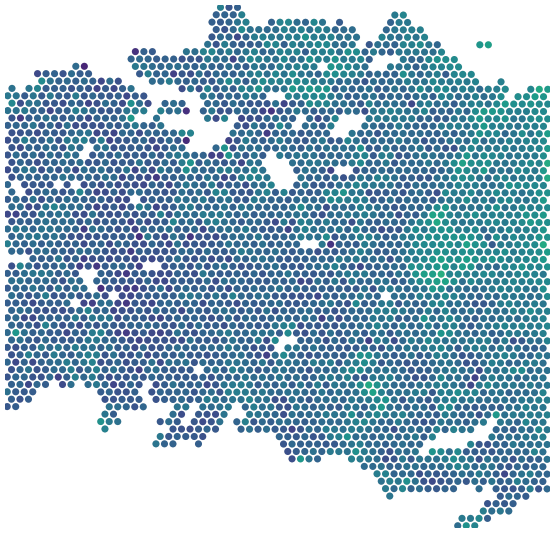
Tgf Beta Signaling



0 0.232

UCell score

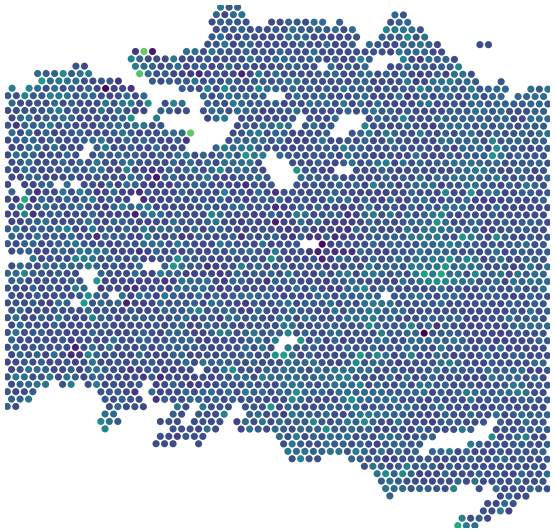
Cholesterol Homeostasis



0 0.21

UCell score

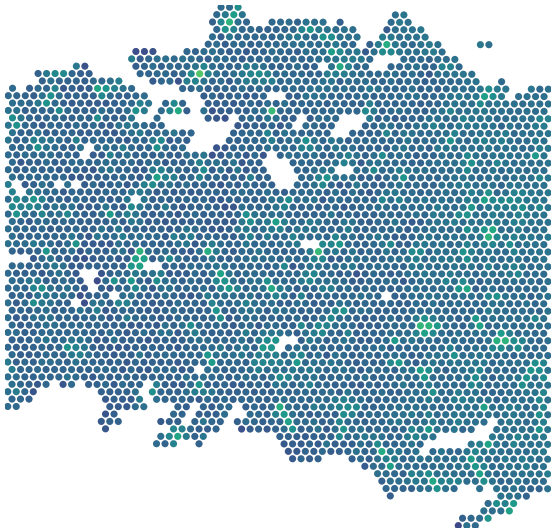
Cholesterol Biosynthesis



0 0.166

UCell score

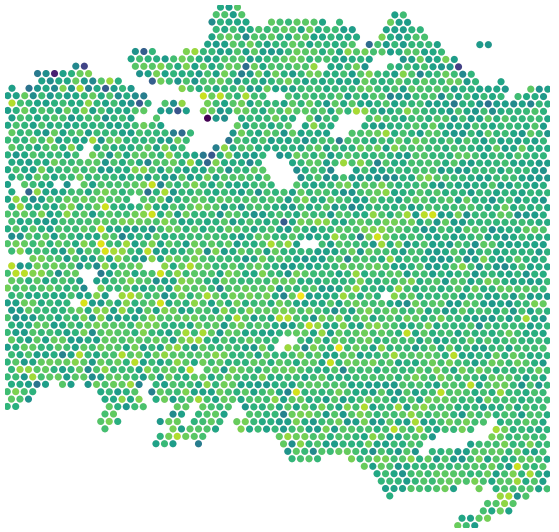
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



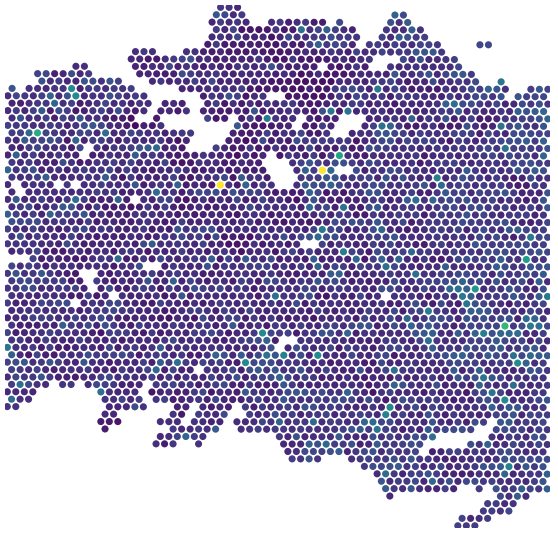
0 0.151

UCell score

GSE285715_T3 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

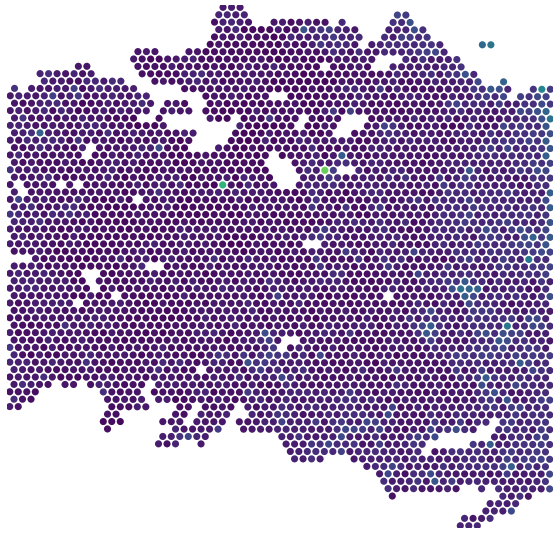
Treg composition



0 0.102

Fraction of all 30

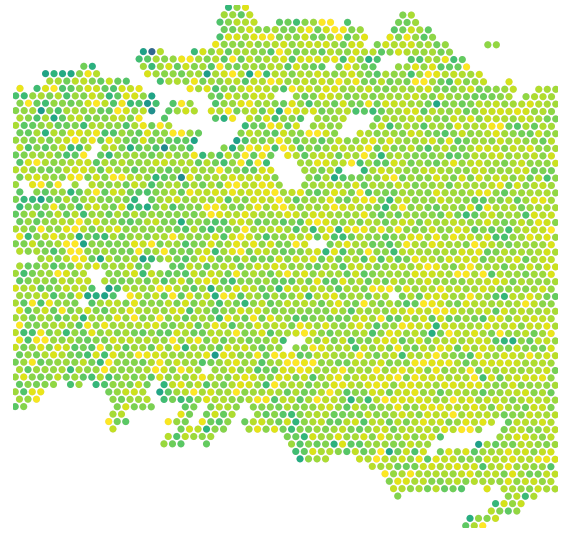
Treg estimated abundance



0 2.85

Posterior mean estimate

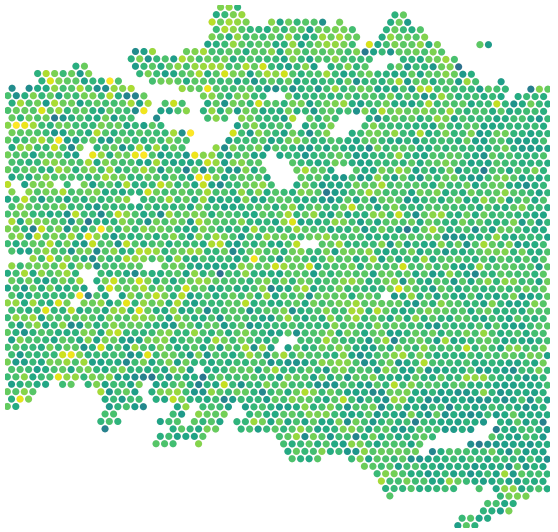
Treg Resting Vs Activated Dn



0 0.0778

UCell score

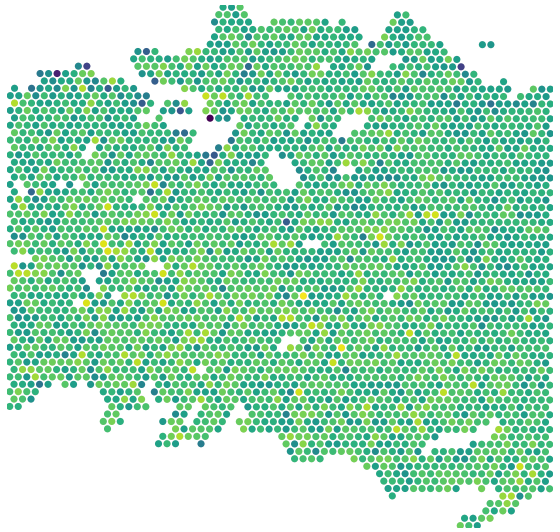
Treg Resting Vs Activated Up



0 0.0609

UCell score

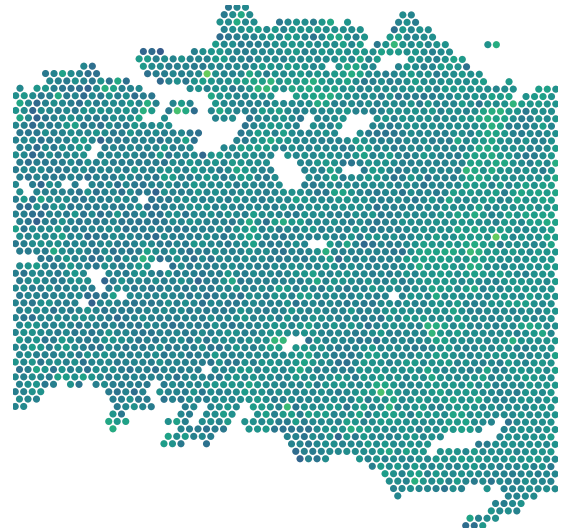
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

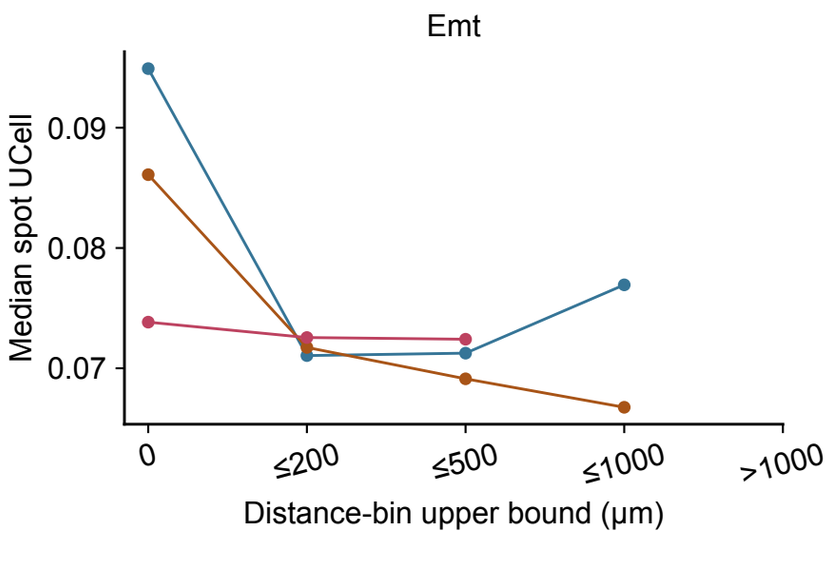


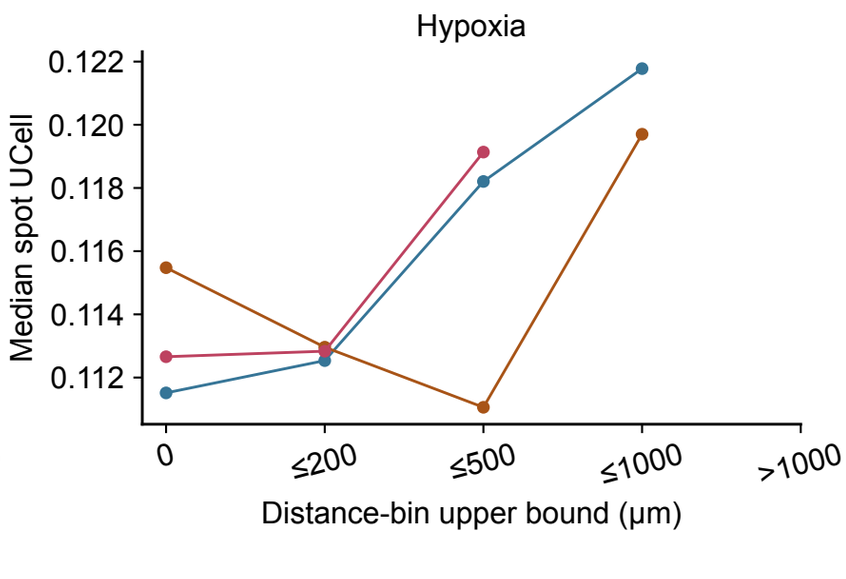
0 0.144

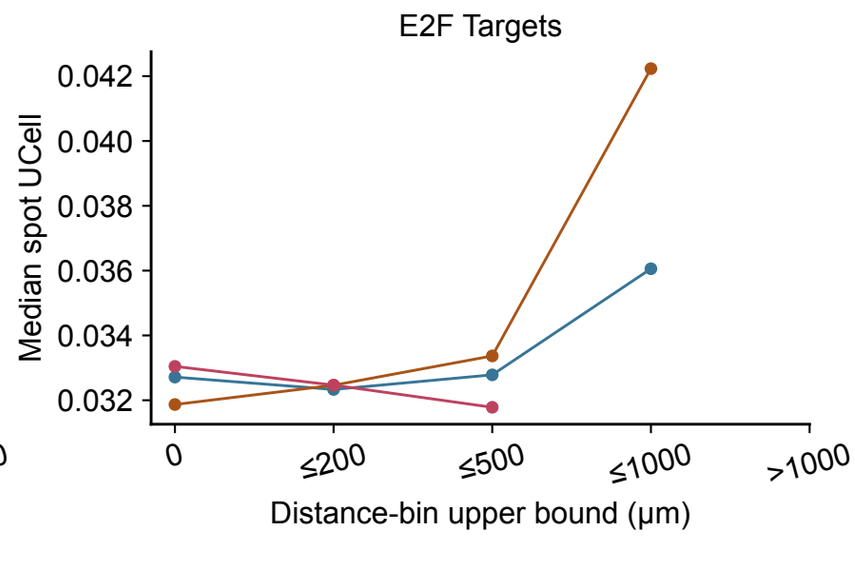
UCell score

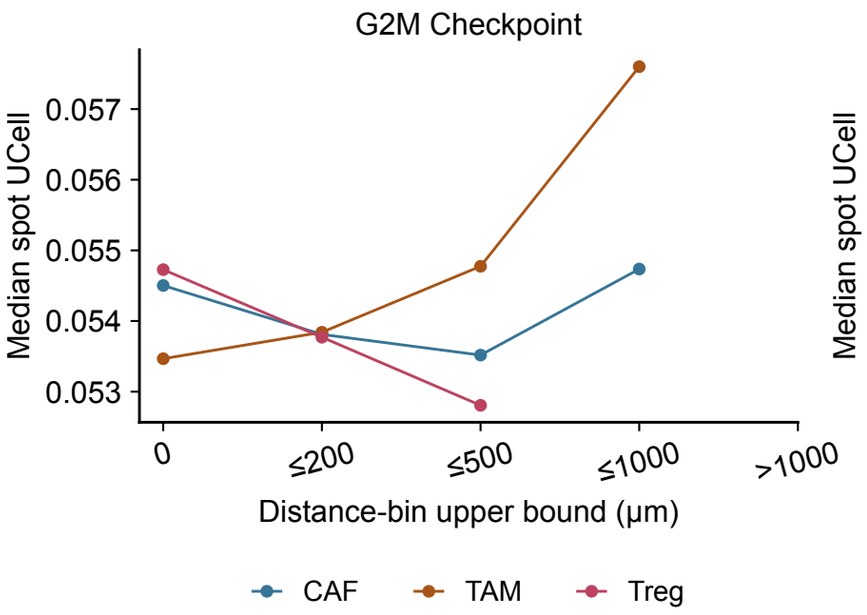
GSE285715_T3 | Programs by target distance

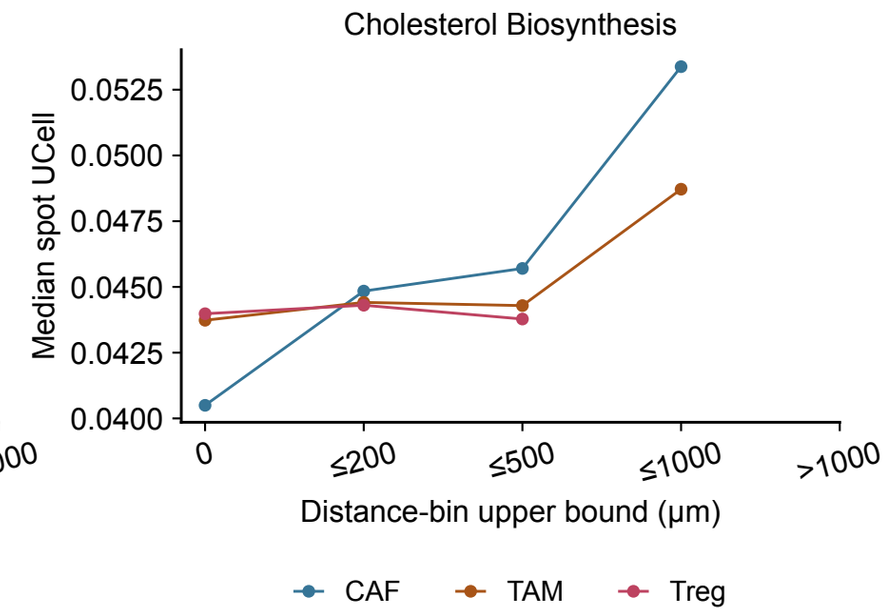
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

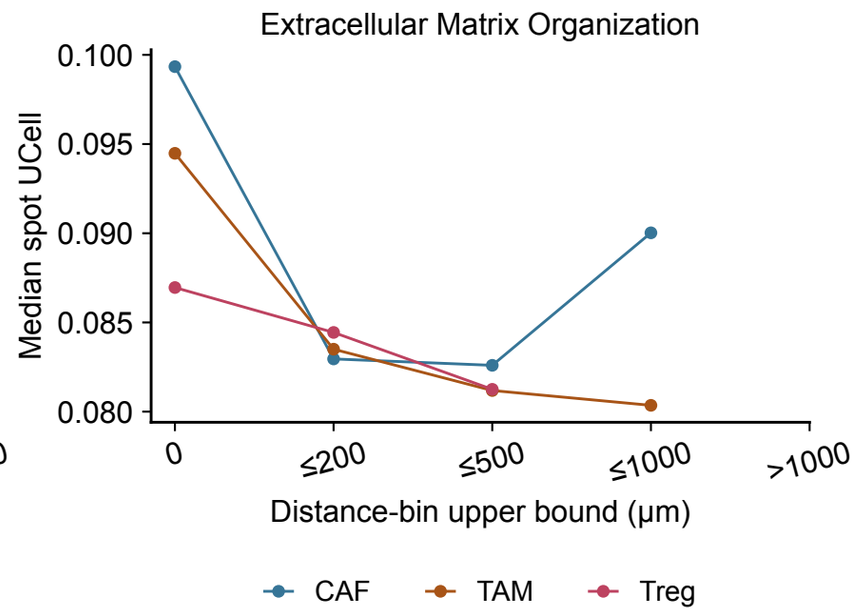












GSE285715_T3 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6